Industrial Automation Wireless

Design Guide

February 2025

Cisco Validated Design

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Industrial Wireless Overview

Executive Summary

Historically, wireless technology in many industrial settings has been limited to non-critical sensing applications and connecting IT devices. With rapid digitization of industrial operations and mobile applications, the need for high-throughput, scalable, reliable, broad wireless connectivity is rising. Wireless connectivity technologies have evolved to support bandwidth-intensive worker productivity applications, reliable mobility for critical assets, and increased data collection from all areas of the plant. All of which significantly boost operational efficiencies and production uptime.

Initially, wireless technologies were designed to serve specific market needs and offered distinct characteristics just for those use cases, such as ubiquitous connectivity for smart devices, cellular communications, wireless sensing applications, and so on. That has changed significantly. Wireless technologies now support reliable low-latency communications, high throughput, and significantly increased density of devices. All of these are useful in many industrial scenarios. Today's wireless technologies also support industrial sensors that generate only low data rates but operate on batteries at low power.

Modern wireless technologies have radically increased the options for connecting devices and applications in industrial zones. In making their technology decisions, organizations must carefully consider different aspects of each technology in the context of end-to-end IP data flow and evaluate them in the organization specific use cases and deployment needs. It is important that they choose the technology best suited for their use case without making compromises on cybersecurity, IP networking, automation, and performance.

This Cisco Validated Design will focus on high-throughput, highly reliable technologies such as WI-FI and Ultra Reliable Wireless Backhaul (URWB) as they could be valid alternatives in many use cases in industrial networking.

Business objectives

Cisco delivers CVDs across a range of technologies and architectures used in a range of industrial verticals. One of the key outcomes of this initiative includes improved plant/factory flexibility by allowing fixed and mobile equipment to be connected via resilient wireless technologies. Another objective involves increasing productivity of plant personnel by automatically discovering network infrastructure and establishing telemetry for monitoring and management purposes. Cisco Validated Designs ensure faster and consistent network deployments, aid in automating networking software and configuration updates, and allows customers to scale. These outcomes will be achieved by deploying industrial wireless in the form of URWB and WI-FI.

Intended Audience

This CVD is intended for anyone deploying Industrial Automation Control Systems (IACS) applications. It is intended to be used by both IT and OT personnel to deploy and optimize secure wireless of industrial system into Enterprise networks. The solution provides industrial automation network and security design guidance for vendors, partners, system implementers, customers, and service providers involved in designing, deploying, or operating production systems. This design guide provides a comprehensive explanation of the Cisco recommended wireless networking for IACS. It includes information about the system architecture, possible deployment models, and best practice guidelines for implementation and optimization.

Key Use Cases and Requirements

Wireless is ideal for connecting remote, mobile, and difficult-to-access applications. Warehouse Operations require a wireless technology to connect machines, sensors, scanners, and video systems with secure, standards-based connectivity that improve operations, supply chain visibility, productivity and safety. Factory Line operations need wireless solutions suitable for mobile AGVs, robots that provide robust connectivity in heavy interference areas due to machinery. The connected workforce within the factory needs reliable wireless to provide voice and video communications to the plant floor from any location at any time.

To provide all the benefits mentioned, automation in manufacturing requires a reliable wireless network that provides:

- **High Bandwidth** to support bandwidth-heavy applications which are becoming more and more common.
- **Low latency** is mandatory for real-time response and not to affect operations and safety.
- **High- availability** to avoid downtime and safety issues
- **Zero-loss handoffs** needed for real-time control of unmanned vehicles and other mobile assets.

Many wireless options

The choice of wireless technology will depend on the use case. It is possible that a mix of wired connectivity along with one or more wireless access technologies is the right solution. This section provides an overview of industrial wireless technologies WI-FI 6/6E and Ultra Reliable Wireless Backhaul.

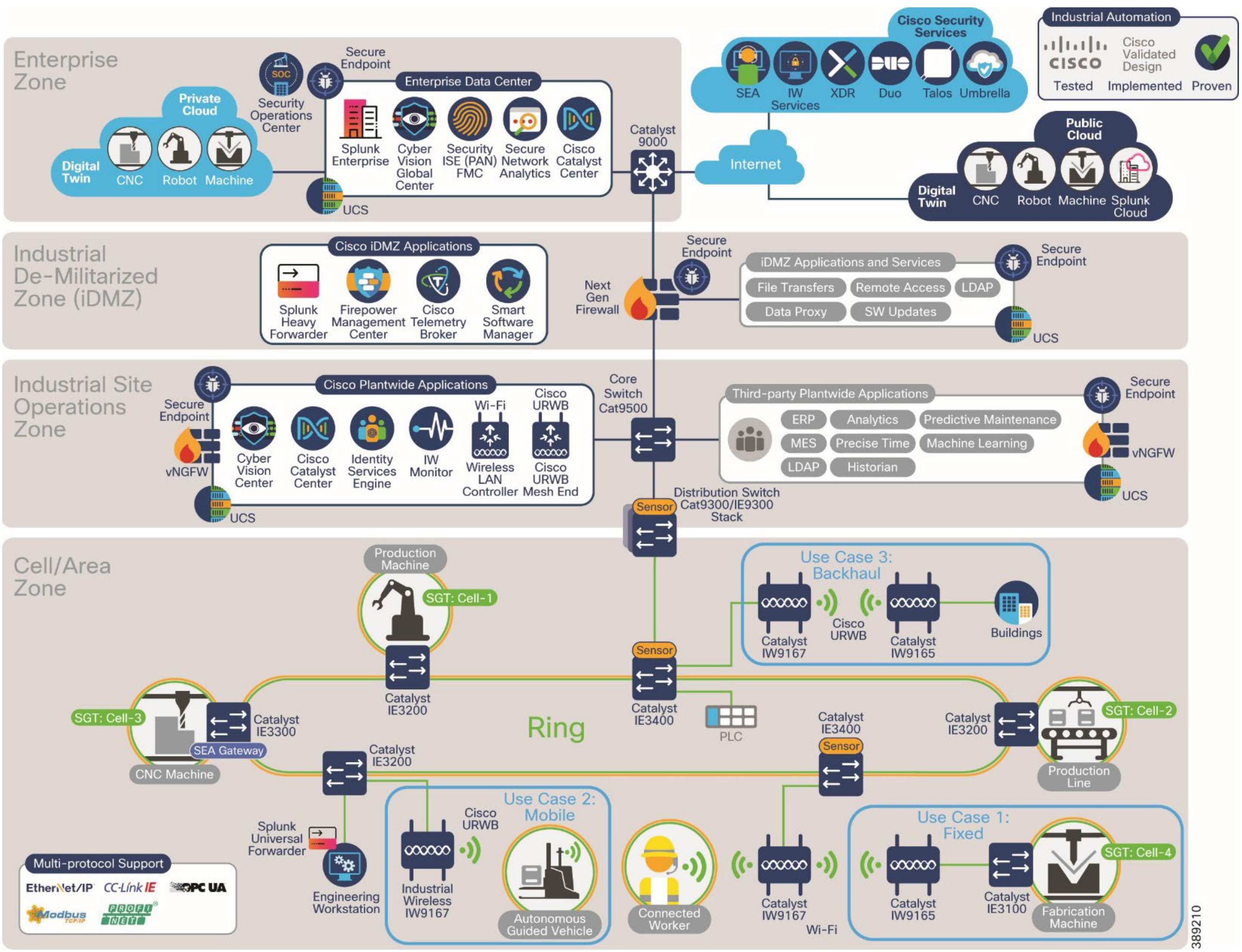
For industrial use cases, WI-FI 6 and especially WI-FI 6E can help accelerate adoption of IoT devices. Operating in the new 6-GHz spectrum, WI-FI 6E offers additional bandwidth and less congestion. The WI-FI 6 and 6E standards also offer power management features that are well suited for battery-operated devices. Spectral efficiency allows dense deployments, and Orthogonal Frequency-Division Multiple Access (OFDMA) allows devices to share the available channel bandwidth with other devices to increase overall capacity. Another advantage of WI-FI 6 is the strong WI-FI Protected Access 3 (WPA3) encryption it offers. WI-FI operates in unlicensed bands but is strictly regulated by countries that define maximum power levels of access points to avoid interference between users.

Ultra Reliable Wireless Backhaul (URWB) is designed to offer reliable wireless connectivity for mission-critical applications, whether they are stationary or mobile. URWB provides low-latency, highly reliable, long-range, and high-bandwidth connections that can handle endpoints moving at high speeds with zero-delay handoffs. It shares the same technology underpinnings as WI-FI which allows URWB to evolve alongside WI-FI. For example, URWB benefits from advances in data rate and additional spectrum in WI-FI 6.

URWB is especially suited for connections requiring 1-Gbps or higher data rates where no fiber is available or where pulling fiber would be prohibitively expensive. Use cases include Layer 2 connectivity to extend a single network between multiple locations, providing connectivity to moving vehicles, and backhaul for remote applications. URWB can be deployed anywhere and can be an excellent alternative to fiber in industrial sites, campuses, or even cities.

Industrial Wireless Reference Architecture

Figure 1 Industrial Wireless Reference Architecture



What's New for Industrial Environments in this CVD

This CVD provides design guidance for integrating an URWB & WI-FI mobility solution to support the factory use cases for mobile IT devices, fixed and mobile industrial assets, and backhaul connectivity. Included are product updates such as the IW9167/5 APs, WI-FI infrastructure management using Catalyst Center, IoT Operations Dashboard, and IW Monitor for monitoring for Industrial Wireless (IW) devices in URWB mode. Figure 1 depicts the reference topology used in validating the wireless use cases in factory.

Solutions Components

For testing and solution validation, the following Cisco infrastructure products were used.

Table 1 Industrial Automation Wireless Components

Component Role	Component	Description
Cell/Area Zone access switches	Cisco Catalyst Industrial Ethernet (IE) 3400 Series Switch and/or Cisco Catalyst Industrial Ethernet (IE) 3300 Series Switch	1Gig REP access ring – IE3400 which serves as the entry point to the network
Distribution Switch	Catalyst 9300-48ux	Distribution 9300 that connects to REP access network in cell/area zone
Network management	Industrial Wireless Service (IW Service) Cisco Catalyst Center (WI-FI)	Network management in the cloud Enterprise WI-FI Network Management
URWB vehicle Mesh Point	Cisco Catalyst IW9165E Rugged Access Point and Wireless Client	AGV mobile unit with on-board client
Onboard vehicle switch	Cisco Catalyst Industrial Ethernet (IE) 3300 Series Switch	Vehicle switch connecting onboard devices such as PLCs, camera, and so on
URWB Infrastructure Mesh Point	Cisco Catalyst IW9167E Heavy Duty Access Point	Infrastructure Mesh Point unit for wireless AGVs
URWB Mesh end	Cisco Catalyst IW9167E Heavy Duty Access Point	Mesh Network demarcation point
URWB IW Monitor	Industrial Wireless Monitor (IW Monitor) VM	Monitoring tool for IW radios residing in DC
Authentication, authorization, and accounting (AAA) network policy administration	Cisco Identity Services Engine (ISE)	AAA and network policy administration
Wireless LAN Controller	Catalyst 9800 -L	Control and provision of APs
Vehicle WGB	Cisco Catalyst IW9165E	Workgroup bridge
Wi-Fi Access Point	Catalyst IW9167E/IW9167I	Wi-Fi Access Points

Figure 2 Cisco Catalyst 6E Industrial Wireless Portfolio

	IW9165E	IW9165D	IW9167
Application	Wireless client for mobile assets	Wireless backhaul for fixed and mobile assets	Wireless backhaul for fixed and mobile assets
Radio	2 x 802.11ax radios (5GHz, 5/6GHz)	2 x 802.11ax radios (5GHz, 5/6GHz)	3 x 802.11ax radios (2.4GHz, 5GHz, 5/6GHz)
Antenna	4 x RP-SMA	Built-in 15dBi directional plus 2 x N-Type (f)	8 x N-Type (f)
Modulation	2x2 MIMO	2x2 MIMO	4x4 MIMO
Wireless Mode	WGB or URWB	URWB	WiFi, WGB, URWB
Ethernet	1 x 2.5Gbps + 1 x 1Gbps RJ45 Optional M12 adapter	1 x 2.5Gbps + 1 x 1Gbps RJ45 Optional M12 adapters	1 x 5Gbps RJ45 + 1 x SFP+ Optional M12 adapters
Expendability	BLE, GNSS, GPIO	BLE, GNSS	BLE, GNSS
Certifications	IP30, EN50155 -40C to +70C	IP67 -50C to +75C	IP67, EN50155 -50C to +75C

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When designing for an Industrial wireless network within factory environments, it is recommended to utilize the Industrial Wireless product line. The Catalyst IW9167E should be used as an infrastructure mode access point providing coverage to AGV and client devices in WI-FI or URWB mode. It has 4x4 radios (2.4GHz, 5GHz, and 5/6GHz) and ports for external antennas which provide flexibility in deployment based on the use case. Not shown above is the IW9167I which comes with a built-in antenna that enables high throughput connectivity to high density WI-FI clients. Additional information on the Catalyst IW9167 can be found on the datasheet.
https://www.cisco.com/c/en/us/products/collateral/wireless/catalyst-iw9167-series/cat-iw9167e-heavy-duty-ap-ds.html

Also shown above is the Catalyst IW9165E model access point applicable for use as a wireless client for mobile assets operating in workgroup bridge mode (WGB) or URWB.

Figure 3 IEC6400 Edge Compute Appliance

Cisco offers the IEC6400 Edge Compute Appliance for URWB networks that allow scalability up to hundreds of IW devices up to 40 Gbps in overall capacity. It works as an aggregation point for all the MPLS-over-the-wireless communications in IW networks.

Table 2 IEC 6400 Physical specs

	IEC6400
Scalability/Core	up to 40 GBps 3rd Gen Intel Xeon Scalable Processors
Ports – Rj45	Dual 10Gbase-T intel x550 Ethernet ports
Ports - Fiber	(Optional) UCS VIC 1455 Quad Port 10/25G SFP28 CNA PCIE
Power Supply	Cisco UCS 1050W AC power for Rack Server

Figure 4 Catalyst 9800 Wireless LAN Controller

In industrial automation wireless network, the WLC resides in the datacenter with other shared services and can address large-scale plant-wide 802.11 wireless needs. AP network licenses are perpetual and are required to connect Aps to WLCs. No licenses are required to boot up a Catalyst 9800 controller, however, to utilized Cisco's Catalyst Center features on the WLC, Catalyst Center software subscriptions are required. For more information see the datasheet.

Solution Hardware and Software Compatibility

Table 3 Cisco products and software versions

Component Role	Hardware Model	Version
Access Infrastructure Network	IE400-8P2S	17.15.1
Core Network - L3	C9300-48UX	17.15.1
Network Management	Cisco Catalyst Center Appliance	2.3.7.0
Authentication, Authorization, and Accounting (AAA) server	Cisco ISE Virtual Appliance	3.2
URWB Mesh Point	IW9167E, IW9165E	17.15.1
URWB Mesh End	IW9167E	17.15.1
URWB IW Monitor	IW Monitor VM	V2.0
Wireless LAN Controller	C9800-L	17.15
Wi-Fi Access Points	IW9167I/E	17.15
Vehicle radio	IW9165E	17.15

References

This solution is based on an integration with other Cisco Solutions. The relevant documents related to this solution include:

- Deploying 802.11 Wireless LAN Technology within a Converged Plantwide Ethernet Architecture Design and Implementation Guide: https://www.cisco.com/c/en/us/td/docs/solutions/Verticals/CPwE/NovCVD/CPwE_WLAN_CVD.html
- WPA3 Deployment Guide: https://www.cisco.com/c/en/us/td/docs/wireless/controller/9800/technical-reference/wpa3-dg.html
- Industrial Automation Reliable wireless for Factory AGV/AMR Environments https://www.cisco.com/c/en/us/td/docs/solutions/Verticals/Industrial_Automation/IA_Verticals/Factory/IA-Factory-CRD1.pdf
- Industrial Automation Security Design Guide 2.0: https://www.cisco.com/c/en/us/td/docs/Technology/industrial-automation-security-design-guide.html
- Industrial Automation Network Design Guide: https://www.cisco.com/c/en/us/td/docs/solutions/Verticals/Industrial_Automation/IA_Horizontal/DG/Industrial-AutomationDG/Industrial-AutomationDG.html
- Cisco IoT Solutions Design Guides: https://www.cisco.com/go/iotcvd
  Renewable Energy – Offshore Wind Farm Design Guide:
- https://www.cisco.com/c/dam/en/us/solutions/collateral/enterprise/design-zone-industry-solutions/wind-farm-design-guide.pdf

The following industrial directives, whitepapers, and publications are also referenced in this guide.

- Choose Wireless That is best-Fit and Not Force-Fit for Your Industrial Network White Paper. https://www.cisco.com/c/en/us/solutions/collateral/internet-of-things/iot-network-connectivity/wireless-operations-ind-network-wp.html
- CIP Safety: Safety networking for today and beyond. https://literature.rockwellautomation.com/idc/groups/literature/documents/wp/safety-wp038_-en-p.pdf

Customers and partners with an appropriate Cisco Account (CCO account) can access additional IoT solutions sales collaterals and technical presentations via IoT Solution hub: https://salesconnect.cisco.com/IoT/s/solutions

Industrial Wireless Use Cases & Requirements

For many years the industrial sector has been heavily reliant on wired Ethernet connectivity. Wired Ethernet has been more resistant to interference, offers more security than wireless, and provides better bandwidth. Over the years wireless technology has increasingly become more important in industrial factories because it helps manufactures improve efficiency, reduce costs, and increase agility. Now with more mobility needs and advancements in wireless to support higher throughput at lower latency, improved security, more manufacturers are compelled to reevaluate the use of wireless in their applications.

Wireless technologies also allow more flexibility in plant floor configurations, help reduce cost, increase productivity, and run operations more efficiently by removing the need for cabling. Cisco has been continually meeting the evolving needs of the factory environment with wireless technology. WI-FI 6/6E and Ultra-Reliable Wireless Backhaul (URWB) address the need for faster and more reliable connectivity. In the following sections we will discuss URWB and WI-FI solutions and their applicable use cases.

Table 4 Attributes to using WI-FI or URWB for industrial use cases

Attribute	Wi-Fi	URWB
Type of Service	Network access via built-in radio on client devices	Network Access with separate attached URWB radio to wired client devices.
Operating characteristics	Short range, high bandwidth, high capacity, high throughput	Medium to long range, high bandwidth, low latency, seamless handoffs, high throughput.
Spectrum	Unlicensed 2.4, 5, or 6GHz	Unlicensed in 5GHz
Range	Indoor/outdoor. Flexible antenna options	Indoor/outdoor. Flexible antenna options
Mobility	Practical for low speeds	Make before break radio link creation at high speed with 0 packet loss and optional Multipath Operations (MPO) for higher resiliency
Ease of Deployment	Easy to deploy. Can use existing infrastructure. WLC RRM channel planning.	Easy to deploy. Requires new installation and training. Requires static radio channel planning.
Automation, provisioning, management	Enterprise. Fully automated, monitored, and assured with modern network management systems (Catalyst Center)	Management through IW Service. No automation currently.

Industrial Wireless Use Cases

Industrial wireless is no longer simply an extension of the Enterprise network into production environments to connect mobile devices used by plant personnel or external vendors/contractors. IW is fully capable of taking on a larger set of use cases as its bandwidth, scale, resiliency and availability have drastically improved. This section will cover a broader set of use cases including

- Mobile IT devices requiring more bandwidth than ever before
- Fixed Industrial Assets operating critical production processes
- Mobile Industrial Assets safely moving inventory, parts and machines around the production environment
- Backhaul communications to connect to remote/distributed assets and environments

Mobile IT Devices

Mobile IT devices in this sense are defined as portable computing devices that can be carried by a person and are designed to operate without any physical connection. Some examples include smartphones, tablets, e-readers, and handheld scanners. Network latency and jitter requirements for these devices are usually less stringent than mission critical applications and protocols. WI-FI is recommended for worker and device connectivity to the infrastructure because of the WI-FI ability to increase network capacity and bandwidth. Using WI-FI 6 and 6E standards in factory also helps offer power management features which are well suited for battery-operated devices. In the end this helps accelerate the adoption of IoT devices.

Fixed Industrial Assets

Fixed position devices in the Industrial space have a permanent operational location which is primarily known as “static”. Fixed position wireless is an alternative to a wired connection for hard-to-reach and remote locations where cabling is too expensive or problematic to install. Some of the usage areas for these devices include process control, machine condition monitoring, fixed environmental monitoring, and energy industries. Within a Manufacturing environment, a common use case is a stand-alone OEM machine or skid that needs to be integrated within the plant-wide architecture over a wireless link. Within the fixed asset space, the wireless link needs to support PLC to PLC or I/O communication using industry protocols such as CIP I/O, CIP Safety, CIP security, PROFINET, or PROFIsafe. URWB unique feature set is designed for providing wireless connectivity for mission-critical applications whether they are stationary or mobile.

Mobile Industrial Assets

Nomadic equipment consists of equipment that changes position during an operation but remains within the same coverage area within a Cell/Area Zone. Common examples of this type of equipment are rotary platforms and turntables, large Automated Storage and Retrieval Systems (ASRS), assembly systems with tracks, and overhead cranes. Nomadic equipment may require rapid changes in orientation and position of the wireless client in respect to the access point which may result in a change in signal.

Mobile (roaming) equipment changes position during an operation by roaming across multiple coverage areas within the Industrial Zone. Some common examples are automatic guided vehicles (AGVs), large ASRS, overhead cranes, and autonomous mobile robots (AMRs). Wireless solutions for AGVs and robots must provide robust connectivity in areas of heavy interference. URWB can provide ultra reliable communication with its multi-path approach thus making it a viable technology suited for autonomous equipment.

Backhaul Connectivity

Within the manufacturing space there may be unique scenarios where networks need to be extended to remote facilities and areas, for example a marshalling yard for automobiles. Applications such as video surveillance & Closed-Circuit TV (CCTV) which provide asset monitoring and telemetry can be deployed at these remote locations where fiber is too expensive or cannot be deployed. URWB is the recommended solution to backhaul the traffic with high-speed connectivity.

Wireless Requirements

In a manufacturing environment, the wireless requirements will vary depending on the use case. The applications used within a factory environment have different characteristics which may require different approaches to wireless design and implementation.

- Wireless in warehouse operations must connect machines, sensors, scanners, and video systems with secure standards-based connectivity to improve operations, supply chain visibility, productivity and safety.
- Support for CIP and PROFINET industrial protocols. Wired PLC(s) need to communicate with PLC on-board the AGV/AMR to send control commands. The default EtherNet/IP Requested Packet Interval for these protocols should be able to run with low latency <50 ms, otherwise timeouts can initiate a system-wide shut down.
- QoS for prioritizing real-time applications over all other kinds of traffic traversing the wired and wireless network.
- Wireless solution for AGVs must provide robust connectivity in heavy RF interference areas due to machinery.
- Reliable voice and video need to be reliable from any location at any time.

Industrial Automation Protocols

This section provides an overview of the two most used industrial automation protocols today, Common Industrial Protocol (CIP) and PROFINET. The following sections will provide a high-level overview of the protocols and their characteristics, as well as design considerations to support the protocols.

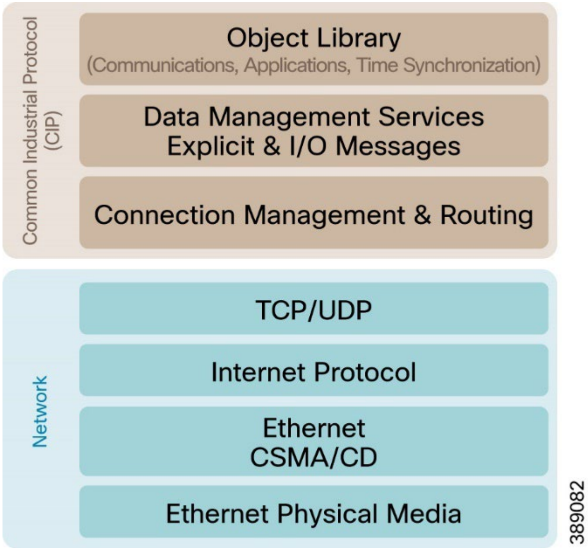
Common Industrial Protocol (CIP)

EtherNet/IP refers to an Ethernet-based industrial communications protocol managed by ODVA Inc, a member-based standards and trade organization. In EtherNet/IP the IP stands for industrial protocol and it relies on the Common Industrial Protocol (CIP). CIP is a media independent protocol using a producer-consumer communication model and is a strictly object-oriented protocol at the upper layers. CIP allows users to integrate automation applications - including control, safety, synchronization, and motion - across all aspects of the business.

Each CIP object has attributes (data), services (commands), connections, and behaviors (relationship between attribute values and services). CIP includes an extensive object library to support general purpose network communications, network services such as file transfer, and typical automation functions such as analog and digital input/output devices, HMI, motion control, and position feedback.

A key feature of CIP is that it defines two types of communication, or messages: explicit and implicit. Explicit messages are used for “as-needed” data (information) and are transmitted via TCP. Implicit messages are used for control data (inputs and outputs) — where high speed and low latency are important — and are transmitted via UDP. The UDP protocol allows messages to be sent in smaller packets and makes it possible to use the producer-consumer model for these critical implicit messages.

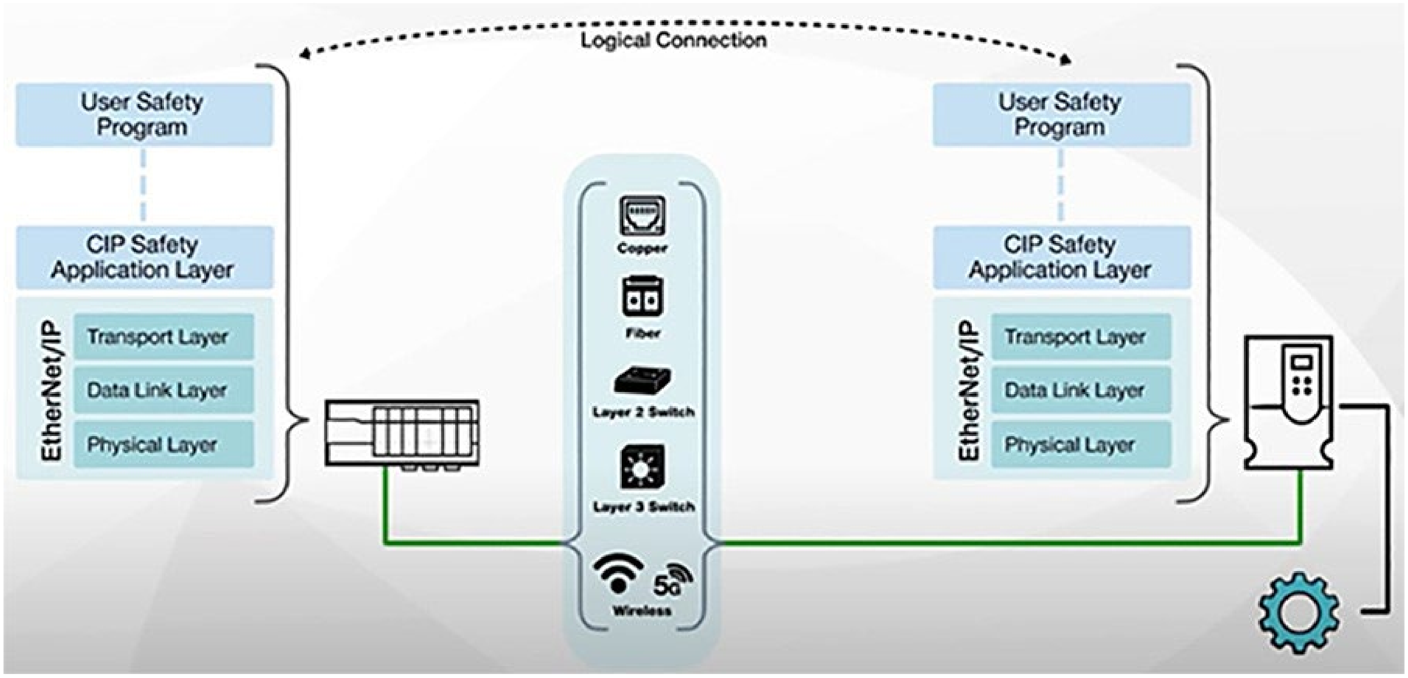
Figure 5 CIP Protocol Stack



CIP Safety

CIP Safety is an extension to the standard capabilities of CIP. It has been certified for use in functional safety applications. CIP Safety devices are connected to the same Ethernet/IP network as the other machine devices. The CIP device is responsible for confirming the integrity of the data, and if an error occurs, it will go into a safe state. There is also a CIP Safety application layer that validates the integrity of the safety data transfers between the safety PLC and the CIP Safety device. There are time stamps and production identifier information that are contained in each packet to ensure the data is from the expected device within the time expected.

Figure 6 CIP Safety Communication Mechanics



CIP Safety over wireless is heavily leveraged in several applications, including factory automation in production environments like those found in automotive assembly plants. In these plants, several different types of conveyance systems, such as AGVs, are used to move vehicles and parts throughout the production cells. Auto manufacturers prioritize the safety of their employees, and the flexibility of the system for future enhancements. This creates the requirement for Functional Safety via CIP Safety, and the wireless link to enable it by meeting certain minimum performance specifications.

In a typical AGV deployment, there are several AGVs traveling hundreds of feet. Each AGV is outfitted with electronics including sensors for collision avoidance, location awareness, and safety, as well as drives and motors for propulsion, I/O, and a safety controller. This AGV system requires a wireless communication connection to the main safety controller. This main controller also serves as the traffic processor for the entire AGV application and thus the system requires control, safety, and diagnostic data to be transmitted wirelessly.

PROFINET

PROFINET is the PROFIBUS International (PI) industrial Ethernet standard designed for automation control communication over Ethernet-based infrastructure.

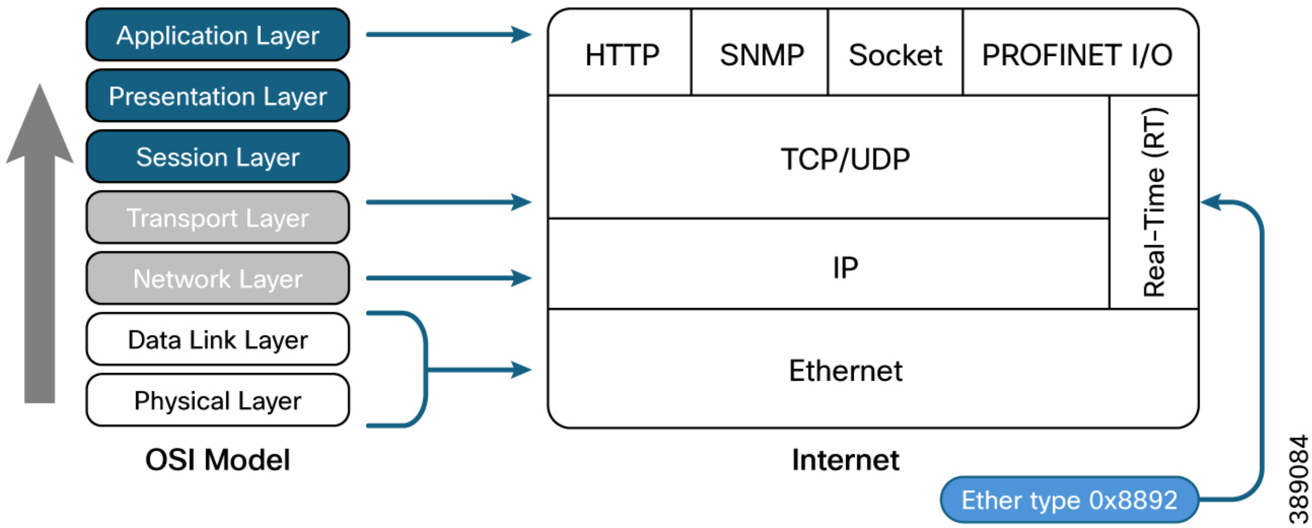
PROFINET devices may require different communication speeds depending on the type of automation process. The PROFINET protocol supports three communication classes, each with a different degree of time sensitivity. These are non-real-time (NRT), Real-time (RT), and Isochronous Real-time (IRT) communication.

NRT, sometimes referred to as TCP/IP communication, is acyclic traffic such as sensory, diagnostic, or maintenance data transferred at best-effort speed. RT communication is cyclic traffic consisting of high-performance process data transmitted over standard networking infrastructure. IRT communication is the highest performing type of deterministic traffic within the PROFINET standard. However, this requires hardware-based bandwidth reservation and network-wide clock synchronization to function.

The PROFINET RT and IRT communication classes involve a cyclic data exchange over standard Ethernet and take place directly on Layer 2 without any TCP/IP overhead to minimize latency. Therefore, it is

essential that any underlying network infrastructure deployed to support RT or IRT PROFINET applications is fully Layer 2 transparent to all connected PROFINET devices.

Figure 7 PROFINET Protocol Stack



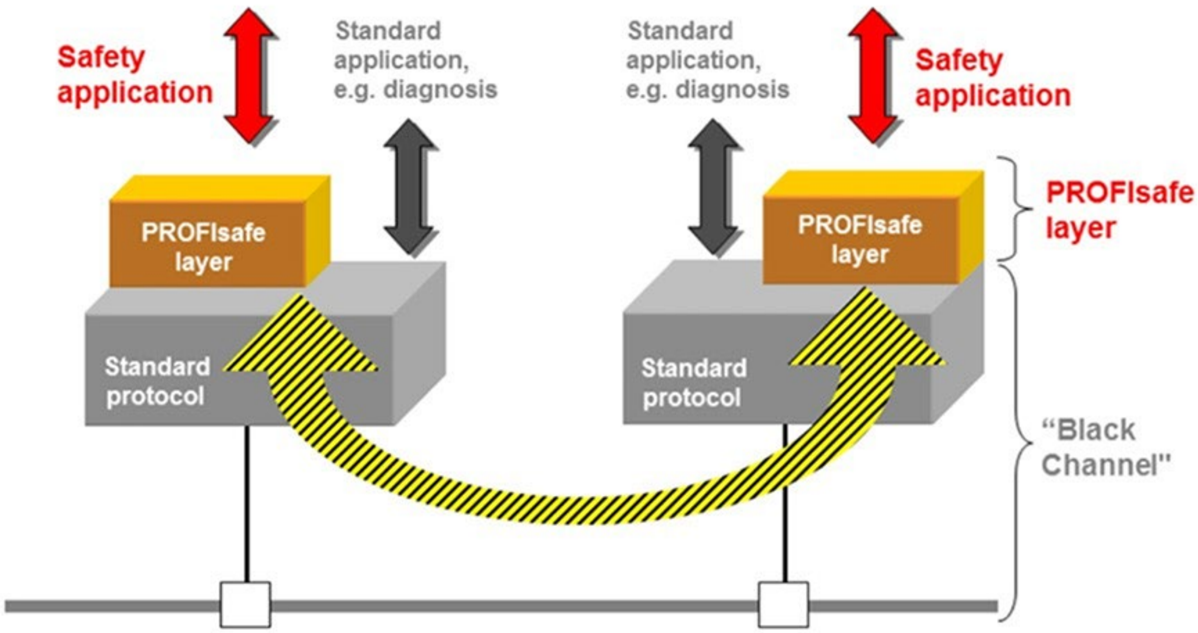
The figure shows PROFINET in relation to the OSI and TCP/IP models. The PROFINET frame is marked with ethertype 0x8892 which is intended for the PROFINET application, thus avoiding the processing time cycling thru the layers.

The performance of PROFINET-based communication is limited to the performance ceiling of the underlying network infrastructure. To provide the flexibility to operate reliably over the different network infrastructure components, the cyclic data exchange rate for PROFINET RT communication can be customized to accommodate any infrastructure limitations or to suit the automation context.

PROFIsafe

PROFIsafe is an additional software layer that provides functional safety over the bus in PROFINET (or PROFIBUS) networks. PROFIsafe will take care of the Functional Safety portion of communications. It ensures the integrity of failsafe signals transmitted between safety devices and a safety controller meeting the relevant safety standards for industrial networks (up to SIL3 according to IEC 61508 / IEC 62061, or Category 4 according to EN 954-1, or PL “e” according to ISO 13849-1).

Figure 8 PROFISAFE Protocol Stack



The diagram above illustrates the difference between PROFIsafe and PROFINET in terms of roles. PROFINET is the overall communication protocol. PROFIsafe is an additional layer that fulfills the functional safety requirements of the application.

PROFIsafe ensures the integrity of failsafe signals transmitted between safety devices and a safety controller meeting the relevant safety standards for Industrial networks. It is even approved for wireless transmission channels such as WI-FI.

Mobility and Roaming

System reliability is critical in obtaining the value of an investment in an AGV/AMR system. Many AGV applications will transmit small bandwidth I/O traffic from each vehicle, but low latency and uninterrupted communication is critical. As the AGV moves about the plant, their wireless links need to roam from one fixed infrastructure radio to another. Roaming requires the wireless link to scan for a radio with a stronger signal once the received signal strength (RSSI) is too weak on the current one and automatically switch to a different radio.

URWB enables seamless roaming through the make before break process which enables a vehicle to maintain end to end connectivity while moving between multiple infrastructure APs. Vehicle radios negotiate with the infrastructure APs and form a new connection to a more favorable AP with better signal before losing its current connection. Other metrics such as infrastructure node traffic load can also be considered as part of the hand off logic.

Within traditional WI-FI the roam process involves a WLAN scanning for a better AP, authenticating to the new AP and then reassociating to the new AP. Fast secure roaming helps avoid a full authentication by skipping the EAP exchange when the client roams between APs. The following table is a list of WI-FI roaming methods supported by Cisco Catalyst 9800 series Wireless Controller Local and FlexConnect deployment modes that can be used in factory environments.

Table 5 Fast secure Roaming supported by Catalyst 9800

Method	Local mode	FlexConnect (connected mode) (central authentication)	FlexConnect (connected mode) (local authentication)	FlexConnect (standalone)
SKC	Not supported	Not supported	Not supported	Not supported
OKC	Supported	Supported	Supported	Supported*
CCKM	Supported	Supported	Supported	Not supported
FT-802.11r/ Adaptive 11r (over the air)	Supported	Supported	Supported	Supported*
FT-8011.r/ Adaptive 11r (over the DS)	Supported	Supported	Not supported	Not supported

The Catalyst 9800 series support the most popular fast secure roaming methods in use today with each method having some advantages and disadvantages. Ultimately it is the client device that decides to roam and for fast roaming to work the client must also support the chosen method within the WLAN infrastructure.

Latency and Jitter

Wireless communication can lead to higher latency and jitter than wired Ethernet for real-time IACS traffic for various reasons such as excessive radio interference, insufficient bandwidth, and network congestion just to name a few.

- Excessive Radio Interference can directly cause latency as when a device experiences a weak disrupted signal due to interference, this leads to delays in data transmission as devices need to re-attempt sending data packets.
- Insufficient bandwidth can directly lead to latency on a network, not having enough network capacity to handle the data traffic can lead to slower speeds and packet delays.
- Network Congestion: can directly cause roaming issues on devices like smartphones or laptops because when a network is overloaded with traffic, throughput will be impacted, prompting the device to switch to a different access point (AP) more frequently, resulting in frequent roaming occurrences and potential packet loss.

Table 6 IACS Application characteristics

IACS Traffic Type	CIP STANDARD	Supported	Constraints
I/O Control **Peer-to-peer Control**	ICIP Class 1 Produced/Consumed Distributed I/O PROFINET RT	Yes	Application should tolerate occasional high latency, jitter and dropped packets; Packet rate restrictions
Safety Control	CIP Safety PROFISafe	Yes	Fast safety reaction times may not be supported
Supervisory Information and diagnostics, peer-to-peer messaging	CIP class 3 (HMI) CIP class 3 (MSG)	Yes	Need to control bandwidth if combined with CIP Class 1 Standard and Safety traffic
Time Synchronization	CIP Sync	No	Accuracy and reliability can be optimized in specific configurations
Motion Control	Integrated Motion on the EtherNet/IP network (CIP Motion, CIP Sync)	No	Not feasible due to higher latency and jitter and poor CIP Sync accuracy

The table above depicts some characteristics of IAC application traffic profiles and their latency constraints. In addition, some important factors to consider:

- Packet rate in the channel must be below the recommended limit for the intended IACS application. Overloading the channel will lead to excessive latency and jitter due to collisions and retransmissions.
- Applying wireless QoS policy is critical to control latency and jitter.
- A very small percentage of packets can be delayed significantly enough to be unusable. The application should be able to tolerate occasional late and lost packets.

- Latency increases with the number of nodes on the channel, due to contention for the channel access. The average latency, however, may not change significantly.
- Very low Requested packet Interval (RPI) (< 10ms) may not be appropriate for wireless applications.

Resiliency

Network resiliency is another critical part of the manufacturing infrastructure. Network resilience in industrial automation refers to the ability for an industrial network to withstand disruptions, recover quickly from failures and maintain operational continuity even when faced with unexpected events like hardware malfunctions, power outages and cyber-attacks. Essentially, it is about building a robust network infrastructure that can keep operations running smoothly despite potential disturbances. Networking solutions that are secure by design can reduce security concerns and help mitigate downtime as more devices are connected to the networks.

Cisco's wireless portfolio whether it be URWB or Wi-Fi, leverages the benefit of the three main pillars it is built upon: Resiliency, Security, Intelligence. Reliable wireless brings greater flexibility to the environment by reducing network downtime when servicing the wireless infrastructure by utilizing the following:

- Use of resilient network topologies and protocols in the wired infrastructure that satisfy the high availability requirements of the wireless IACS applications and AGV systems such as redundant star and ring topologies for network switches, REP and EtherChannel.
- Use of redundant network switches for the distribution layer, such as the Cisco Catalyst 9300, which utilizes stack Wise technology.
- Access points can provide the same by installing multiple APs that provide overlapping coverage of all required areas within the cell/Area Zone Level 0 – 2.
- Monitor and regulate the amount of data that is transmitted over the wireless channel.
- Using spectrum analysis tools to proactively detect and mitigate interferers and unauthorized transmission. Create and enforce spectrum policies at the site.
- The WLC provides redundancy using an active/hot-standby configuration. To provide the backup/failure performance, it is recommended that the controller be connected via port aggregate links (EtherChannel) which provides physical redundant connectivity between the WLC and the wired network.
- URWB is equipped with enhancements to help devices maintain their connection while roaming, maintain multiple connections and replicate packets to reduce and eliminate data loss.

Cyber Security Requirements

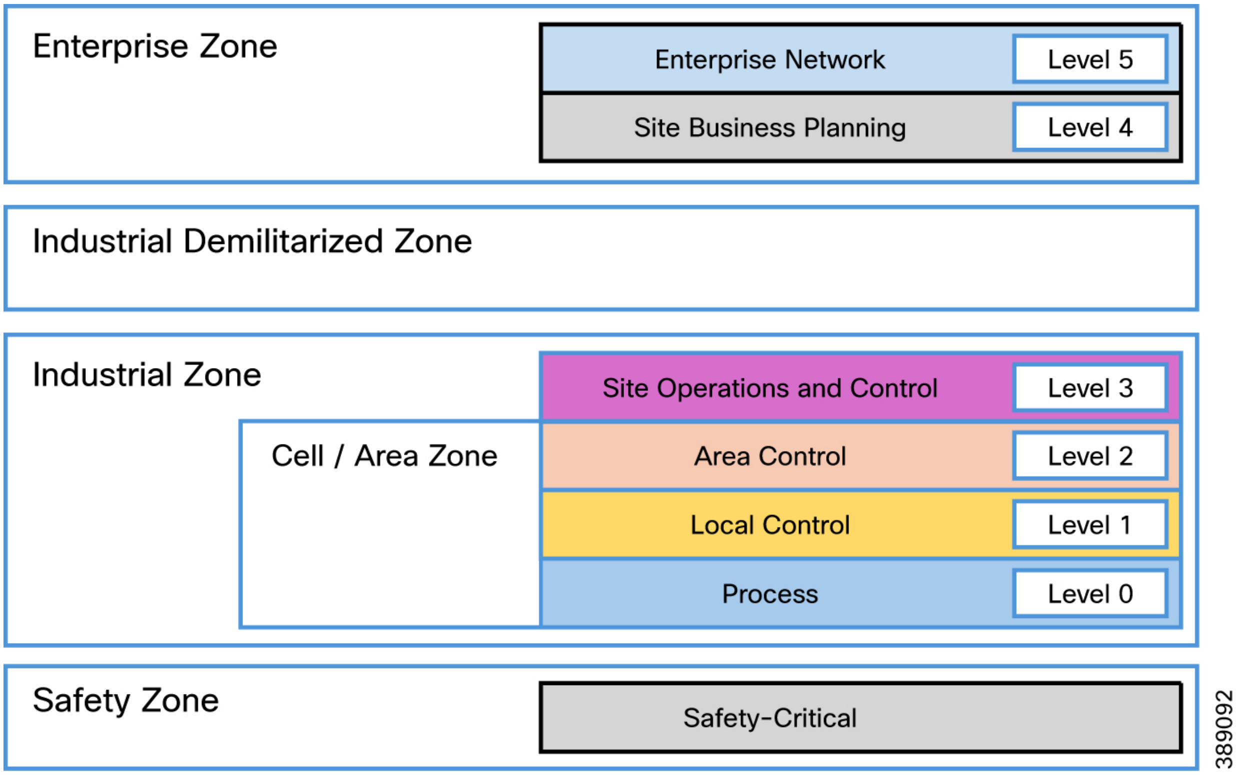
Protecting industrial automation and control systems (IACS) from cyber threats is critical and should be a top priority. Cisco industrial security architecture simplifies complexity across the network by implementing a model that focuses on the use cases an organization must secure.

The asset network operator must adopt the following fundamental principles to ensure secure systems:

- Visibility of all devices in the plant networks: Traditionally, enterprise devices such as laptops, mobile phones, printers, and scanners are identified by the enterprise management systems when these devices access the network. This visibility can be extended to all devices on plant network and is provided by Cisco Cyber Vision. (For more information see the Cyber Vision data sheet.)
- Segmentation and zoning of the network: Segmentation is a process of bounding the reachability of a device and zoning is defining a layer where all members in that zone have identical security functions. Designing zones in a network is an organized method for managing device access within a zone and controlling communication flows across zones. Segmenting devices further reduces the risk of an infection spreading if a device is subjected to malware.

- Identification and restricting data flows. All devices in a plant network (operational network) and enterprise (IT-managed network) must be identified, authenticated, and authorized. The network must enforce a policy when users and IACS assets attach to the network.
- Detection of network anomalies: Any unusual behavior in network activity must be detected and examined to determine if the change is intentional or due to a malfunction of a device. Detecting network anomalies as soon as possible gives plant operators the ability to remediate abnormalities in the network quickly, which can help reduce possible downtime.
- Detection and mitigation of malware: Unusual behavior by an infected device must be detected immediately, and the security tools should allow remediation actions for an infected device.
- Implementation of appropriate firewalls: Traditional firewalls are not typically built for industrial environments. There is a need for a firewall that can perform deep packet inspection on industrial protocols to identify anomalies in IACS traffic flows.
- Hardening of the networking assets and infrastructure: This critical consideration includes securing key management and control protocols, such as using SSH instead of Telnet for remote sessions, using simple network management protocol (SNMP) V3, and using HTTPS instead of HTTP for device GUIs, wireless rogue policies for wireless networks.
- Monitoring of automation and control protocols: It is important to monitor the IACS protocols for anomalies and abuse.
- Adhering to security standards: In the 1990s, the Purdue Reference Model and ISA 95 created a strong emphasis on architecture that uses segmented levels between various parts of a control system. This approach was further developed in ISA99 and with IEC 62443, which brought focus to risk assessment and business processes. Any security risk assessment identifies which systems are defined as critical control systems, non-critical control systems, and non-control systems.

Figure 9 Logical Cyber Security Industrial Automation Framework



As mentioned previously, one of the main principles of cyber security is network segmentation into zones of IAC applications and devices that have similar security requirements. This process is the next step after operators have gained visibility of all assets within the network. The above figure is a reference to the Purdue Model for control Hierarchy that segments devices and equipment into hierarchical functions.

The **Industrial zone** is important because all the IACS applications, AGVs, devices and controllers critical to monitoring and controlling plan floor operations are in this zone. This zone requires logical segmentation and protection from zones 4 and 5.

The **safety zone** is perhaps the most critical zone in the factory IAC environment. In this zone safety networks are isolated from the rest of the IAC and have color coded hardware and are subject to stringent standards. They often interact with other IACs devices on the same physical infrastructure and require security controls to protect them from malicious actors looking to cause harm.

The **Cell/Area zone** is the foundation of the industrial automation architecture supporting levels 0-2 of the Purdue model. This area will fundamentally be a layer 2 access network, subnet, broadcast domain, virtual local area network (VLAN), or service set identifier (SSID). PLCs communicate with their actuators, and other IAC devices within this level.

Site Operations and Control represents the highest level of the IACS network and completes the segments of the industrial zone. These are carpeted spaces with heating, ventilation and air conditioning (HVAC). Applications are primarily based on standard computing equipment and operating systems, so they are more likely to communicate with standard ethernet and IP networking protocols.

Enterprise Zone is where the traditional IT systems exist. These functioning systems can include wired and wireless access to the enterprise network for services such as internet access, email, SAP, etc. Services are important in this zone but are not viewed as critical to the IACS zone operations. Access to the IACS network from an external zone must be managed and controlled through the industrial demilitarized zone (IDMZ) to maintain security and stability.

The **Industrial DMZ** is deployed within plants to separate the enterprise networks and the operational domain of the plant environment. Network access is usually not permitted between the enterprise and the plant; however, data and services are usually required to be shared between zones.

Segmentation between the enterprise and industrial network is the foundation, however risk of breach remains as malware can still be introduced into the network. The ISA/IEC 62443 also defines a set of principles to be followed in industrial environments:

- Least Privilege: to give users/devices only the rights they need to perform their work, to prevent unwanted access to data or programs and to block or slow an attack if an account is compromised
- Defense in Depth: multiple layered defense techniques to delay or prevent a cyberattack in the industrial network
- Risk Analysis: address risk related to production infrastructure, production capacity (downtime), impact on people (injury, death), and the environment (pollution)

Based on these principles the ISA/IEC 62443 recommends segmenting the functional levels of an industrial network using the different networking technologies listed:

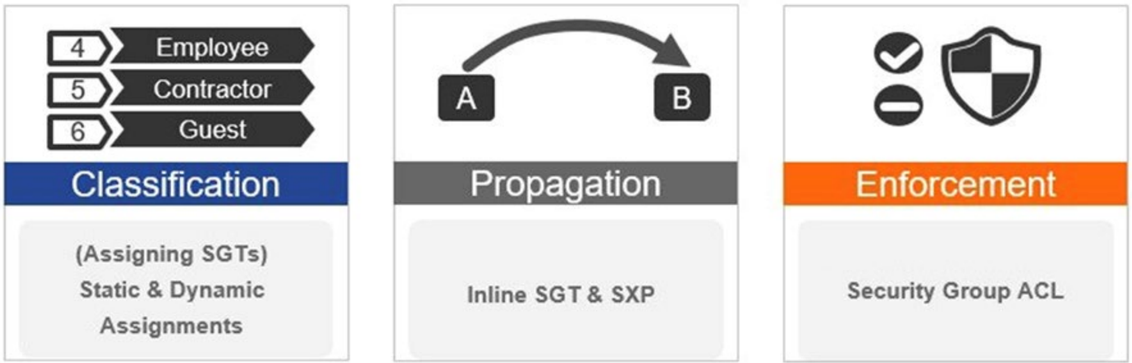
- VLANS
- VRF-lite
- Access Control List
- Stateful Firewall
- TrustSec

To implement segmentation within the OT environment, two segmentation methods can be used:

- Macro-segmentation: Network infrastructure is divided into functional modules. Policy can be applied to larger functional zones based on subnet, VLAN or other network information. For example, assets in interlocking require no communication with assets in video surveillance.
- Micro-segmentation: This can be described as segmentation within a VLAN segment. Using a Cisco TrustSec role or SGT can be used as the means to describe permissions on the network, allowing the interaction of different systems to be determined by comparing SGT values. This avoids the need for additional VLAN provisioning, keeping the access network design simple while avoiding VLAN proliferation and configuration tasks as the number of roles grow.

The Cisco TrustSec (CTS) architecture provides an end-to-end secure network where each entity is authenticated and trusted by its neighbors and communication links secured that help ensure data confidentiality, authenticity and integrity protection. CTS works with ISE to create a consistent and unified set of polices across the network.

Figure 10 TrustSec Components



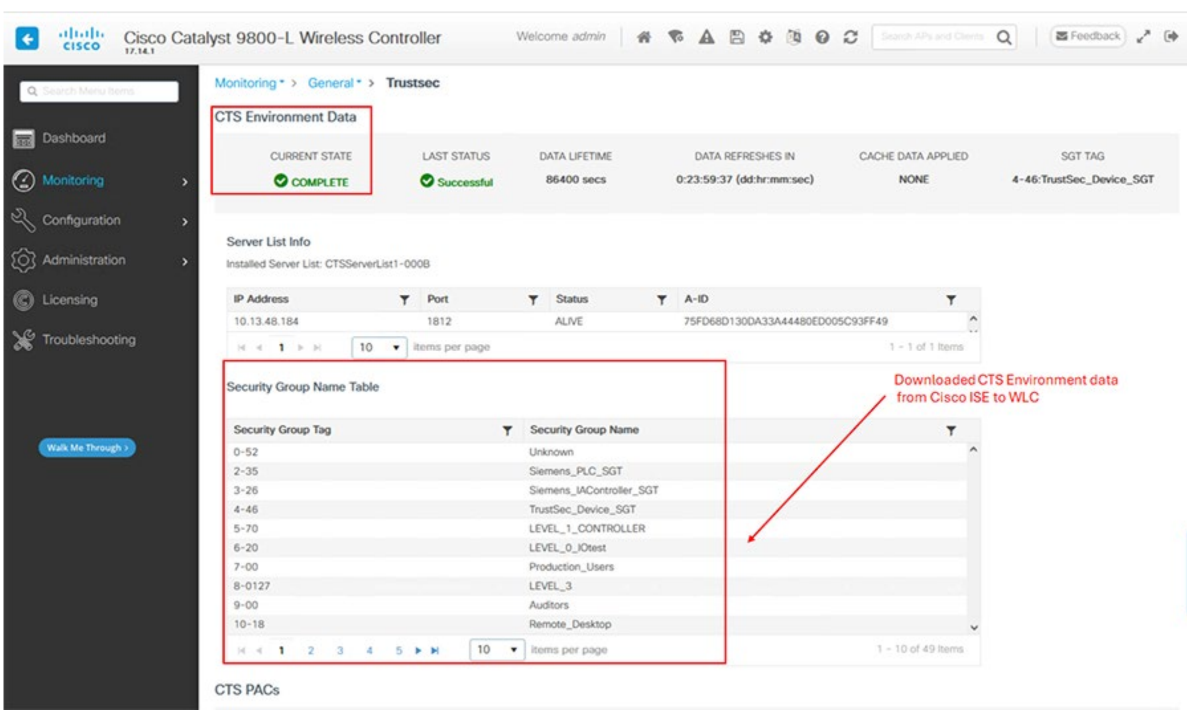
Every end point that enters the TrustSec domain gets classified by Cisco Identity Services Engine (ISE) based on end user identity based on role or device-type and is associated with a unique tag called an SGT (security Group Tag). Security Group Tags (SGTs) are shared with the device that requested the client authentication after successful authentication. This feature allows grouping of clients based on client identity attributes which helps reduce the number of Access Control Entities (ACE).

When devices partake in the CTS, they are required to be authenticated and trusted. The devices connected must go through an enrollment process where the devices obtain the credentials that it specifically needs for CTS authentication and obtains the general environment information. In the context of WI-FI, the WLC device enrollment is initiated by the controller as part of the PAC (protected access credentials) provisioning with ISE.

Environment Data

CTS environment data is a set of information or attribute that help the device to perform CTS related functions. In the plant network the WLC acquires this data from ISE when the device first joins the TrustSec domain.

Figure11 TrustSec enabled WLC



Inline Tagging

Inline tagging functionality is a transport mechanism by which a WLC or an access point understand the source SGT (S-SGT). It covers the following two types:

Central switching: For centrally switched packets, WLC performs inline tagging for all packets sourced from wireless clients that reside on the WLC by tagging it with Cisco Meta Data (CMD) tag. For packets inbound from the DS, inline tagging also involves WLC will strip the packet of the header and send it to the AP over CAPWAP for the AP to learn the S-SGT tag. SGACL enforcement will happen at the AP.

Local switching: For transmitting locally switched traffic, the AP performs inline tagging for packets sourced from clients that reside on the AP. When receiving traffic, AP will handle both locally switched and centrally switched packets and use S-SGT tag for packets and apply the SGACL policy.

With wireless TrustSec enabled on the WLC, the choice of also enabling and configuring SXP to exchange tags with the switches is optional and both modes, SXP speaker mode and inline tagging, are supported. It is recommended to follow the guidance provided within the Industrial Automation Security design guide which is based on ISA/IEC 62443 for building a secure industrial network.

Cisco Identity Service Engine (ISE) does not natively contain profiling services for industrial OT devices. Cisco Cyber Vision can provide more information to ISE based on how much it can profile the end point. When devices and components are placed inside groups, the group tag is also shared with ISE via pxGrid. After the group tag has been shared with ISE, a change of authorization (CoA) is sent to the access switch that the OT asset is connected to. This results in the asset reauthenticating with ISE, ultimately matching the new AA policy defined for this asset.

Develop an Incident Investigation and Response Plan

The last step of the journey is to develop an incident investigation and response plan to reduce Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR). Solutions like Security Information and Event Management (SIEM) and Security Orchestration, Automation, and Response (SOAR) emphasize logs and analysis. The Extended Detection and Response (XDR) solution delivers a unified security incident detection and response platform that automatically collects and correlates data from multiple security components. It provides automation and orchestration capabilities to maximize operational efficiency. Additional tools such as Cisco SecureX, SecureX Ribbon, Threat Response and Orchestration can be utilized to gain contextual awareness into potential threats from various data sources. For more information, please see the Industrial Automation Security design guide.

General wireless technology recommendations

The performance of the wireless network whether it is URWB or Wi-Fi, depends greatly on the RF coverage design and selection of RF parameters such as channels, data rates, and transmit power. The following sections provide an overview of RF recommendations and best practices for wireless IACS applications.

Radio Frequency Design

Certain RF parameters, such as allowed data rates and 802.11n features, should be configured differently for IACS applications using EtherNet/IP data than for a typical enterprise application.

- Configure 6, 12, 24, 54 Mbps data rates as the basic (required) rates in the IACS WLAN.
- For applications with real-time EtherNet/IP traffic (I/O, Produced/Consumed, CIP Safety) a recommended MCS data rate would typically fall within the range of MCS 6 to MCS 9. Although higher MCS values provide faster data rates, they require better signal strength and less interference.

- 802.11ac and 802.11ax data rates can be enabled for WLANs with lower priority EtherNet/IP traffic (HMI, maintenance, Class 3 messaging) or non-IACS client devices and applications.
- Do not use channel bonding (40 MHz bandwidth). Use of wider channels consume the available bandwidth without improving EtherNet/IP performance.
- WI-FI 6E WLANs support AP discovery with In-band and out of band mechanisms. When using in band there are two passive methods of discovery FILS (Fast Initial Link setup) and Unsolicited Probe Response (UPR). Only one can be configured on the WLAN.
- For WLANs operating in 6E only, If FILS is configured, the 6 GHz AP broadcasts an announcement discovery frame approximately every 20 milliseconds which consumes less airtime and reduces probe request overhead.
- When UPR is configured the 6 GHz AP broadcasts a full probe response frame every 20 milliseconds which helps avoid probe storms.
- Preferred Scanning Channel (PSC) is the 3rd discovery method but is active in the approach. This is the only method where clients can send probe requests. With this method the clients can only send probe request on every 4th 20 MHz channel thus reducing the scan list from 59 channels to 15. PSC can be configured in RF profiles.

Wireless Spectrum

The main considerations when allocating wireless spectrum to an IACS application are availability of the channels, bandwidth requirements of the application, and existing and potential sources of interference.

- Use only 5 GHz frequency band for critical IACS applications such as I/O, peer to peer and safety control. The 2.4 GHz band is not recommended for these applications because of limited number of channels, widespread utilization of the band, and much higher chance of interference. Use 2.4 GHz and or 6GHz band for personnel access and non-critical applications on the plant floor.
- Avoid using Dynamic Frequency Selection (DFS) channels in 5 GHz band (channels 52-144) for IACS applications because of potential radar interference.
- Refer to the local regulatory authority, product documentation and the Cisco website for the most recent compliance information and channel availability in a particular country.
- Determine the number of available channels and existing bandwidth utilization in each channel. It is critical to have a spectrum management policy and to coordinate spectrum allocation between IT, OEM, and control engineers.
- Avoid sharing spectrum between different applications, especially under different management and with unknown bandwidth utilization. It is recommended to allocate channels exclusively to the IACS application.
- Calculate required or worst-case packet rate over the wireless media for the application. Determine how many channels are necessary to cover the requirements based on the recommended packet rate limit.
- Perform detailed RF spectrum survey at the site. Adequate time should be spent analyzing the channels to detect intermittent interference throughout the site. Spectrum analysis is critical for IACS applications with high packet rate and low latency requirements.
- Many sources of interference are intermittent, and new sources may appear over time. It is important to proactively monitor for radio interference and rogue sources before and after the deployment. Properly defined and enforced spectrum policy on site is critical for interference prevention.

Wireless Coverage

The main goal when designing wireless coverage for an IACS application is to provide adequate signal strength for wireless clients throughout the Cell/Area zone and to be able to support the required data rate. In addition, wireless cell size should be controlled to achieve desired number of clients per AP, and to minimize co-channel interference between cells.

Determine the maximum number and locations of the wireless clients, including future expansions.

Identify redundancy requirements for coverage, i.e., if two (or more) APs should be seen from any point to provide for failures.

Perform professional site survey to determine the number and locations of the APs that can cover the area with required level of redundancy. The site survey should also determine the appropriate antenna types and verify link performance and supported data rates.

- A good SNR (Signal-to-Noise Ratio) is generally considered to be around 25dB, meaning that the signal strength should be significantly higher than the background noise level. Lower values can indicate potential issues with connection quality and performance. Design the wireless coverage to maintain an minimums RSSI of ~67 dBm, Signal to Noise (SNR) 25 dB and supported data rate of 54Mbps.
- The transmit power of the AP typically matches the transmit power of client adapters or WGBs. For URWB configure transmit power manually for each radio device to provide adequate coverage.
- Change the transmit power from the maximum to reduce signal propagation outside the intended area and to minimize co-channel interference (CCI) onsite.
- Use dynamic channel allocation in the WLAN. Determine if wireless channels must be reused based on the spectrum availability. Channel allocation scheme should provide maximum distance separation between cells using the same channels.
- Do not reuse the channels for wireless cells operating with high utilization and high client count, unless complete signal separation can be achieved.
- If a channel is reused and CCI is expected, the available bandwidth is essentially shared between the wireless cells. The total packet rate should be calculated including every application using the channel.
- If possible, use non-adjacent channels in 5 GHz band for overlapping wireless cells (for example, 36 and 44).
- In Traditional WI-FI, Dynamic Channel Assignment (DCA) will manage the channels dynamically and adjust as needed over time and changing conditions. However, if this is a new installation, or if you have made major changes to DCA such as changing channel widths or adding new APs, you can restart the DCA process. This initializes an aggressive search mode (startup) and provides an optimized starting channel plan.
    - DCA restart should not be performed without change management approval for wireless networks involving real-time based applications.

Antennas

Factory environments can provide challenges for access point placement due to their unique layouts, therefore choosing the right type of antennas for infrastructure access points is critical. The Catalyst

IW9167E is equipped with 8 antenna ports across 3 radio interfaces offering a variety of options for multiple use cases. The Catalyst IW9165E is equipped with 4 antenna ports across 2 radio interfaces. This section covers the recommended antenna types for the Catalyst IW9167E and Catalyst IW9165E.

Figure 12 IW9167E Antenna Port Mapping

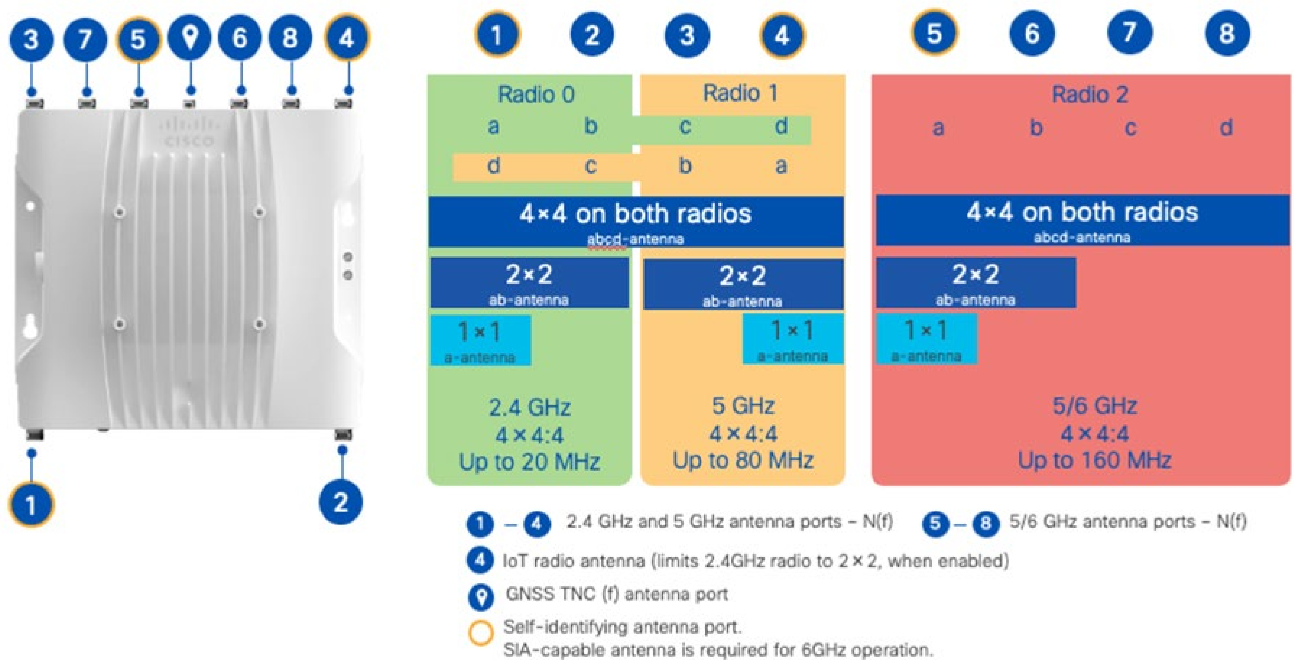
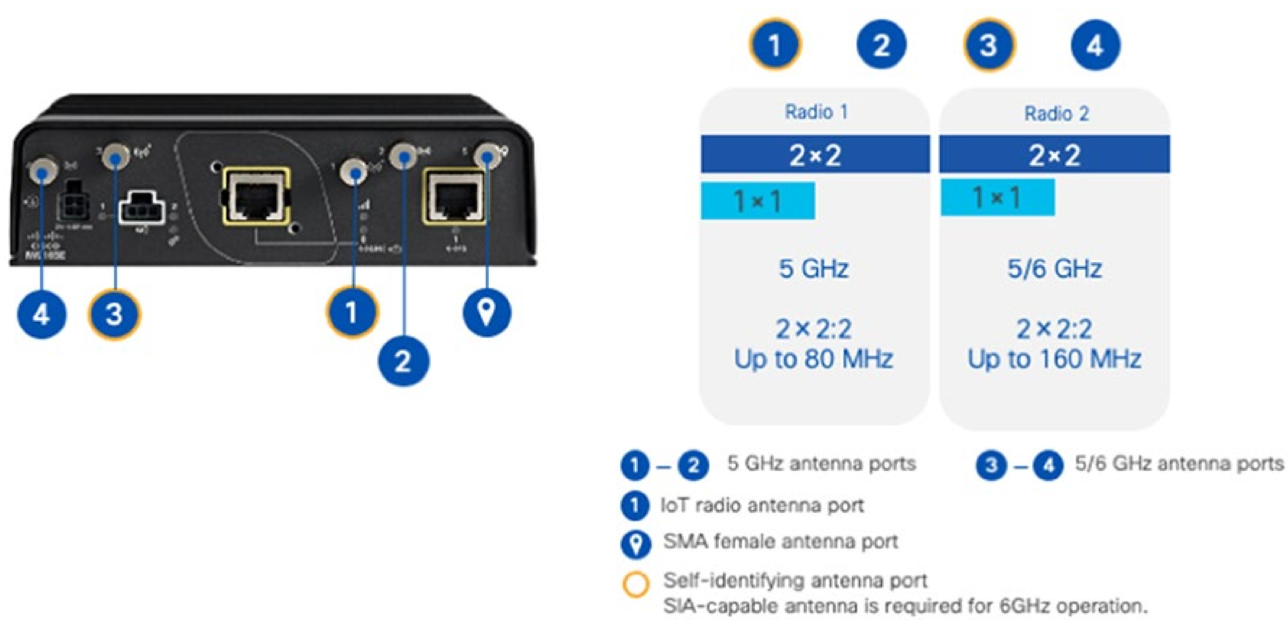
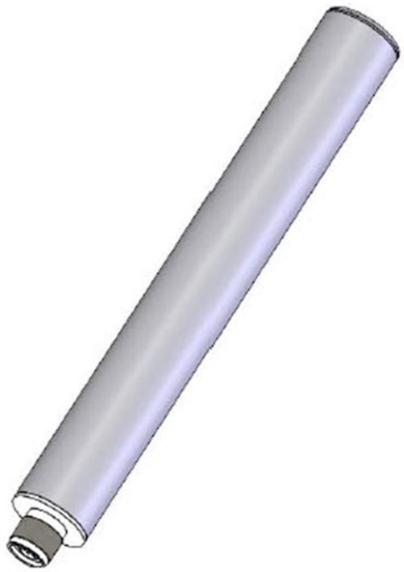


Figure 13 IW9165E Antenna Port Mapping



Omnidirectional antennas are recommended for situations where consistent coverage across a broad area is needed without focusing on a specific direction. This is particularly useful in open spaces or when the device location might change frequently. The Cisco 2.4/5/6 GHz Tri-Band Omnidirectional Antenna is especially designed to support exterior and interior mounted wireless systems. The integrated male N(m) connector is embedded in the antenna base cap for direct mounting to the chosen AP. The ingress protection rating of this antenna element permits either upright or inverted orientation in outdoor locations. The antenna may also be pole-mounted when separation from the AP is required for optimum positioning.

Figure 14 IW-ANT-OMH-2567-N & IW-ANT-OMV-2567-N



The IW-ANT-OMV-2567-N is the vertically polarized model omni which mirrors its counterpart the IW-ANT-OMH-2567-N in design with minor exceptions such as polarity and elevation plane beamwidth. Vertically polarized antennas can apply to use cases where wall mounted AP are deployed and ground wave propagation is needed for short to medium range, whereas horizontal polarized antennas can be more suited for ceiling mounted use cases.

The horizontally polarized omnidirectional base station antenna (IW-ANT-OMV-55-N) design utilizes a linear array, encapsulated in a heavy-duty fiberglass radome with a thick-walled mounting base for reliable, long-term use. This rugged design withstands harsh environments, making the antenna ideal for Industrial Wireless and Military applications. The antennas in this series are DC grounded for ESD protection of radio components. An alternate to this model is the IW-ANT-OMH-55 (not shown).

Figure 15 IW-ANT-OMV-55-N

The IW-ANT-OMM-53-N delivers enhanced connectivity for AGV/AMRs and unmanned systems even in highly obstructed environments. It has a multi-polarized three-dimensional element design and provides maximum reliability and signal stability.

Figure 16 IW-ANT-OMM-53-N

For scenarios where RF coverage needs to be targeted to a specific area, (ex, industrial storage shelving) a directional antenna such as the Industrial Wireless 2 port High Gain Panel Antenna (IW-ANT-PNL5615-NS=) can be utilized for fixed point to point communications.

Figure 17 IW-ANT-PNL5615-NS

In addition to the omni and directional models' antennas described previously other model antennas can be deployed in accordance with the coverage use case. More specific details on the antenna models can be found at the Cisco Industrial Wireless Access Point Antenna Guide.

Cisco Ultra Reliable Wireless Backhaul

Cisco URWB Overview

Cisco Ultra Reliable Wireless Backhaul (Cisco URWB) is designed to offer reliable wireless connectivity for mission-critical applications, whether they are stationary or mobile. Cisco URWB provides low-latency, highly reliable, long-range, and high-bandwidth connections that can handle endpoints moving at high speeds with zero-delay handoffs. URWB is easy to deploy, manage, troubleshoot, and provides full control of the network. Reliable, scalable, and suitable for the most demanding industrial wireless applications, URWB is a leading-edge solution for vehicle connectivity for mission-critical applications.

Operating in unlicensed frequencies (ISM bands), Cisco URWB can be deployed anywhere and can be an excellent alternative to fiber in industrial sites, campuses, or even cities. Using the same frequencies as WI-FI means that customers do not have to pay for use of the spectrum while taking advantage of the high throughput. Sharing the same technology underpinnings as Wi-Fi allows Cisco URWB to evolve alongside Wi-Fi. For example, Cisco URWB benefits from advances in data rate and additional spectrum in Wi-Fi 6 and 6E, and from further progress envisioned for W-Fi 7 and beyond.

Key differentiating features of Cisco URWB include:

- Multiprotocol Label Switching (MPLS) based. Cisco URWB uses a proprietary wireless based MPLS to “route” traffic between Cisco URWB nodes. The protocol allows Cisco URWB to work on a variety of wireless topologies: point-to-point, point-to-multipoint, mesh with fixed and mobile nodes
- Fluidity for mobile use cases. Fluidity supports high-speed mobile hand-offs to reduce or eliminate communication disruption as asset move around the factory environment
- Multi-Path Operations (MPO) can deliver uninterrupted connectivity to fast-moving devices by sending high priority traffic via redundant paths.

MPLS Overlay

URWB uses a proprietary wireless based MPLS transmission protocol to discover and create label-switched paths (LSPs) between Mesh-point radios and mesh-ends(s). MPLS provides an end-to-end packet delivery service operating between levels 2 and 3 of the OSI network stack. It relies on label identifiers, rather than the network destination address as in traditional IP routing, to determine the sequence of nodes to be traversed to reach the end of the path. In essence, the MPLS creates an “overlay” on top of the wired and wireless links.

This Cisco URWB MPLS protocol is typically not configurable by end-users and does not interoperate with other Cisco MPLS-based features and capabilities.

Cisco URWB Components

This section covers the components that make up the URWB architecture. An URWB node can operate as Mesh Points with wireless interfaces enabled or disabled or Mesh ends/Gateways with wireless interfaces enabled or disabled.

Mesh-End

In the Cisco URWB network, the Mesh end is the demarcation point between the wireless mesh network and the wired switch network. Packets destined for the wired network will have the MPLS label popped off while packets coming into the wireless network will have the MPLS label imposed. In terms of MPLS, the Mesh end is a Label Edge Router (LER) that holds all the label switch paths (LSPs) to all the other radios in its database. The ME in the factory environment would operate at the Distribution stack or the Data Center.

A logical Mesh End (ME) can be redundant consisting of two physical Mesh Ends (IEC6400) or radios (IW9167E) with Fast Failover enabled.

Note: The URWB solution has the capability to automatically select the gateway/radio with the lowest Mesh-ID to become the ME, as a best-practice it is highly recommended to configure the role of Mesh End and Mesh Point(s) manually within the deployment to have more deterministic convergence in case of failure within the network.

Mesh Point

The Mesh Point swaps MPLS labels on the received traffic when it is destined for another device in the mesh network. They serve as the primary RF path between the AGV and the wired side network. The vehicle radio is a special kind of Mesh Point that imposes an MPLS label on the incoming data from the onboard device and removes the label before putting it onto the wired network.

A wireless mesh is made of a Mesh End and one or more Mesh Points. The wireless mesh is kept separate from other wireless meshes by having a unique passphrase on the Mesh End and associated Mesh Points.

For this solution the IW9165E is ideal for operating as a Mesh Point configured in vehicle mode for AGV/AMR deployments. For infrastructure Mesh Points the IW9167E is the recommended unit to provide coverage for the URWB network within the Cell/Area zones. The IW9165D can also be used as a Mesh Point to provide connectivity in point-to-point backhaul situations with its directional antenna.

URWB Topology and Operational Modes

URWB wireless meshing technology supports a range of topologies for fixed and mobile deployments. For fixed topologies, URWB supports Point-to-Point, Point-to-Multipoint, and Serial backhaul.

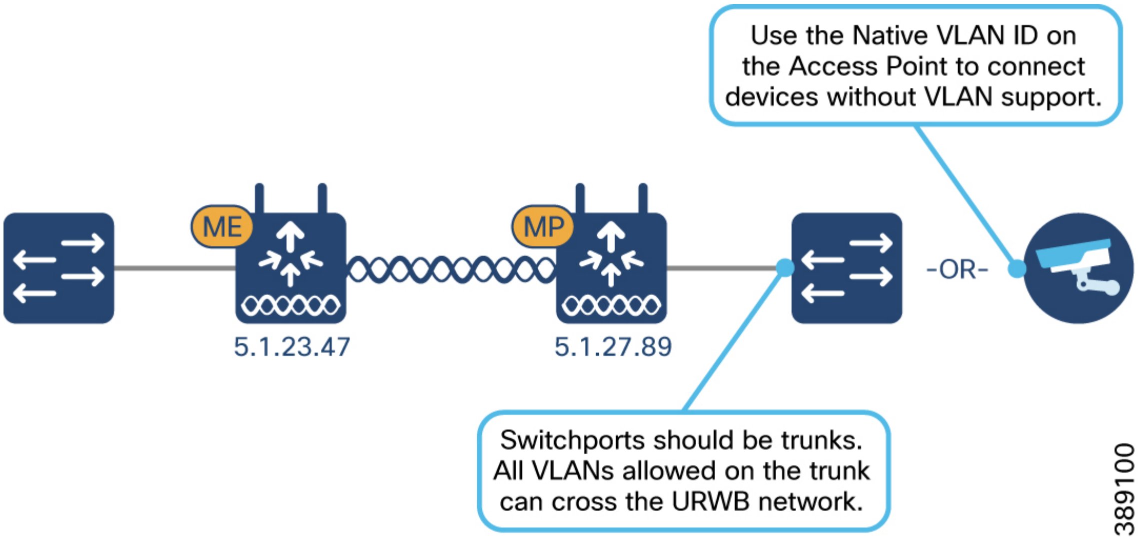
Mobility: Connectivity between radios deployed on vehicles and fixed structures or just between vehicles is generally used to connect Automated Guided Vehicles (AGVs), autonomous mobile robots (AMRs) and other moving assets. In this case, the Cisco URWB radio is set to the Fluidity mode.

Fixed infrastructure: Connectivity between radios attached to fixed structures – poles, towers, buildings, etc. – is generally used to support wireless backhaul, physical surveillance, and more. The layout and data communication can be implemented through Point-To-Point (PTP), Point-To-Multipoint (PTMP) and mesh topologies. In this case, the Cisco URWB radio is set to the Fluidmax mode.

Each operational mode is described in the sections that follow.

Fixed Point-to-Point

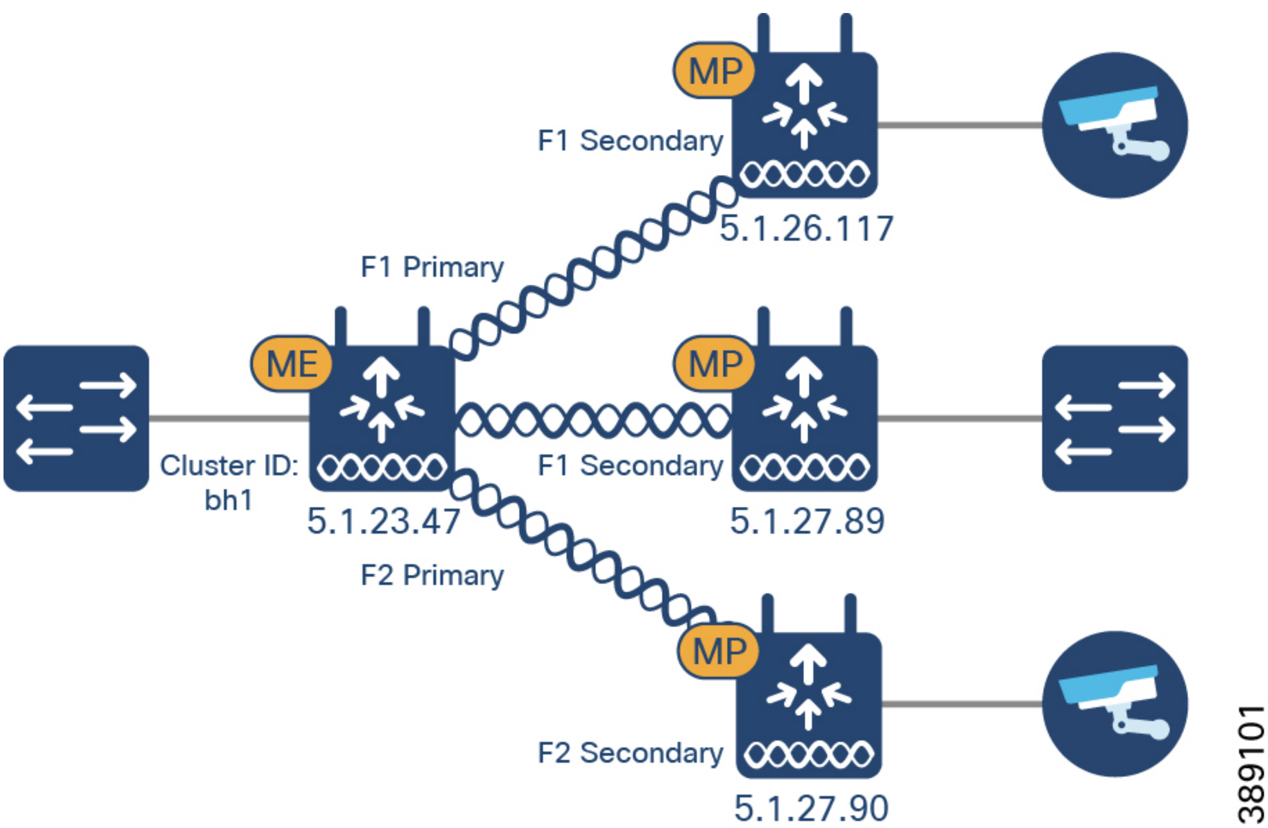
Figure 18 Fixed point to point topologies



Operational Mode – Fluidmax for fixed Point to Multi-Point Topologies

As stated previously URWB units can operate in a fixed deployment mode for point to point and point to multipoint topologies. This is particularly useful for backhauling communication from remote locations, such as a marshalling yard, parking lot or between buildings. Example discussions follow.

Figure 19 Fixed point to multi-point topologies



Fluidmax is a URWB technology that allows the radio to function within the Point-to-Multipoint networks without having to physically replace hardware. It enables high performance with flexibility in fixed scenarios. Within Fluidmax a node can be set as primary or secondary. If a unit is set as primary, it will dictate the operating frequency of the mesh cluster of which it is a primary unit. When a unit is set to secondary and the autoscan feature is enabled the unit will scan the spectrum of available frequencies for a primary unit that shares its cluster ID. The frequency selection feature will be disabled.

Operational Mode – Fluidity for Mobil topologies

Fluidity enables a Fluidity Vehicle (FV) node that is moving between multiple Fluidity Infrastructure (FI) nodes to maintain end-to-end connectivity with seamless handoffs between FI nodes. FV radios negotiate with the FI node and form a new wireless connection to a more favorable FI node with better signal quality before breaking or losing their currently active wireless connections.

Figure 20 Mobility: Fluidity Network topologies (small)

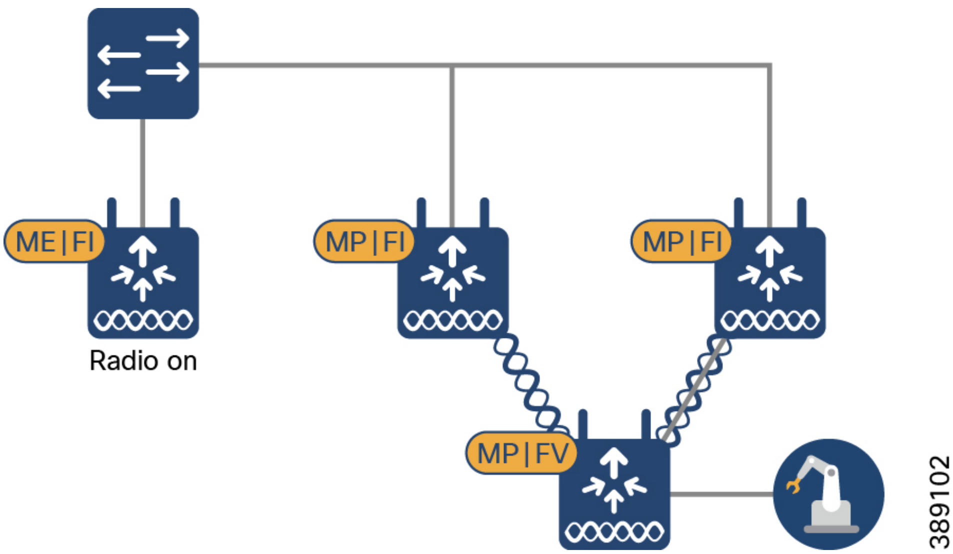
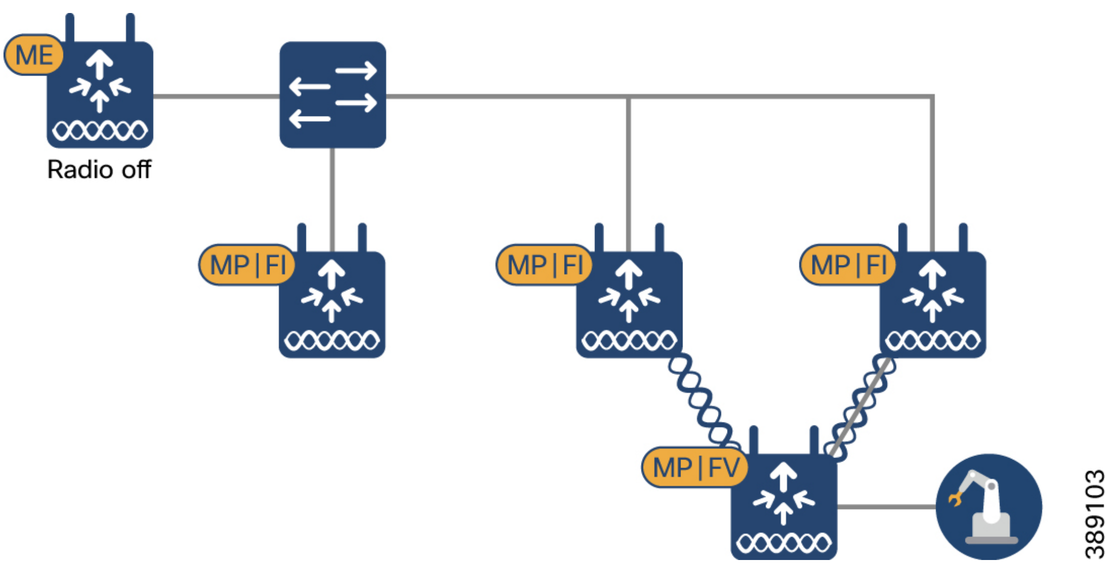


Figure 21 Mobility Fluidity Network topologies (large)

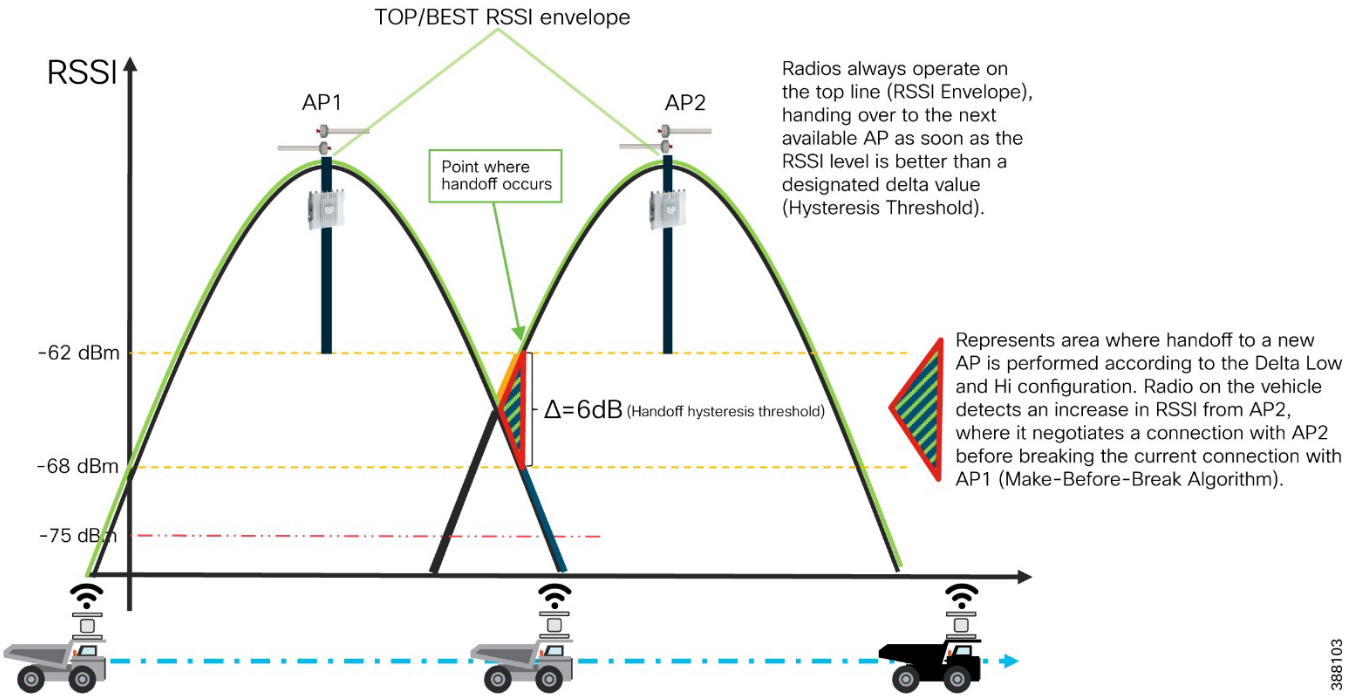


URWB Fluidity Handoff Decision

Within standard WI-FI based communication, a handoff is triggered by the WI-FI client based on pre-configured static thresholds like RSSI and/or SNR. For example, a WI-FI client might be configured to trigger a handoff when its RSSI value drops below -75 dBm. URWB on the other hand uses a dynamic handoff decision algorithm.

As can be seen in the figure below, the vehicle radio always operates on the top line (RSSI Envelope), handing over from the currently connected radio to the next available radio as soon as the difference in RSSI meets the configured hysteresis threshold.

Figure 22 Fluidity Dynamic Handoff Decision



Fluidity Advanced Handoff Tuning for Vehicle Radio Units

The URWB solution provides certain advanced handoff parameters for vehicle nodes that can be tuned depending on the RF environment to achieve optimal handoffs.

The RSSI zone threshold and Handoff hysteresis threshold features provide safeguards against unwanted handoffs - in other words, against unreasonably long periods of time between received signal strength from the connected unit falling too low, and a handoff request from a relief unit.

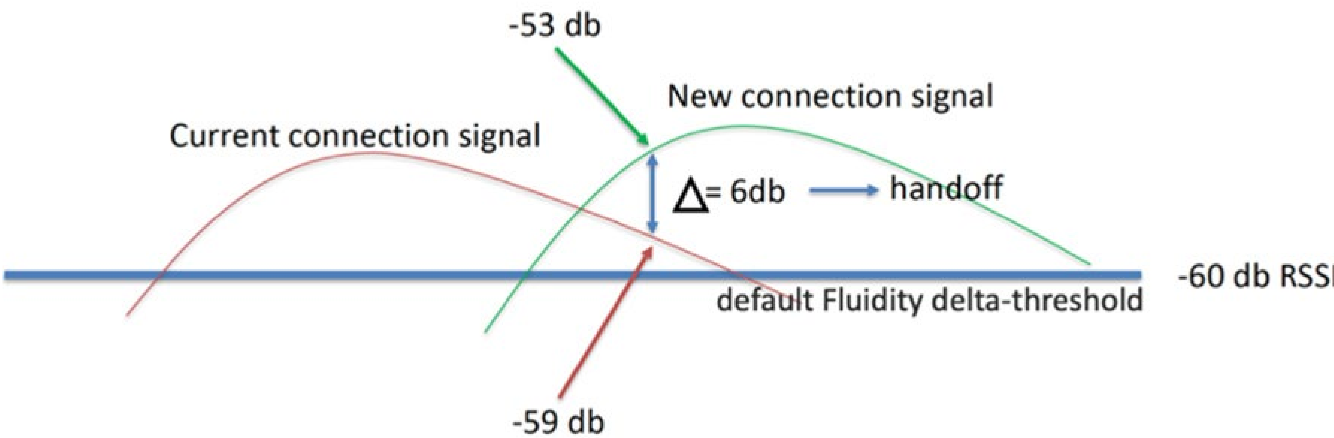
The relationship between these three settings governs whether a handoff will take place from one unit to another, based on a difference in comparative signal amplitude values over a period.

The RSSI low/high zone threshold sets the border between the low and high RSSI zones. In this case, as represented by the two graphs below, the -60 dB level marks the border between the low and high RSSI zones.

The threshold value is always expressed as SNR, with -95 dBm as the reference value, and is always expressed as a value greater than 0. The default value is 35. This equates to -60 dBm.

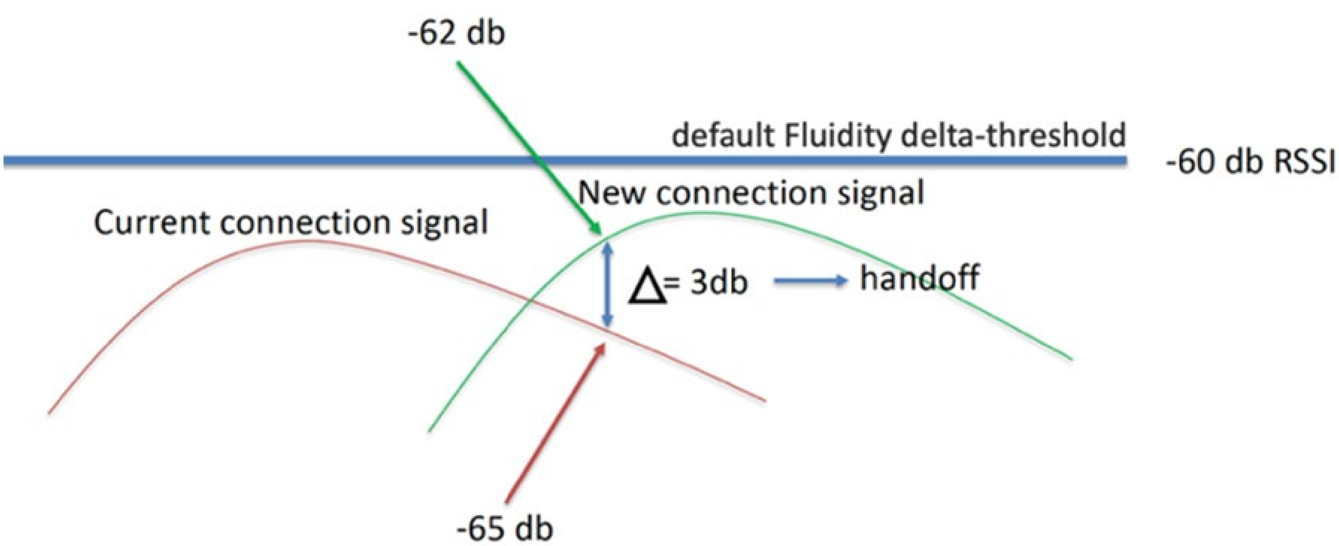
The default Fluidity delta-threshold is -60dBm. The default delta-high threshold is 6 dBm. What this means is that in good RF environments where the signal strength is higher than -60 dBm, the vehicle radio will only attempt a handoff to another wayside infrastructure radio if it provides a signal that is at least 6 dBm higher than what it is receiving from its currently connected wayside infrastructure radio. If the delta value is lower than 6, no handoff will occur at that time.

Figure 23 Fluidity delta-high example



The default delta-low threshold is 3 dBm. What this means is that in poor RF environments where the signal strength is lower than -60 dBm, the vehicle radio will attempt a handoff to another wayside infrastructure radio if it provides a signal that is at least 3 dBm higher than what it is receiving from its currently connected wayside infrastructure radio. If the delta value is lower than 3, no handoff will occur at that time.

Figure 24 Fluidity delta-low example



Note: The Fluidity delta-threshold, the delta-high and delta-low values are all configurable to values different from the default using either IW-Service or the radio CLI if needed for your RF environment tuning.

Multipath operations – (MPO)

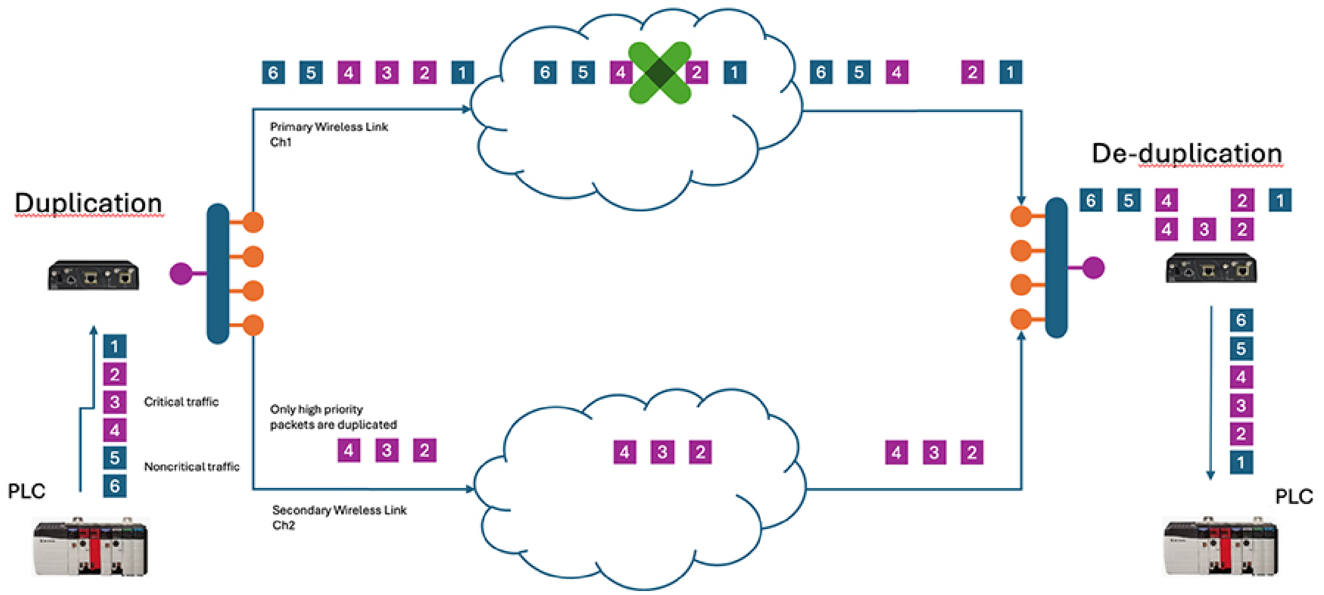
Multipath Operation (MPO) enhances reliability by sending duplicate copies of packets across multiple wireless paths. This patented technology duplicates high priority traffic up to 4 times to increase availability, reduce latency, and lower the effects of interference.

MPO establishes multiple label switched paths (LSPs) between a mobile system and the backend infrastructure of a wireless network. The multiple LSPs enable high priority packets to be sent through redundant paths to reduce packet loss.

For MPO, a duplicate packet is sent through several wireless channels (to all available wireless paths). This helps to ensure reliability, and the spatial diversity of the receiving access points greatly increases the chances of at least one of the copies to be received correctly. Deduplication is another aspect of MPO to remove any duplicates of a packet that are received along the different wireless paths.

As a result, the delivered packets have sequence numbers assigned to them, thereby allowing the deduplication algorithm to eliminate copies of any packets that it has already received. This is demonstrated in the following diagram.

Figure 25 MPO Deduplication



Cisco URWB Design Considerations

This section describes the design options available for deploying URWB within the manufacturing space. The key design options and considerations include

- Backhaul (Fluidmax) considerations
- Mobile (Fluidity) considerations
- Frequency Considerations
- Multicast Traffic
- Security
- Resiliency and High Availability
- Management and Monitoring

Backhaul (FluidMax)

- When designing for Fixed Point to Point and Multipoint networks it is important to understand the environmental factors that apply to the area of deployment. Short guard intervals (400ns) are typically used in environments with minimal multipath while longer intervals (800ns) are used to improve signal reliability in environments with high multipath interference. Longer guard intervals also improve link reliability for longer range outdoor deployments and help prevent inter-symbol interference in those environments as well. To use 802.11ax data rates high efficiency must be enabled.
- Fluidmax should only be used on Point-to-multipoint networks as this enables RTS/CTS(right to send/clear to send) on the radio interface. For all nodes that are unique to a point to multipoint topology should share the same cluster ID and passphrase.

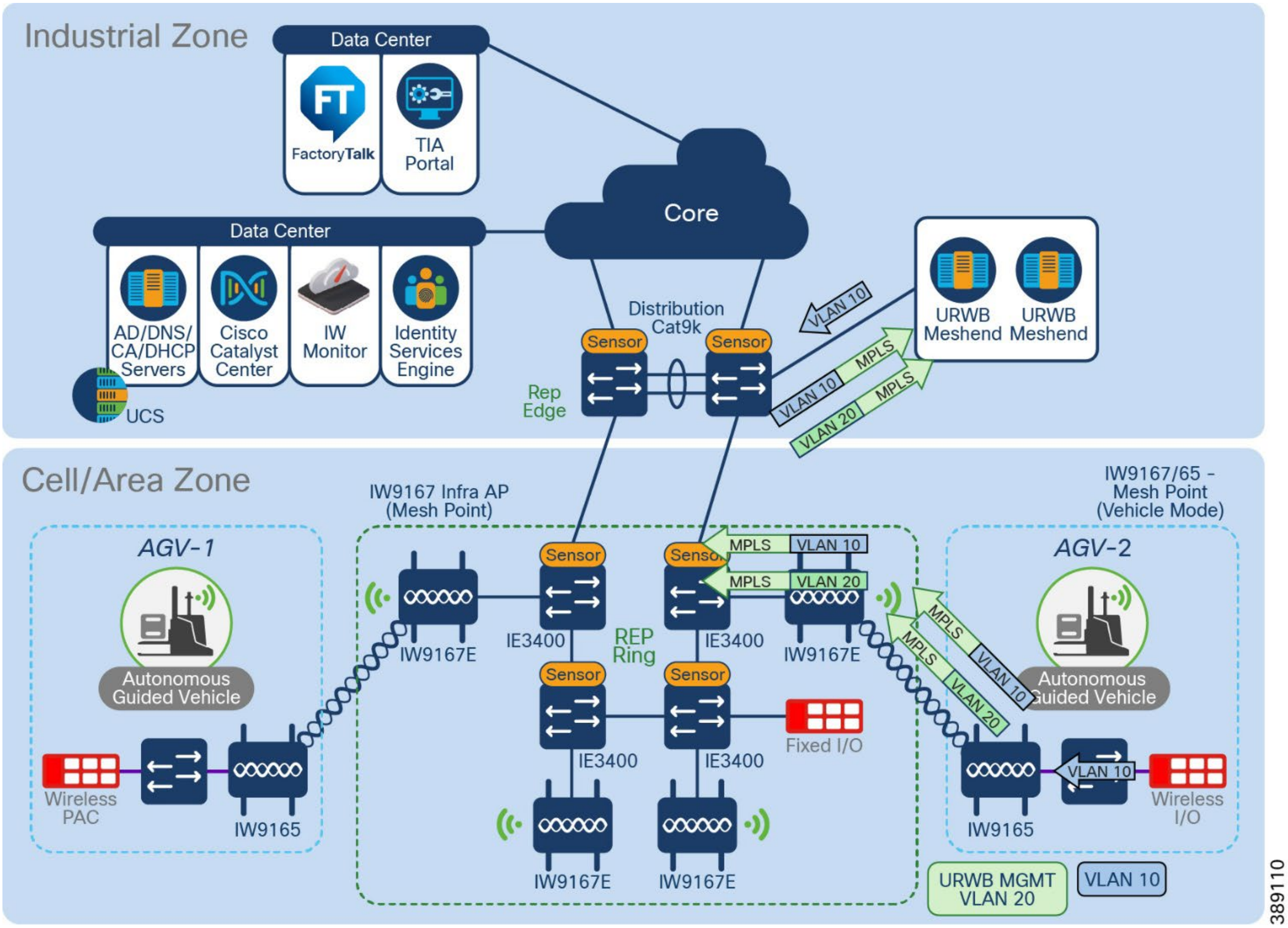
Mobile (Fluidity) considerations

When using Fluidity with a L2 design all the radios in the mesh network are in the same broadcast domain and the data VLANs are trunked on the switch ports. This design usually is deployed for smaller networks where roaming across L3 boundaries isn’t a requirement. L3 Fluidity is required when a Mesh Point must roam between L3 boundaries. In this type of design, the Mesh Point will roam to another cluster of Mesh Points on a different subnet with a local Mesh End. Traffic is then tunneled backed in the form of a L2TP tunnel to the Global gateway.

When using Fluidity with a L2 design there is one Mesh End and enough Mesh Points to provide the RF coverage for the service area. A prerequisite for L2 Fluidity is that all the URWB devices (mesh-end gateways, access radios, and mobile radios) need to be within the same VLAN/IP subnet/L2 broadcast domain and configured with the same passphrase. The URWB radio on the vehicle is configured in “Vehicle” mode.

The following is a L2 fluidity example used to validate the design.

Figure 26 L2 Fluidity

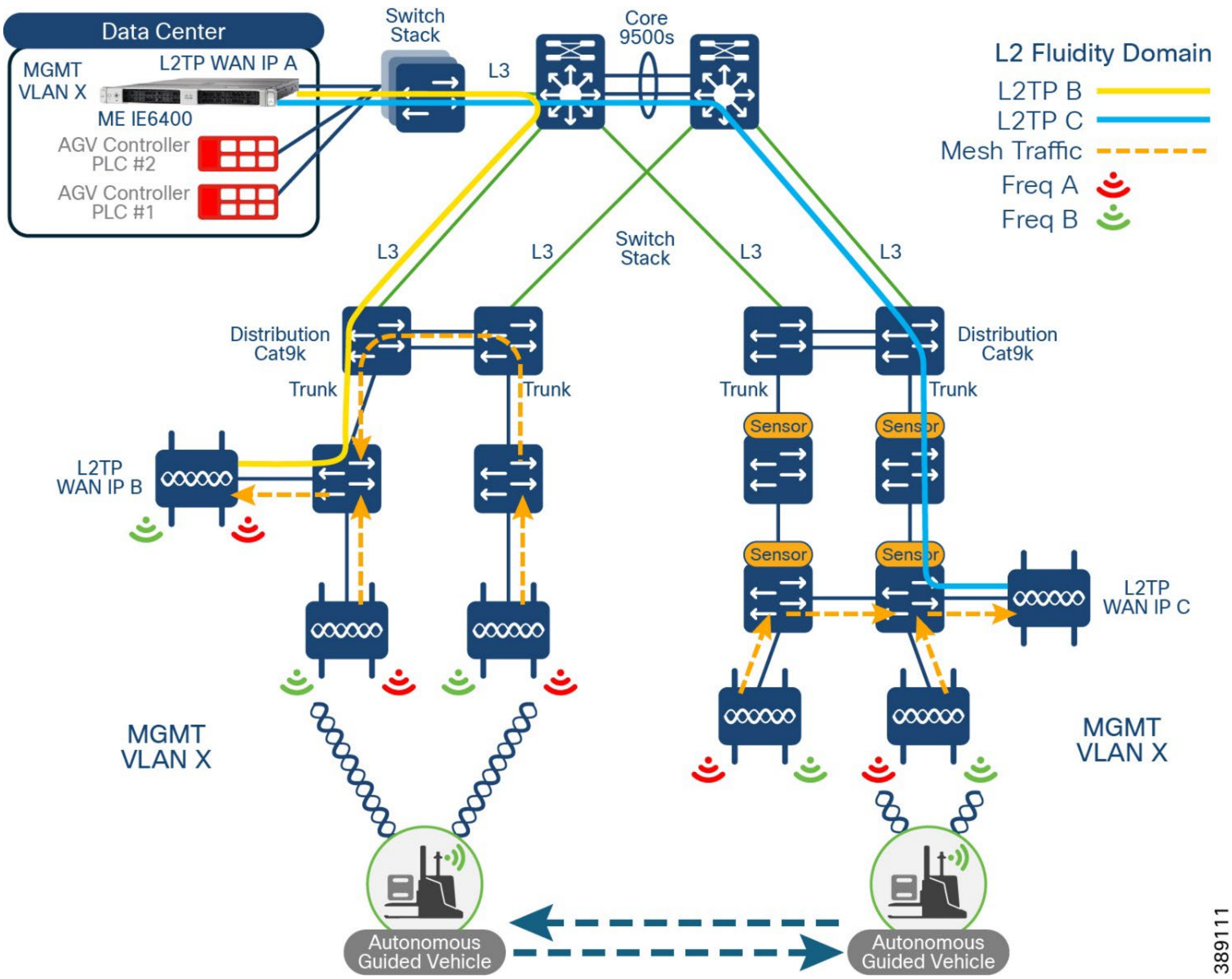


Note: L2 Fluidity can also be implemented without the use of the ring topology as depicted above. In that case the URWB access radios can be connected to the access layer switches within the deployment.

The deployment consists of a redundant pair of mesh ends. The mesh ends can either be URWB radios for smaller deployments (IW9167E) or IEC6400 gateways for larger deployments.

Centralized L2 Fluidity with L2TP and IEC 6400

Figure 27 Layer 2 Fluidity with L2TP Centralized



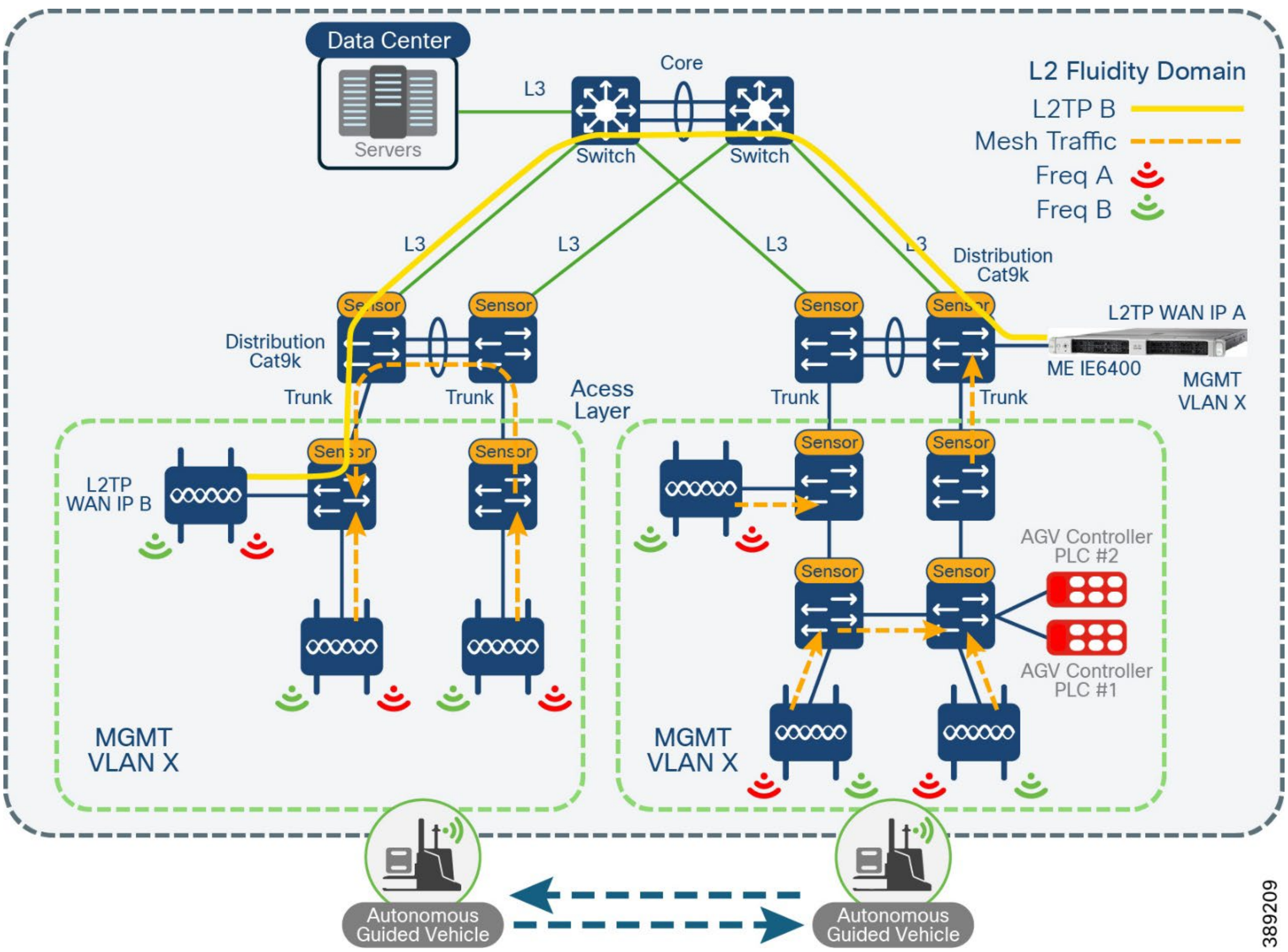
Within a Layer 3 Fluidity design, URWB devices belong to different L2 broadcast domains which are then organized into separate L3 routing domains. In order words there is a local mesh End for each l2 broadcast domain and traffic from the APs is sent through a L2TP tunnel to the global gateway to be handled further. This is typically implemented when AGVs need to cross subnet boundaries into various cell areas. To support inter-cell roaming for layer 2 fluidity URWB allows for L2TP tunnels to be established across L3 routing.

The IEC6400 Mesh end in this case, would reside within the datacenter Industrial zone and have a L2TP tunnel connection to infrastructure APs in the cell access area. All infrastructure AP providing coverage to the Cells should have identical URWB management VLAN configurations just as a normal l2 fluidity. The L2TP Remote Peer IP address of the ME should be configured on two APs for redundancy within each cell area. That means if there are multiple layer 3 domains or distribution Switches then a L2TP tunnel will need to be setup for each cell. The other infrastructure APs within that cell will communicate through the primary established segment radio to the ME. The Mesh Point with the lowest tunnel IP will be active while the others will be idle. Another component of using L2TP is that the URWB management VLAN will need to reside on the switches within the access and distribution layer. Traffic from an AGV is tunneled through to the ME where the MPLS label is removed, and the traffic is forwarded to the destination. In the diagram above, both the ME and the AGV I/O controller reside locally on the same switch in the Data Center.

Distributed L2 Fluidity with L2TP and IEC 6400

Within a Distributed L2 Fluidity design. The IEC6400 ME resides locally at the Distribution layer and L2TP tunnels are still established across L3 boundaries. All infrastructure AP providing coverage to the Cells should have identical URWB management VLAN configurations just as a normal l2 fluidity. L2TP Remote Peer IP addresses should be configured on two APs for redundancy in within the cell area. The other infrastructure APs within that cell will communicate through the primary established segment radio to the ME. Layer 2 adjacency is provided as the AGV I/O traffic is destined to a controller connected locally on the same distribution switch as the ME. The mobile unit can roam across the factory seamlessly. The following is an example of a distributed topology.

Figure 28. Distributed Layer 2 Fluidity with L2TP



Frequency Considerations

URWB is designed to support single frequency and multi-frequesncy.

URWB Single Frequency Design

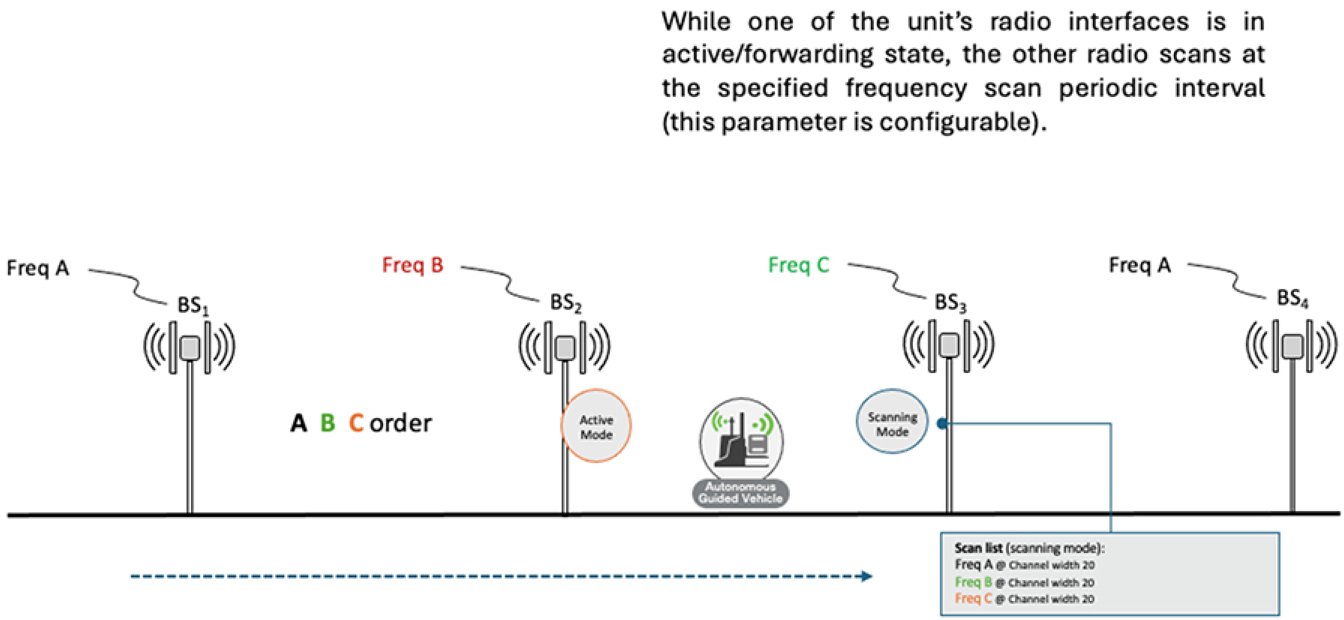
In the case of a URWB single-frequency design, all the URWB radios infrastructure and vehicles operate on the same frequency which is statically configured. An advantage of this design is that the vehicle radios don’t need to perform any off-channel scanning. Additionally, the deployment can use the cleanest channel available (determined as an outcome of the site spectrum analysis) and interference on other neighboring channel does not affect the system, hence providing reliable wireless connectivity. Single frequency designs are recommended for small deployments.

Dynamic Multi-Frequency Design

The dynamic multi-frequency design is like traditional WI-FI design where adjacent infrastructure APs are on a different frequency (Freq-A, Freq-B, Freq-C, Freq-A, Freq-B, Freq-C and so on). URWB infrastructure APs have two radios interfaces that can be configured for different frequencies.

To implement a Dynamic Multi-Frequency design a change needs to be made on the vehicle end. Due to the nature of the design, the vehicle radio now needs to channel scan to look for an AP providing a better signal while roaming across the plant floor. This functionality is termed " off-channel scanning." To perform off-channel scanning, the vehicle AP servicing client traffic can continue servicing client on the active radio while the second radio will actively scan for a better AP. This eliminates the need to have an additional on-board radio. The fluidity channel scan list should be configured on each vehicle node and the scan list will allow the radio to scan only the channels configured within the list.

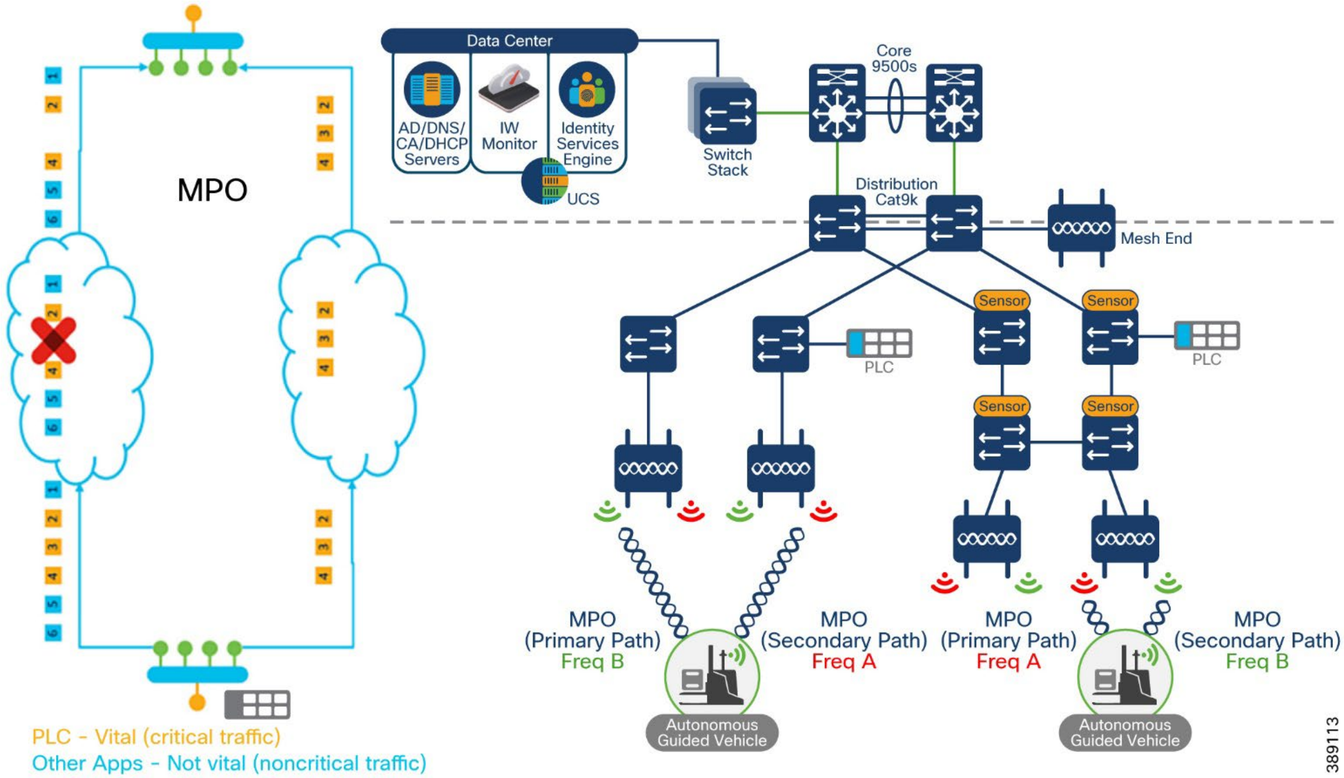
Figure 29. Dynamic Multi-Frequency



Dual Frequency Design with MPO

The dual frequency design is recommended for IAC application traffic that is deemed high priority and sensitive to latency and packet loss. With this design infrastructure APs is configured with 2 frequencies (Freq A, B) along with the vehicle mode Mesh Points. This provides redundant paths for the critical IAC applications as the mobile device moves across the plant. MPO must be configured on the Mesh End and all vehicle radios and QoS must be enabled. The following diagram illustrates an MPO design.

Figure 30 Multi-path Operations (MPO)



Quality of Service (QoS)

The URWB radios support eight levels of priority when classifying traffic. When inspecting a packet, the. DSCP/TOS value is translated into one of eight queues based on the three most significant digits. This translation follows.

Figure 31 QoS on URWB

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Priority			X	X	X	X	X
0	0	0	Priority 0 (lowest)				
0	0	1					
0	1	0	.				
0	1	1					
1	0	0	.				
1	0	1					
1	1	0	.				
1	1	1	Priority 7 (highest)				

388809

This priority value is copied into the MPLS EXP bits which is maintained throughout the mesh network until it leaves the mesh. When this data is transmitted over the wireless interface, the eight priorities are further mapped into four access categories based on the 802.11e WMM standard.

In a L2 Fluidity network, a VLAN header is also appended to the packet. The priority value that was copied into MPLS EXP is also copied into the 802.1p field of the VLAN header as the COS (Class of Service) value. When this traffic reaches the wayside network from the AGV/AMR, the VLAN header is maintained until it reaches the Mesh End. This means that all switches in the path can match on the COS value to differentiate between traffic of differing priorities coming from the mobile radio.

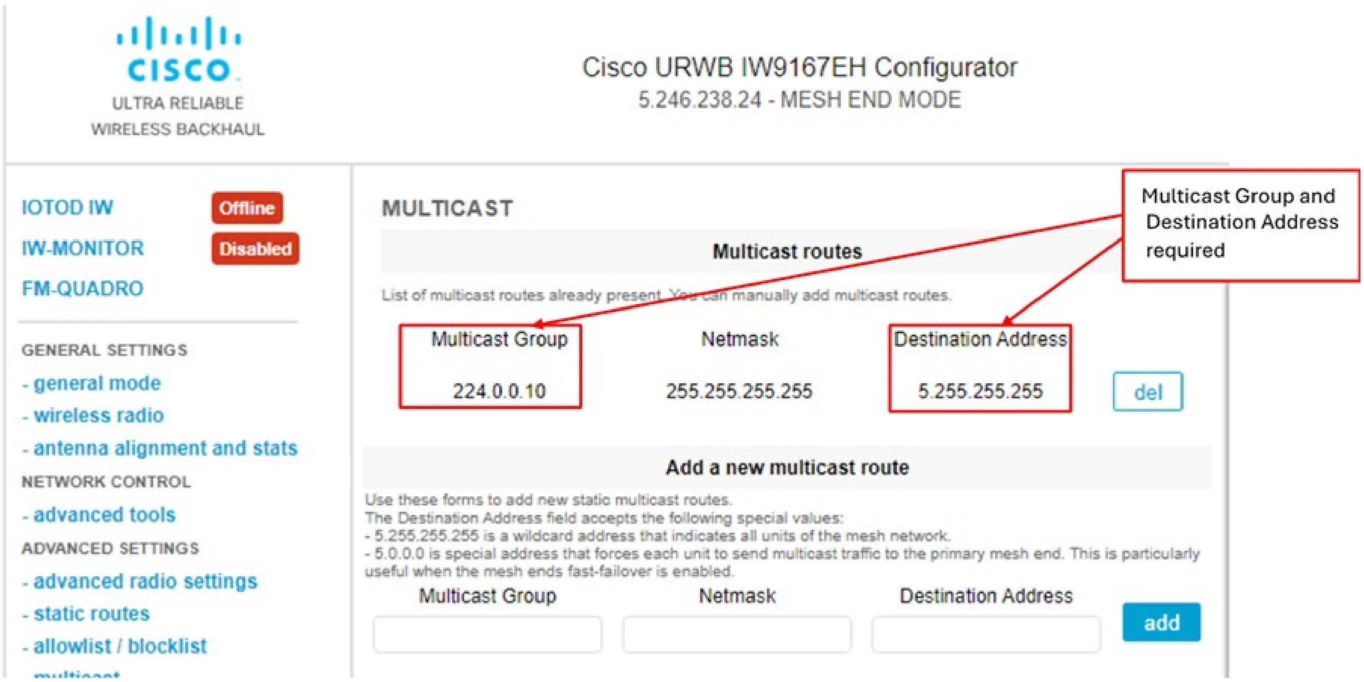
It is crucial that PLC traffic (CIP, CIP Safety, PROFINET, PROFIsafe) is tagged with CS6 on the vehicle and **infrastructure switches. Data traffic is not recommended to be used in QoS queue 7 as that is intended for internal control and signaling. If the application requires high throughput, for example high resolution video, then the traffic should be tagged with CS5 which gets mapped to VI. The VI buffer by default will aggregate traffic before it sends out the wireless interface.**

Multicast

Multicast is a group communication method in which data transmissions are addressed simultaneously to more than destination device. It is a common communication model in many Industrial Automation and Control protocols like Profinet, CIP, CC_LinkIE, and others. As an example, CCTV cameras connected to the URWB Mesh Point unit may forward multicast camera traffic to the closest mesh end in the wireless network. It may also be convenient to forward traffic from one Mesh Point to another depending on the network configuration.

By default, Mesh ends do not forward multicast traffic to the wireless network with the exception being universal plug and play (UPnP) and Internet Group Management Protocol (IGMP) traffic. For traffic intended to a Mesh Point unit, all multicast flow information must be specified in the settings on the Mesh end unit. Two things must be defined on the mesh end: the multicast group (ex 224.1.1.0/24 indicates all multicast groups from 224.1.1.1 through 224.1.1.254) and the destination address. This field will consist of one of more URWB unit id numbers in the form of 5.x.x.x. Destination address wild cards can also be used, for example 5.255.255.255 represents all URWB units in the network.

Figure 32 Multicast on URWB



Security

URWB offers advanced encryption standard (AES) traffic encryption. To support equivalent wireless security applicable to WPA a key rotation mechanism is implemented. A Key Controller can be enabled to control the algorithm used to generate an encryption key. Key validity time can be defined by properly setting a key Rotation timer.

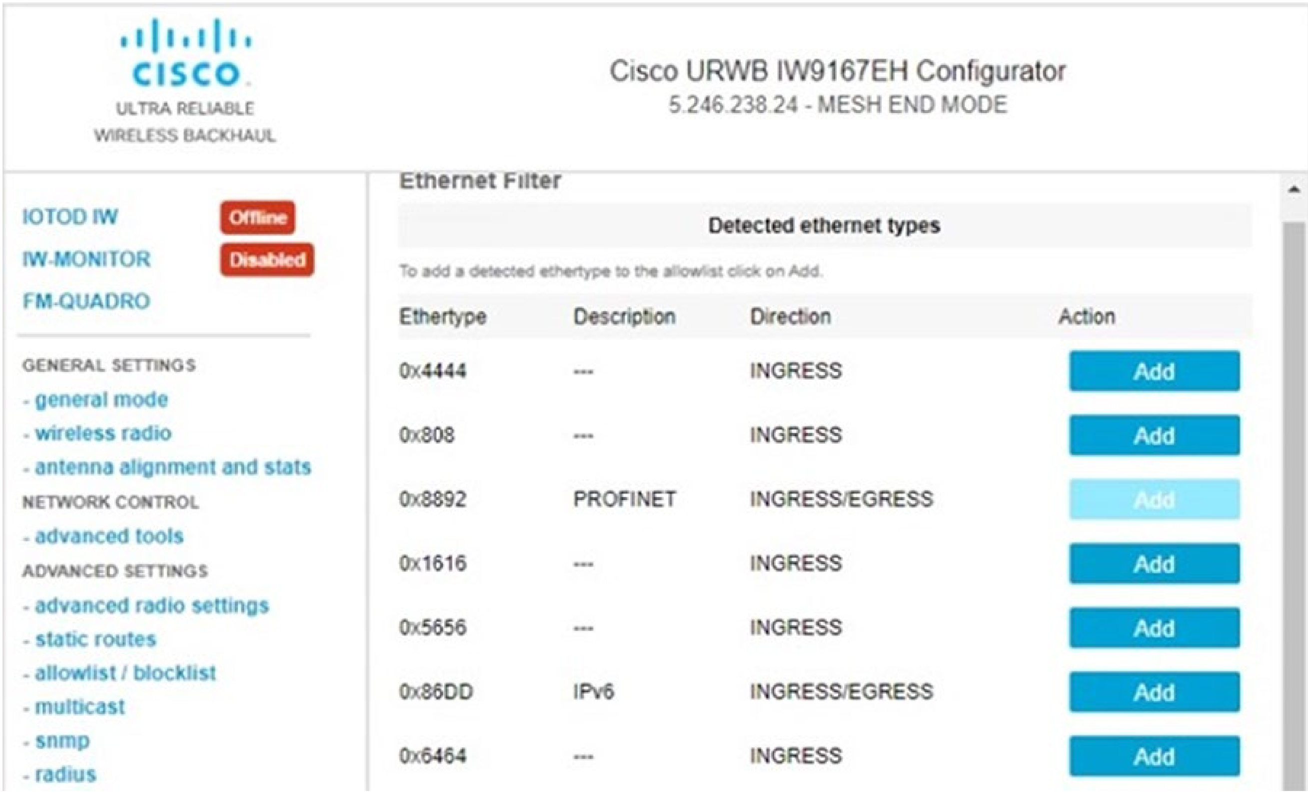
This setting controls whether AES encryption is applied to all wireless outgoing data packets:

- Without AES encryption enabled, only control and management traffic are encrypted (radio signaling is encrypted by default)
- With AES encryption enables, the data payload traffic is also encrypted
- AES-CCMP 128 is used to encrypt both data plane traffic between Radio Access Points and management plane traffic to reach the Radio Access Points.

Layer 2 Transparency Ethernet Filter

As previously mentioned URWB supports industry standard protocols such as CIP and PROFINET. These protocols are detected via Layer-2 mesh transparency and are reported automatically, by default the PROFINET ethertype is allowed. The layer 2 transparency feature allows forwarding non-ipv4 layer 2 protocols across the network. Additional protocols can be added and removed if they are required to traverse the wireless network. Ether type filtering can be done via the web UI or the command line of the node.

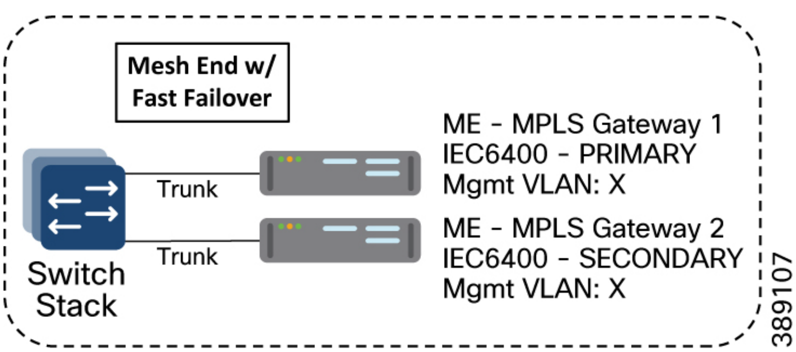
Figure 33 Layer 2 Ethernet Filter



High Availability

Fast Failover is a function providing high-availability and protection against network failures. The feature provides uninterrupted service for mission-critical applications where safety and/or operations would otherwise be compromised by failure of a single radio or gateway device. Because of the MPLS based technology, URWB fast failover can achieve device failovers within 500 mSec on both L2 and L3 networks. When enabled, the active/backup radios will share the same passphrase and the Mesh End with the lowest Mesh-ID will become the primary while the other remains the secondary. This feature also supports preemption with delay to restore the radio network back to a known working state upon recovery.

Figure 34 Mesh end Redundancy at the distribution stack



It is highly recommended to apply the MPLS fast failover for all Mesh ends in the URWB network. Once configured it is completely autonomous and ensures stable and reliable connectivity without intervention.

Primary Mesh-end Failure

When the primary mesh-end unit fails for any reason, a timeout expires on the secondary after not receiving keepalives for a configurable interval (typically between 50 - 200 mSec). At that point, the secondary becomes the new active mesh end taking over the role of primary and executes the following actions:

1. Issues a Primary Change command to inform all other units on the same wired network that the primary has changed. The message is propagated to mobile units as well using an efficient distribution protocol.
2. Updates the internal MAC and MPLS forwarding tables. This step is performed using a patented fast-rerouting technique that provides seamless performance.
3. Sends gratuitous ARPs for the on-board devices on its ethernet/fiber port. This forces the network switch to update its MAC forwarding table (CAM) so that it will send traffic for on-board destinations through the port connected to the new primary.

When the other units receive the Primary Change command from the secondary, they perform the internal seamless fast-rerouting procedure (step 2), so the traffic can be immediately forwarded with no additional delay and signaling required. This allows for fast network reconvergence, with an effective end-to-end service disruption below 500 mSec.

Primary Mesh-End Recovery

When the primary mesh-end recovers, it initially scans the network for the presence of an active secondary mesh end. If the detection is positive, then the unit enters an inhibition mode where the secondary remains the current edge point of the infrastructure for a certain amount of time (default 70 seconds). During this grace period, the primary receives updates from the currently active secondary mesh end and acquires full knowledge about the state of the network. If preemption is enabled, it switches to being the primary node following the same procedure described above for failover.

Redundancy for L2 Fluidity with L2TP

Layer 2 Fluidity Mesh ends configured with L2TP can also be redundant. Each Mesh end should be configured with MPLS fast failover to enable fast convergence in case of a hardware or network link failure. With the system in normal condition (all devices up and running), this is the expected scenario between the Mesh End and each L2TP configured Mesh Point:

L2TP Tunnel between Primary Mesh End and Mesh Point – CONN

L2TP Tunnel between Primary Mesh End and Secondary Mesh Point – IDLE

L2TP Tunnel between Secondary Mesh End and Primary Mesh Point – IDLE

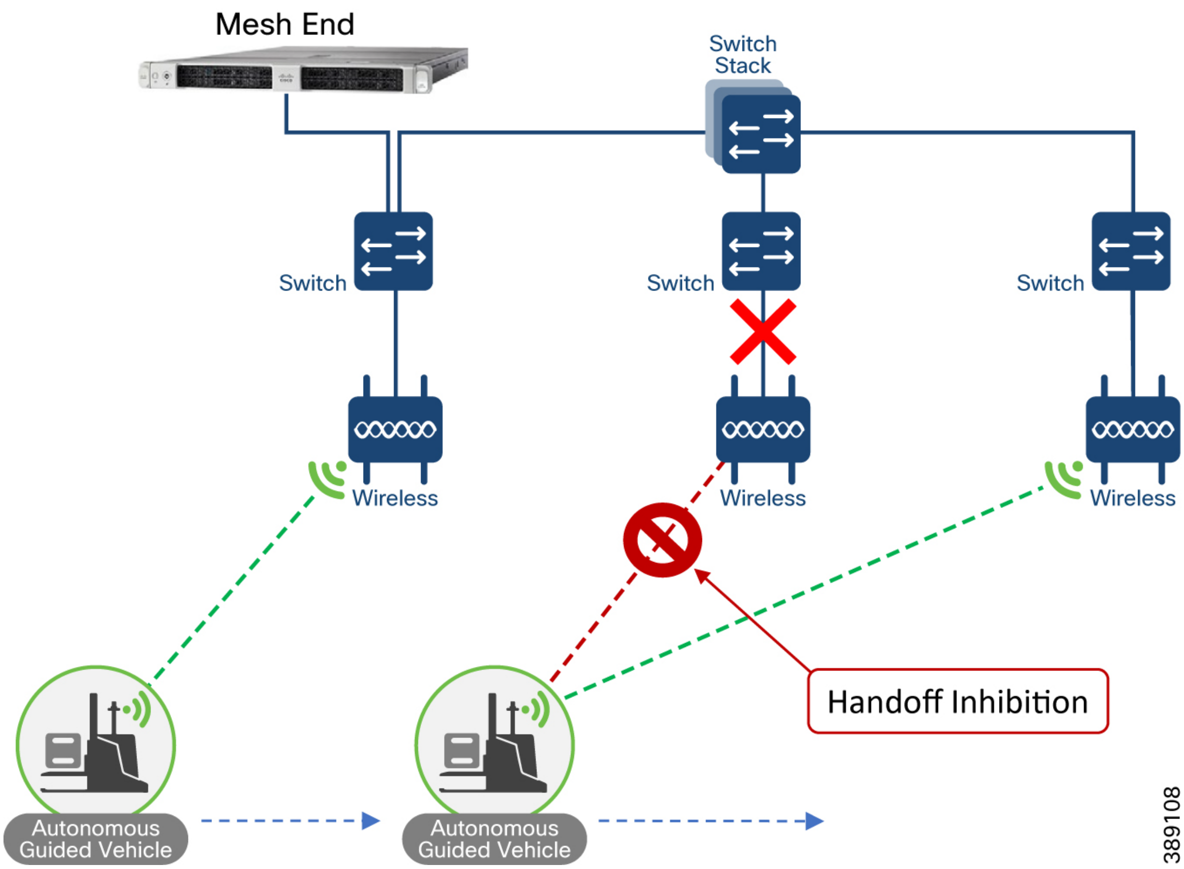
L2TP Tunnel between Secondary Mesh End and Secondary Mesh Point – IDLE

Mesh End Failure: In case of failure of Primary Mesh End, the L2TP tunnels between itself and the L2TP Mesh point peer will become IDLE, while the L2TP tunnels between the secondary Mesh End (elected the new Primary) and the Mesh Point will become CONN. For further details please see the Implementation Guide for configuration guidance.

Link Backhaul Check – Handoff Inhibition

Leveraging the Link Backhaul Check feature, an access node detects a carrier loss on its Ethernet/Fiber port hence losing its ability to deliver mobility traffic to the mesh end. The affected node immediately advertises its status as ‘Unavailable’, by transmitting a ‘handoff inhibition’ message over the wireless channel. Upon receiving the ‘handoff inhibition’ message any existing mobile radios connected to this node will try to search for another access radio to connect to. All mobile nodes currently connected to this unavailable access radio will find and connect to an alternative access node within a few hundred milliseconds, typically within < 400 mSec. Also, any handoff attempts from any other mobile radios to this affected access radio will be rejected. It is highly recommended to enable the Link Backhaul Check feature on the access radios within an AGV/AMR deployment.

Figure 35 Link Backhaul Check - Inhibition

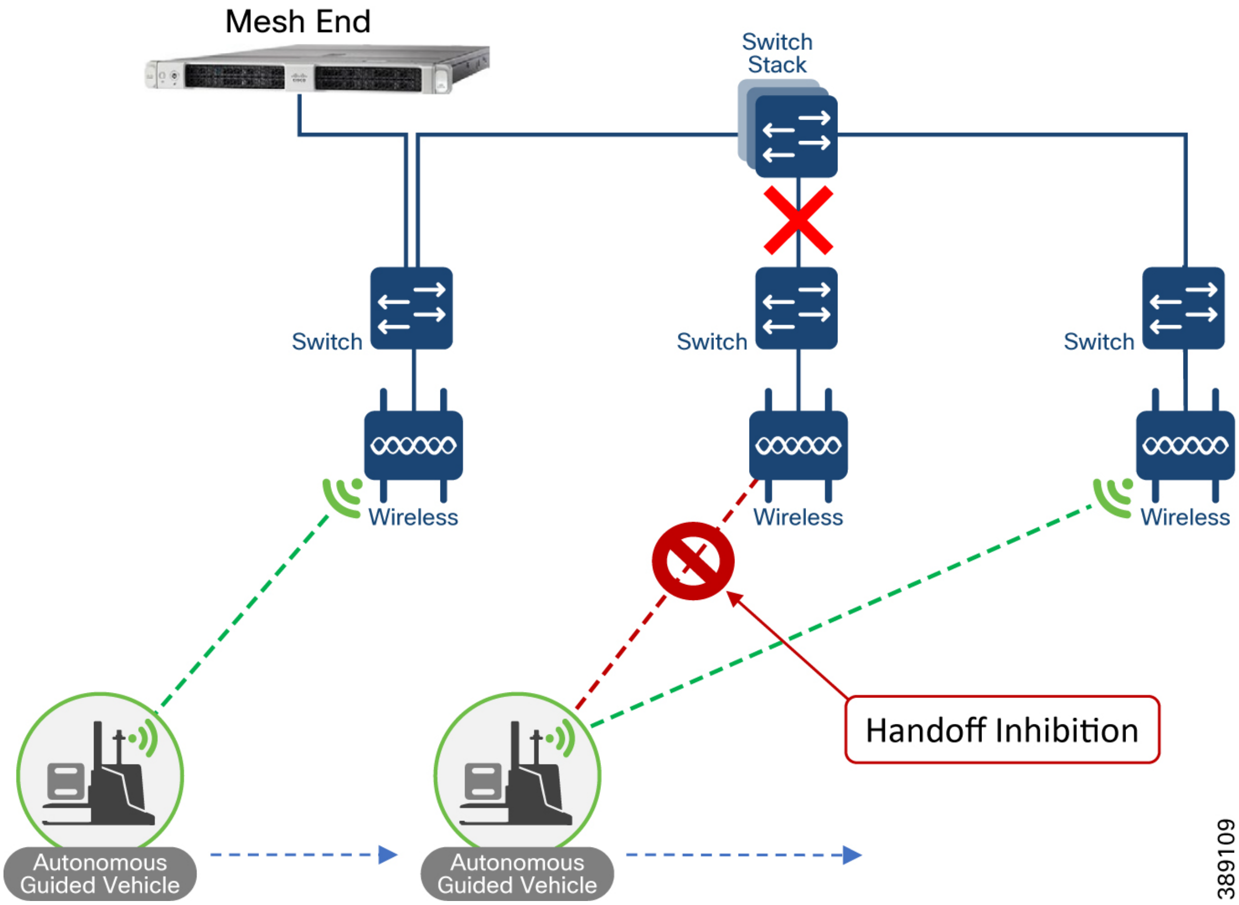


In the figure above it is shown that the link between the infrastructure radio and the directly connected switch is down. Assuming that the radio is not powered using PoE but via an external power source the radio is still up and providing good wireless connectivity to the vehicles. However, since the wired link is down and the radio is not able to forward traffic to the wired network, the radio goes into handoff inhibition mode and will reject any handoff attempts from vehicles.

Mesh-End Backhaul Check – Handoff Inhibition

Leveraging the Mesh-End Backhaul Check feature, an access node detects that it is not able to reach the active mesh end. This failure is triggered when L2 MAC reachability is lot to the active mesh end for 250 mSec. The affected node immediately advertises its status as 'Unavailable', by transmitting a 'handoff inhibition' message over the wireless channel. Upon receiving the 'handoff inhibition' message any existing mobile radios connected to this node will try and search for another access radio to connect to. All mobile nodes currently connected to this unavailable access radio will find and connect to an alternative access node within a few hundred milliseconds, typically within < 400 mSec. Also, any handoff attempts from any other mobile radios to this affected access radio will be rejected. It is highly recommended to enable the Mesh-End Backhaul Check feature on the access radios.

Figure 36 Mesh-End Backhaul Check - handoff Inhibition



In the figure above it is shown that the directly connected switch of a URWB access loses its wired connectivity to the core switch, the infrastructure radio is powered on and providing good coverage and connectivity to vehicles but since the radio is not able to forward traffic to the mesh end located within the control room, it will go into handoff inhibition mode and reject any handoff attempts from vehicles.

Management & Monitoring

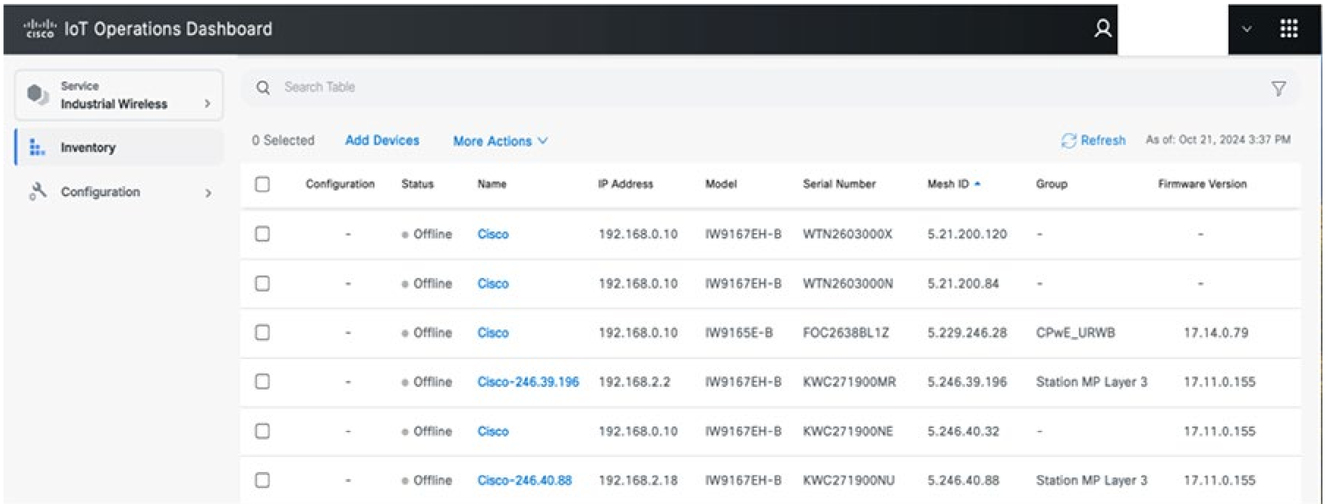
Industrial Wireless Service for URWB Management

Industrial Wireless Service in IOT Operations Dashboard (IoT OD) is an OT service used for configuring, provisioning, and monitoring URWB devices in a centralized location. The URWB devices can be onboarded through the wizard and all configurations and image versions are managed through the Dashboard. These configurations can then be manually pushed down to the URWB devices which will load the config, upgrade the firmware if necessary, and then automatically reboot with the new configuration.

Additionally, configuration templates can be created to bulk configure devices with similar configuration parameters instead of maintaining a separate configuration per device. The dashboard also checks that the configuration running on the URWB device matches what is configured in the dashboard ensuring consistency across the mesh network.

By default, all URWB devices are configured to connect to the Dashboard which eases the process of provisioning new devices. If the URWB devices are unable to connect to the cloud dashboard, the service can still be used offline. When the devices are onboarded into the dashboard, they will show as offline if they have not connected to the Internet. All configurations can then be created and downloaded as a file. This file can be manually uploaded to the radio from the local GUI where it will apply the config and update the image if necessary. The radio will then reboot with the updated config. This is an effective way to reliably configure the URWB devices in case the mesh network is air-gapped or otherwise disconnected from the Internet.

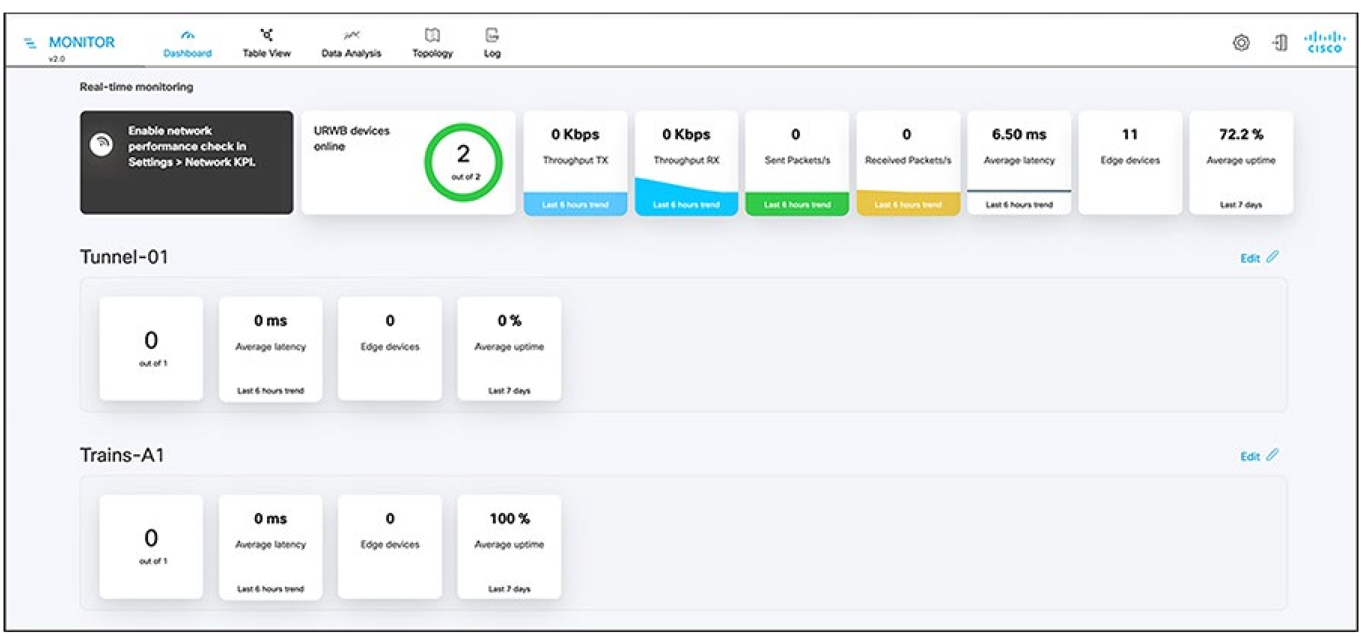
Figure 37 IW service on IOT Operations Dashboard



Industrial Wireless Monitoring (IW Monitor)

IW Monitor is an on-prem application that monitors the mesh network real-time status and performance based on customizable KPIs without relying on SNMP. It can graphically display all URWB devices and the connections between them to see a bird's-eye view of the mesh network. This can also be overlaid on a scale map of the service area to show where devices are physically located. The application is delivered as a Docker container instead of a dedicated appliance and requires reachability to all URWB devices. Within the factory environment IW monitor resides at the Industrial Zone within the Data Center.

Figure 38 IW-Monitor Dashboard



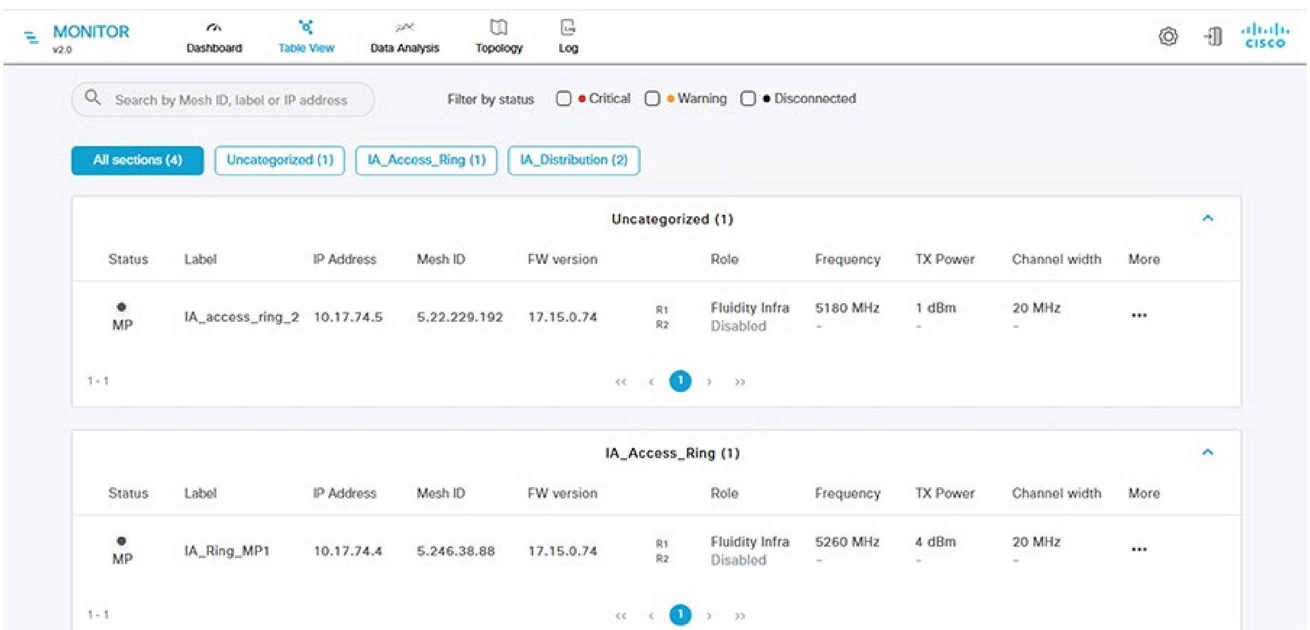
Features and benefits:

- On-premises monitoring tool for URWB networks
- Wizard setup for quick and easy installation and deployment of IW-Monitor
- Real-time dashboard displaying uptime, throughput, latency, jitter, and other network KPIs
- Customizable section view to easily check groups of radios
- Customizable monitoring alerts for prompt response
- Radio-by-radio data logging with a minimum sampling interval of 300 mSec
- Real-time information display for quick and accurate troubleshooting
- Side-by-side comparison of radio KPIs over time and over vehicle position
- Alerts/events can be forwarded to a Syslog server
- Radio KPIs such as RSSI, LER, PER, etc. can be exported to a CSV file for graphing

IW-Monitor Table View

The table view allows customers to condense sections of the network into a tabular view, isolating specific radio configurations and performance statistics. During troubleshooting, this drastically reduces the time needed to understand system performance on a radio-by-radio basis.

Figure 39 IW-Monitor Table View



Industrial Wireless with WI-FI

Industrial WI-FI Overview

The factory WLAN provides ubiquitous data and voice connectivity for employees, wireless internet access for guests, and connectivity for IoT devices. This section details the design considerations for use cases applicable for WI-FI 6/6E deployments. We will look at the infrastructure components involved in deploying the factory WLAN such as the Catalyst 9800 WLC, access points, workgroup bridges, and Management using Catalyst Center.

Wireless LAN Controller

The Wireless LAN Controller (WLC) is a highly scalable and flexible platform that enables system-wide services for mission-critical wireless networking in medium- to large-sized industrial environments. The WLC should provide the following:

- Support for RF visibility and protection
- Sub-second stateful failover of all access points and clients from the primary to the hot-standby controller
- Larger mobility domain for more simultaneous client associations
- Intelligent RF control plane for self-configuration, self-healing and self-optimization
- Efficient roaming that improves application performance as required by IACS applications, whether within a Cell/Area Zone (Levels 0-2) or between multiple Cell/Area Zones
- Full Control and Provisioning of Wireless Access Points (CAPWAP) access-point-to-controller encryption
- Support for rogue access point detection and denial-of-service attack detection

Access Points

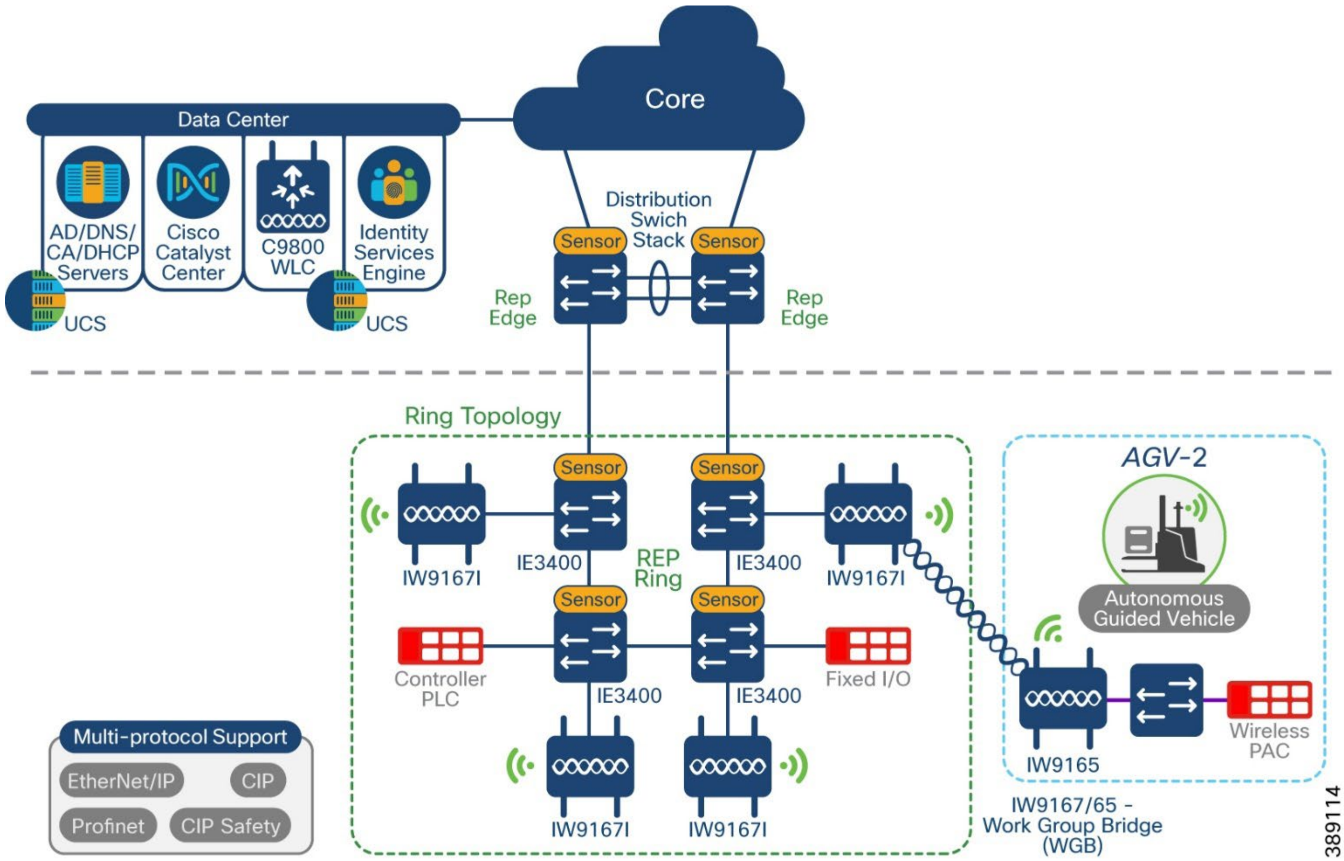
In a Unified WLAN architecture WLAN functionality is split between light weight access points and the WLC. Access points can operate in local or FlexConnect mode. A local mode AP encapsulates user data within the CAPWAP (control and provisioning of wireless access points) tunnel and sends the traffic to the WLC to get switched back on the network. In FlexConnect wireless access points, the client data traffic is switched locally and authentication can be done locally or centrally.

Work-group Bridge Clients

Workgroup Bridge mode (WGB) is an Access point mode that provides wireless connectivity to the wired clients. The clients are connected via ethernet port of the WGB AP or via switch. The WGB client then works like a bridge between the wired network and the wireless segment by learning the MAC addresses of its wired clients on the Ethernet. The WGB then shares these identifies with the WLC through the infrastructure AP using the Internet Access Point Protocol (IAPP) messaging.

Within the IW portfolio the IW9165E AP can serve as a WGB and is designed to add reliable wireless to static, moving vehicles and machines.

Figure 40 Workgroup Bridge connection example



Management

Cisco Catalyst Center offers centralized, intuitive management that makes it fast and easy to design, provision, and apply policies across your network environment. The Cisco Catalyst Center GUI provides network visibility and uses network insights to optimize network performance and deliver the improved user and application experience. A lack of network health visibility to network administrators and manual maintenance tasks like software upgrades and configuration changes are some of the common network challenges in enterprise networks.

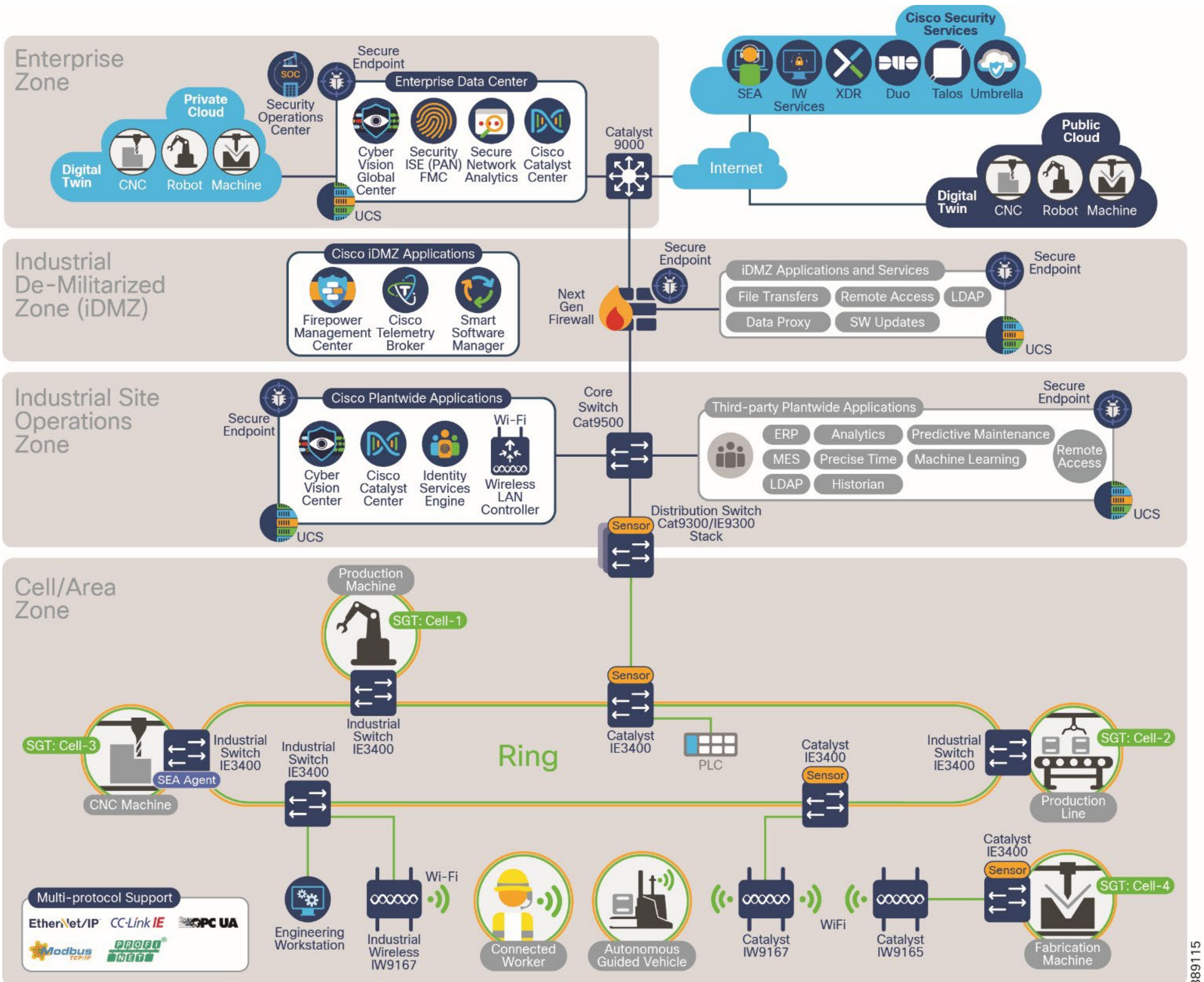
Benefits of Cisco Catalyst Center include:

- Simplified deployment and automation of network maintenance and configuration tasks—Cisco Catalyst Center automation provides Zero-touch device provisioning, software image management, device replacement flows, and network provisioning tasks to facilitate device deployment, configuration, and maintenance at scale. Additionally, compliance checks are provided to guarantee the network is compliant with business intent.
- Network monitoring and analytics for proactive remediation—Cisco Catalyst Center Assurance enables every point on the network to become a sensor, sending continuous telemetry on application performance and user connectivity in real time. This, coupled with automatic path-trace visibility and guided remediation, means network issues are resolved in minutes—before they become problems.
- Consistent security policies for endpoints connecting to the network—The proposed architecture uses Cisco Catalyst Center, Cisco Identity Services Engine (ISE), and Cisco Cyber Vision to enhance the visibility of assets and interactions and create security policy to segment the network.

Industrial WI-FI Architecture

Wireless IACS equipment may stay within a single Cell/Area Zone and remain associated to a single AP until powered down or connected. AGV/AMRs may roam across the Industrial Zone and associate to multiple APs while remaining operational. In addition to wireless equipment, manufacturers should consider providing onsite wireless access for trusted partners and employees. The following WLAN architecture can address the wireless needs of the plant.

Figure 41 WLAN Architecture



The previous diagram depicts the Unified WLAN utilizing the Perdue model within the factory. The WLC is currently placed at the Industrial Site operations Zone. Access points are located within the Cell Area Zone where they will provide wireless connectivity to clients throughout the plant. Although the diagram depicts a ring topology, star topologies are supported as well. The Industrial Wireless architecture validates two implementation modes in the Unified WLAN, centralized switching and FlexConnect mode.

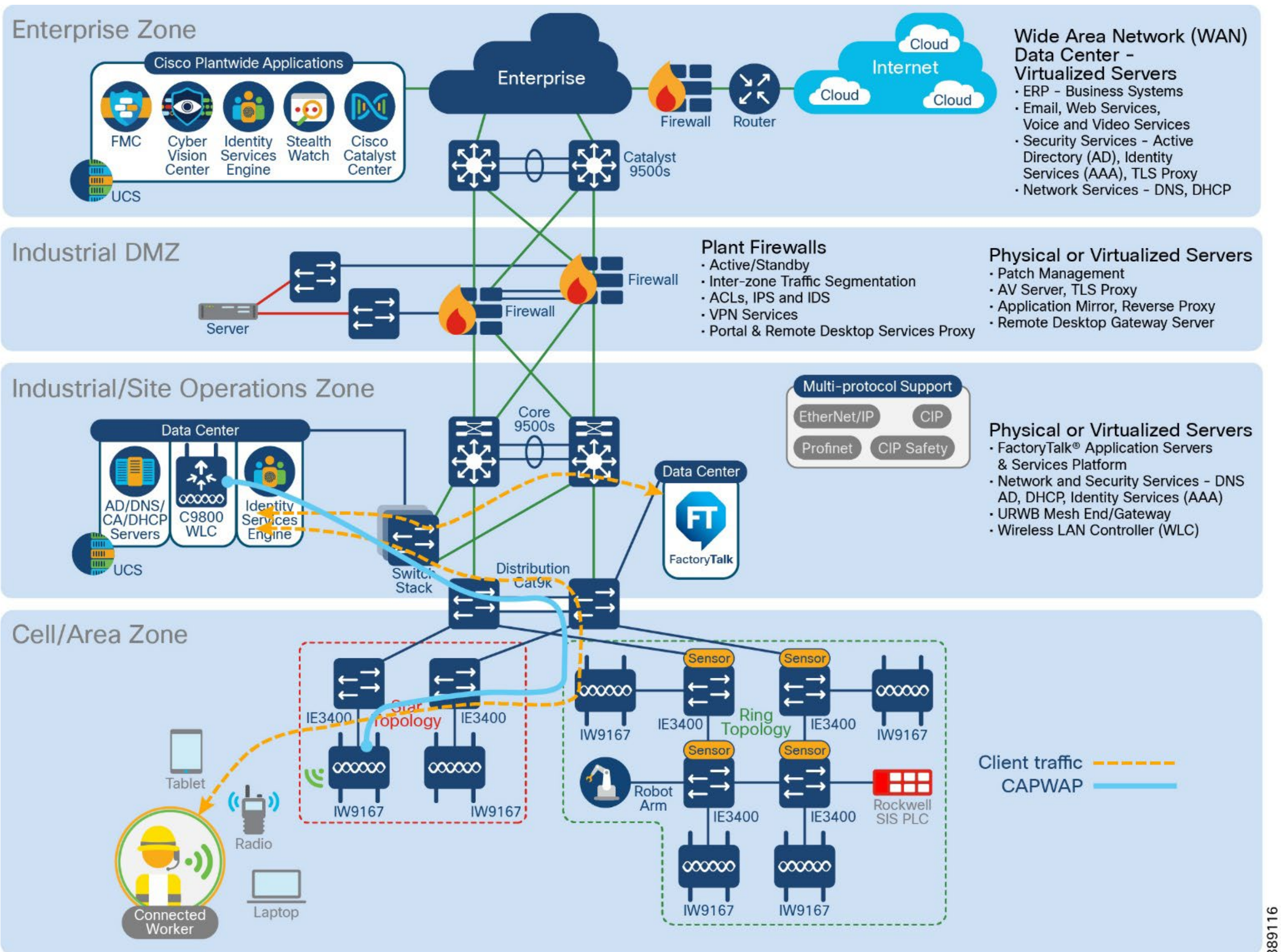
Centralized Data Switching

In the Unified WLAN design with centralized data switching, all data and control traffic are backhauled from the lightweight access point to the WLC before being terminated and placed on the wired network.

The advantages of this design are centralized access control of all traffic from a single point within the wired network and less complexity for wireless roaming in large scale environments.

The disadvantages of this design are the potential for scalability bottlenecks at the wireless controllers or the network infrastructure connecting to the wireless controllers and inefficient data flow when source and destination are close to each other (in the same Cell/Area Zone). This is especially important for latency sensitive data such as real-time control.

Figure 42 Local mode with centrally-switched traffic example



Centrally switched WLANs can apply to the many scenarios such as:

- Connectivity to mobile IT devices for employees
- Plant wide Inter cell roaming for mobile IT assets (WGBs)
- Industrial Site/operation zone application access

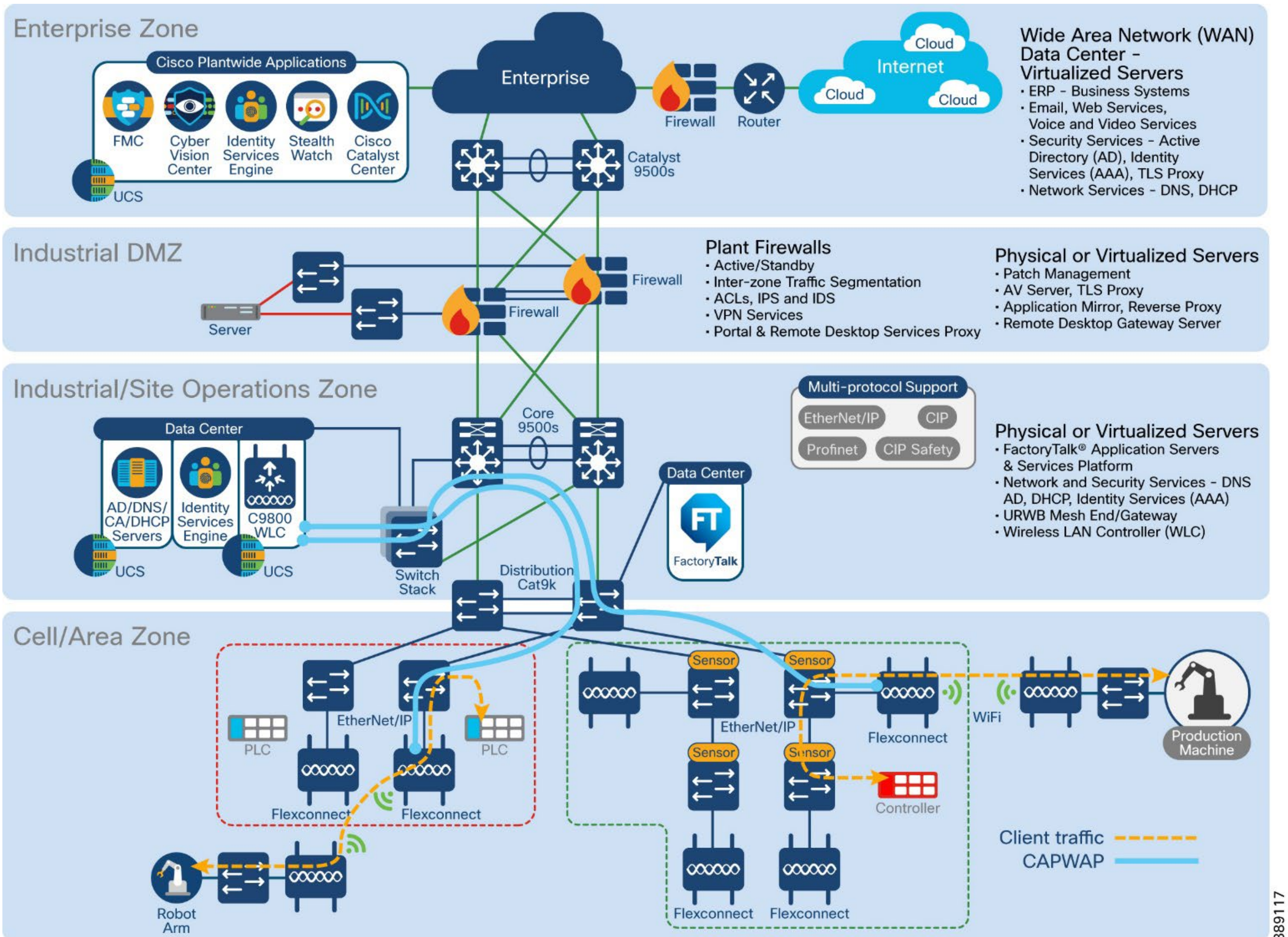
FlexConnect Mode

This mode provides more flexibility in deploying a wireless LAN. For example, the wireless LAN may be configured to authenticate plant personnel using a centralized AAA server, but once the user is authenticated the traffic is switched locally on the IES in the Cell/Area Zone.

FlexConnect mode with local switching functionality eliminates the need for data to go all the way back to the WLC in Level 3 Site Operations when access to local resources within the Cell/Area Zone is a requirement. This reduces the Round-Trip Time (RTT) delay for access to IACS applications on plant floor devices, increasing application performance.

The following topology diagram demonstrates the use of FlexConnect for deploying WI-FI with IACS applications for this solution.

Figure 43 FlexConnect topology



In the previous image, a fixed PAC or IoT device communicates with more or more PAC/IO devices behind a WGB. The traffic is sent through the wireless interface on the WGB to the infrastructure AP where the data traffic terminates locally and is forwarded to the destination. As mentioned previously this helps with latency.

The decision to use FlexConnect versus centralized switching model on the CAPWAP/SSID depends on the application requirements and most prevalent wireless data flows. For example, if mobile users only must access application servers in Level 3 Site Operations, centralized switching makes more sense.

If the use case is plant-wide Fast Roaming for mobile equipment such as overhead cranes for moving large objects between Cell/Area zones or AGV/AMR, then centrally switched model can be utilized. It should operate within a specifically designed SSID/VLAN that spans multiple zones.

Industrial WI-FI Design considerations

This section details the wireless access design considerations that are to be considered when deploying WI-FI as the solution for mobile IT assets, factory personnel, and IoT device (static and mobile) connectivity.

Wireless Access to the Industrial Zone

Plant personal and trusted partners may require access within the Industrial zone to:

- Commission new equipment faster
- Monitor parameters of IACS devices for proactive maintenance
- Collect information, troubleshoot problems and apply configuration changes to help minimize downtime.
- Access level 3 site operation servers from mobile human-machine interface (HMI) and thin client devices.

The access may be further restricted by filtering network protocols that are allowed for the mobile device. The following access policies have been considered:

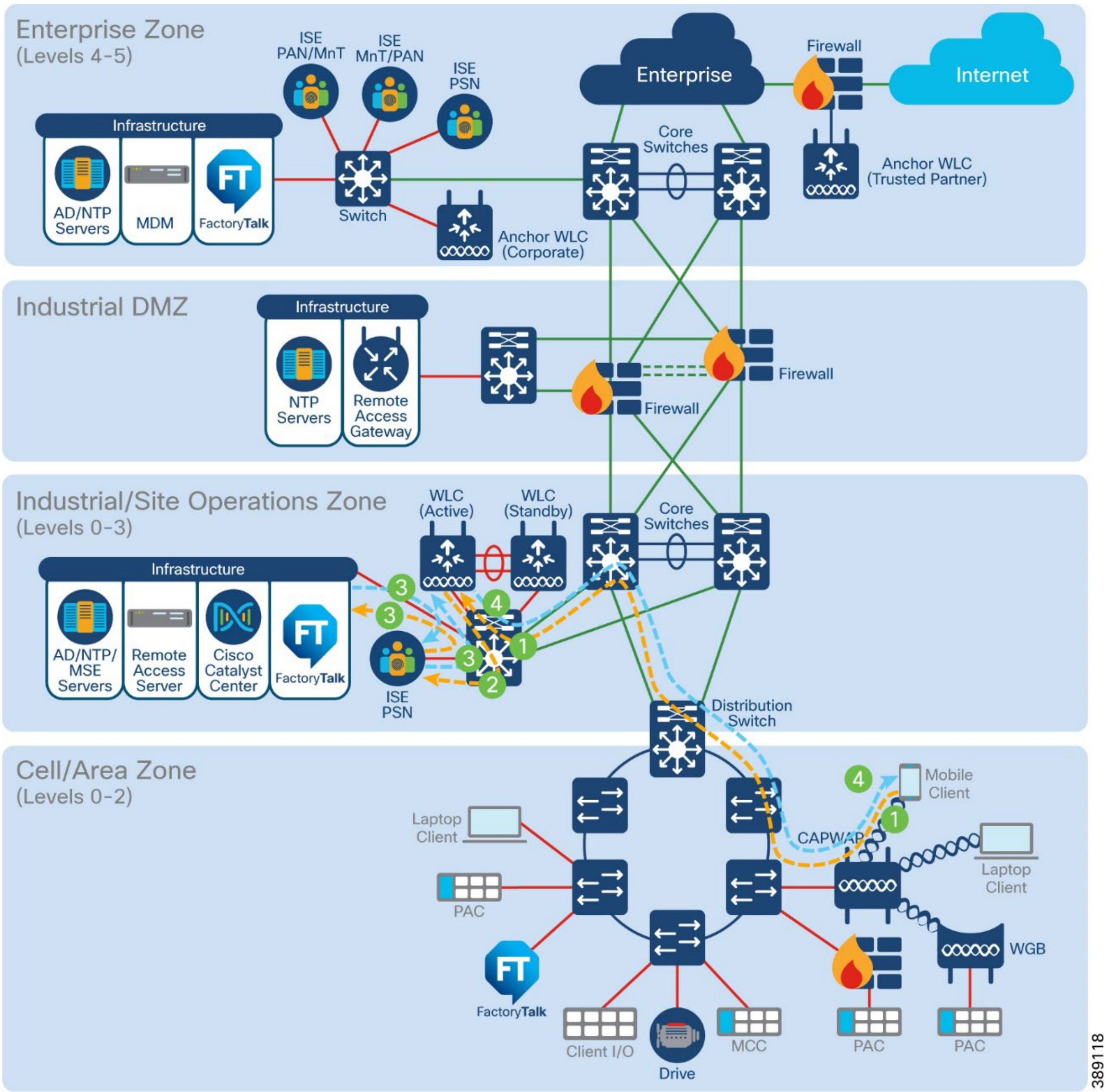
- Plant Personnel with full access to the Industrial Zone IACS equipment or Level 3 Site Operations
- Plant Personnel or Trusted Partner with partial access to the IACS equipment or Level 3 Site Operations limited to specific machines, production areas, or servers
- Plant Personnel or Trusted Partner access to the Remote Access Server using Remote Desktop Services for all IACS applications

Plant personnel access can be permitted for corporate-issued mobile devices or for BYOD (Bring your own device) if the manufacturer's policy allows BYOD in the network. The architecture should implement profiling and posturing of mobile devices using Cisco ISE and Mobile device Management (MDM) solutions.

Trusted partner access to the Industrial Zone assets should only be allowed from corporate-issued mobile devices that are dedicated for that purpose. Mobile devices owned by trusted partners, vendors and guests should not be given access to the Industrial Zone directly.

A user connecting to the wireless network using the Industrial Employee SSID will be directed by the AP to the Industrial WLC located in the Level 3 Site Operations. The Industrial WLC, using the Cisco ISE Industrial PSN as the authentication server, will validate the user credentials and certificate and will provide appropriate access to the Industrial Zone.

Figure 44 Wireless Access to Industrial Zone



Wireless Access to the Enterprise Zone

Plant personnel or corporate users may require access to the Enterprise network from the Industrial Zone to be able to:

- User corporate applications such as email and ERP systems as part of the normal workflow during production
- Access the internet and manufacturer's internal sites for IACS firmware downloads, documentation, and technical support.

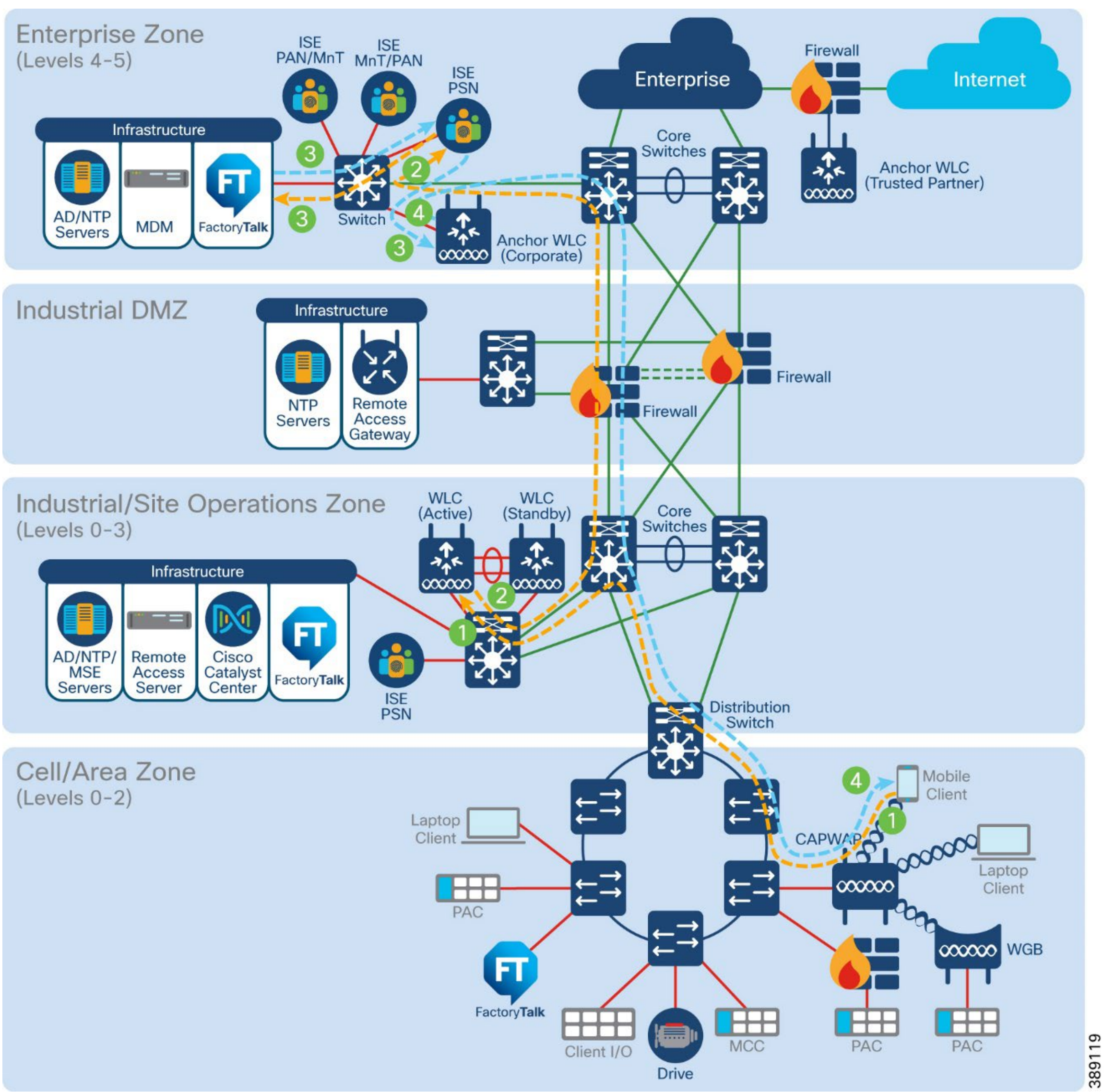
The Industrial Wireless Architecture should provide secure access to the Enterprise Zone for authorized users without comprising the security of the Industrial Zone assets. Authorization policies defined in the Enterprise Zone apply to the user's access, including Internet access. Therefore, the following access policies should be considered.

- Plant Personnel or Corporate User with full access to the Enterprise Zone assets and to the Internet as permitted for a corporate user
- Plant Personnel or Corporate User with partial access to the Enterprise Zone limited to specific application servers
- Corporate User with access to the Remote Desktop Gateway (RDG) in the IDMZ for remote connectivity to the appropriate assets in the Industrial Zone

Plant personnel or corporate user access to the Enterprise Zone can be allowed for corporate-issued mobile devices or for BYOD (personal or newly issued devices) if the manufacturer's policy allows BYOD in the network. The architecture should implement profiling and posturing of BYOD using Cisco ISE and MDM solutions.

The wireless user data is placed in a secure EoIP (Ethernet over IP) tunnel between WLCs in both zones. The IDMZ firewall configuration allows the tunnel traffic between WLCs. As a result, the user's traffic is isolated from the Industrial Zone network traffic and terminated in the Enterprise Zone. From a user perspective, wireless connectivity looks as if a user is attached to the Enterprise wireless network.

Figure 45 Wireless Access to Enterprise Zone Plant personnel



As indicated within the preceding Figure a mobile device connects to the Plant SSID, logs in and sends an 802.1x request which gets tunneled from the AP to the local WLC in the industrial Zone. The industrial WLC forwards the RADIUS authentication request on behalf of the client to the enterprise ISE PSN via IPsec tunnel. Next the PSN checks whether the device is BYOD or corporate issued and applies policies based on such. After devices are showing as compliant traffic flows from the resources in the enterprise zone through the EoIP tunnel and Industrial WLC to the corporate user mobile device and so on.

Wireless Access for Trusted Partners and Guests

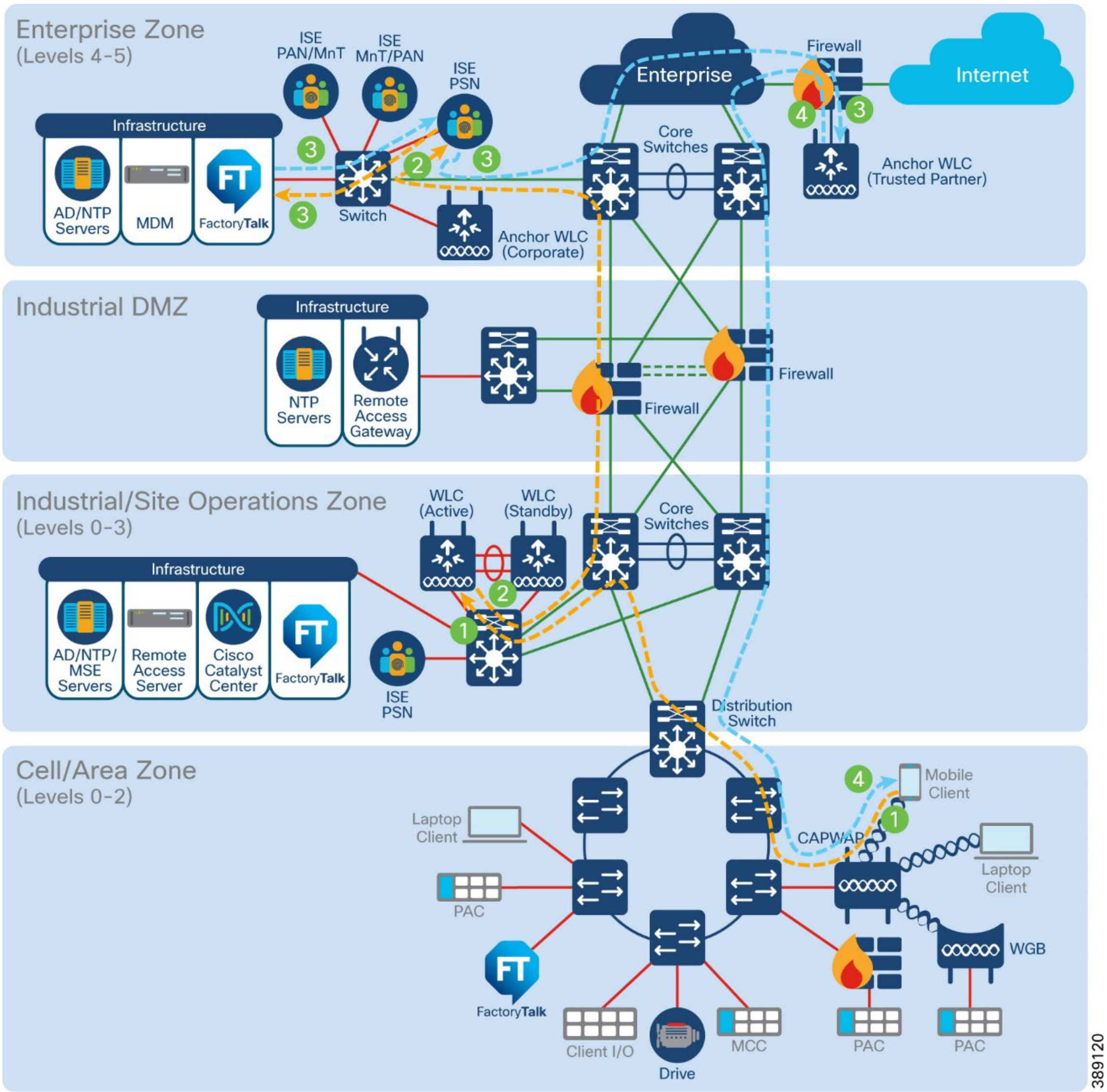
Trusted partners and guests may require access to the Enterprise Guest DMZ and the Internet to be able to:

- Establish VPN access to their company's corporate network as part of the normal workflow during commissioning and maintenance of the vendor's equipment.
- Access the Internet for IACS firmware downloads, documentation, and technical support.
- Establish secure remote access via VPN through IDMZ to the IACS equipment managed by the vendor

By default, authorization policies defined for guest users in the Enterprise Guest DMZ apply here, including access to the internet for and VPN portals for trusted partners. The contrast to typical guest access in Enterprise is that authentication is done by AD user account. Internet access may also be limited to whitelisted websites only. In addition, the follow access policies should be considered.

- Trusted partner or guest user with access to the Internet as permitted for a corporate guest user
- Trusted partner access to the VPN portal in the Enterprise DMZ for secure remote access to specific IACS equipment as permitted by manufacturer's security policy

Figure 46 Wireless Trusted Partner access flow



Within the topology, the use case is very similar to enterprise access as the wireless user's data is placed in a secure EoIP tunnel between the industrial WLC and the Guest WLC within the Enterprise Guest DMZ. The IMDZ firewall configuration allows the tunnel traffic between WLCs resulting in user traffic being isolated from the industrial Zone network traffic and terminated in the guest DMZ.

Design Considerations for WGB clients

This section outlines the design consideration of using WGB as clients for static and mobile use cases.

The WGB can associate only with Cisco lightweight access points unless it is configured as a universal Work Group Bridge then the uWGB can associate to a third-party access point.

WGBs can bridge STP packets. The wired network segments need to be properly configured to handle STP packets otherwise connection issues can occur. If administrator needs to disable bridging of STP between the wired segments by the WGB, we recommend disabling STP on the directly connected switches in the wireless network.

With Layer 3 roaming, if you plug a wired client into the WGB network after the WGB has roamed to another controller (for example, to a foreign controller), the wired client's IP address displays only on the anchor controller, not on the foreign controller.

When you DE authenticate a WGB record from a controller, all the WGB wired clients' entries are also deleted.

Associating a WGB to a WLAN that is configured for Adaptive 802.11r is not supported.

FlexConnect design considerations

- FlexConnect APs can be used to support static use cases or intra cell roaming. For seamless roaming to be supported on FlexConnect APs within the same roaming domain must utilize the same site tag. The max is 300 Aps and 3000 clients per site tag.
- Connect the FlexConnect AP to the 802.1Q trunk port on the switch.
- When connecting with a native VLAN on the AP, the native VLAN configuration on the Layer 2 must match the configuration on the AP.
- Ensure that the native VLAN is the same across all APs in the same location and site tag.

Wireless LAN QoS

In an Industrial wireless deployment, QoS settings for the access points are configured on the WLC and pushed down to the access points for consistency and ease of configuration. The default parameters for Cisco APs is optimized for applications most sensitive to latency, jitter and packet loss.

- It is recommended to use Platinum QoS level for IACs WLANs as this provides the best performance. The IACS traffic will be assigned the appropriate wireless QoS category (Best effort, Video, Voice) based on the QoS marking in the data packets.
- It is recommended to configure the Low latency profile on the WGB client that supports automation and IAC applications.
- For voice SSIDs it is recommended to use the "Fastlane" auto-QoS profile (and not the voice profile). This will enable the platinum QoS policy settings.
- For Guest SSId SSID it is recommended to use the QoS policy Bronze
- Regarding EDCA settings, remember that these settings are global per radio and not per SSID. There is no single recommended value for all networks, so it is important to test different values. For factory environments with voice and video traffic, it is a good idea to set the EDCA to "optimized-video-voice".

Multicast

This section provides design considerations for use of multicast over wireless in the IA network

- Multicast Frames are not acknowledged and not repeated if lost. This increases packet loss and the chance of time outs.
- Although multicast delivery is supported by the 802.11 standard It is recommended to use unicast with wireless I/O connections.

SSID/WLAN

This section gives SSID/WLAN related design recommendations.

- It is recommended that you enable broadcast SSID option to have the best client interoperability. Hiding the SSID does not provide additional security as it is always possible to obtain the SSID name by doing simple attacks, and it has secondary side effects, such as slower association for some client types (for example, Apple iOS).
- Application Visibility and Control (AVC) classifies applications using Cisco's deep packet inspection (DPI) techniques with the Network-Based Application Recognition (NBAR) engine and provides application-level visibility into and control of the WI-FI network. AVC inspection may have a performance impact of up to 30%, therefore it should be avoided on wireless controller setups that are running close to the maximum forwarding capacity.
- 802.11k standard allows clients to request neighbor reports containing information about known neighbor Aps that are candidates for roaming. It is recommended to enable this feature with dual-band reporting as this allows the client to receive a list of 2.4/5 GHz Aps upon request. In the end this logic reduces scan times and saves battery power.

  Note: Do not enable the dual-list option if using single-band clients or for deployment scenarios using primarily 5 GHz.
- Passive clients are clients that do not send DHCP nor ARP packets after authentication is complete. In other words, the client doesn't talk unless it is talked to. A printer with a static IP sitting idle is a good example. For this type of client to become operable and be able to receive and send traffic you need to configure the WLC to enable the passive client feature which instructs the controller to disable the IP learn timeout that would prevent the client from going to run state.

  Note: if traffic is centrally switched, then ARP broadcast needs to be enabled on the client VLAN

- If traffic is locally switched, then you need to disable ARP proxy under the Flex Profile so that ARP traffic can reach the client.

WI-FI 6E WLAN Security

WPA3 is the 3rd and latest iteration of the WI-FI Protected Access standard developed by the WI-FI Alliance and replaces the previous standard WPA2. This security feature protects against such dangerous attacks as Denial of Service (DoS), honeypots, and eavesdropping. By default, WPA3 uses 128-bit encryption, but it also introduces an optionally configurable SuiteB-192 bit192-bit cryptographic strength encryption using GMCP-256, which gives additional protection to any network transmitting sensitive data.

WPA3 Enterprise

- The WPA3-Enterprise is highly preferred and recommended to be used in sectors where network security is most critical.

- WPA3+WPA2 Enterprise mixed-mode configuration can be used in environments where legacy clients such as legacy scanners that can only support up to WPA2 while newer clients can support WPA3.
- WPA2+WPA3 enterprise transition mode can operate with pure WPA3 for the 6GHz band which allows users to enable WPA2+WPA3 on the same WLAN. In this mode 2.4/5 GHZ can use the parameter and only WPA3 relevant configurations will be pushed on the 6GHz band.

WPA3 Personal

- WPA3 Personal uses 128-bit cryptographic-strength encryption with password-based authentication method through Simultaneous Authentication of Equals (SAE) for user authentication purposes. WPA3 Personal heightens network security against offline dictionary attacks by limiting password guesses and requiring users to interact with a live network every time they do so. This persuades attempts at a brute force attack.
- WPA3-Personal Transition Mode also known as WPA2+WPA3 Personal mixed, is used like WPA3 Enterprise transition in which clients which are capable of only supporting WPA2 while others can support up to WPA3. This mode should only be used when necessary as it is recommended to use only WPA3 and not mix.
- As stated previously transition mode can operate with pure WPA3 for the 6GHz band within the same WLAN. Starting from 17.12.1 software version Cisco supports WPA2+WPA3 being used on 2.4/5 GHz and only WPA3 relevant configs are pushed on the 6GHz band when the WLAN has both configurations.

Network Controller and Access Point Settings

This section covers the recommended design considerations for the uplink ports connected the wireless infrastructure.

The C9800 WLC functions as a layer 2 host from a network perspective, therefore it doesn’t participate in Spanning Tree. To speed up convergence time, it is recommended that you enable PortFast or PortFast trunk configuration for the uplink ports where the C9800 is connected.

In any environment it is important to avoid unnecessary work by the WLC data plane and prevent network loops, therefore it is advised to only allow the required VLANs. Specifically, this includes the wireless management interface VLAN and any centrally switched client VLANs, all other VLANs should be pruned.

ARP proxy allows the controller to forward ARP traffic by changing the destination MAC from broadcast to unicast. This is the recommended setting as it helps save battery life on wireless devices because the WLC will answer ARP on the device’s behalf.

WI-FI Security settings design considerations

This section covers the security considerations when deploying the WI-FI infrastructure.

- Trust point is a certificate authority (CA) that you trust. This is used for providing certificate management for various functions and protocols such as HTTPS, Secure Shell (SSH) and so on. It is recommended to statically assign the trust point used for HTTPS GUI access to the WLC.
- For increased security configure 802.1x authentication between the AP and the switch. The AP acts as an 802.1X supplicant and is authenticated by the switch using EAP-FAST, EAP-PEAP, or EAP-TLS (Extensible Authentication Protocol [EAP] - Flexible Authentication via Secure Tunneling [FAST],

Protected EAP [PEAP], or Transport Layer Security [TLS])

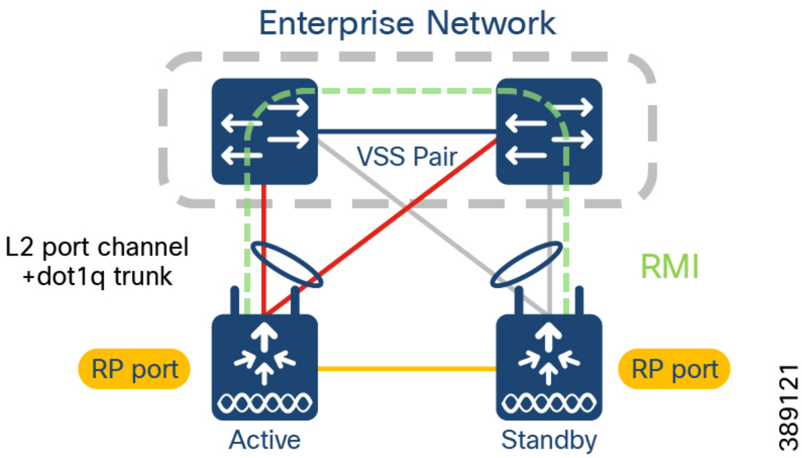
- Aironet IE is used by Cisco devices for better connectivity and troubleshooting providing information such as the AP name, load, and number of associated clients within the probe responses on the WLAN sent by the AP. This setting is recommended when using voice devices (8821 phones) and workgroup bridges.
- It is advisable to always have client exclusion configured on all WLANs. Client exclusion can act as a protective mechanism for the AAA servers, as it will stop authentication request floods that could be triggered by misconfigured clients.

High Availability design considerations

Implementing high availability within the wireless infrastructure involves multiple components and functionality deployed throughout the overall network infrastructure, which itself must be designed for high availability. This section discusses the high availability design considerations with respect to implementation of wireless controller platforms.

- Cisco WLCs support AP stateful switchover and client stateful switchover, they work hand and hand as High Availability and SSO. HA SSO is the recommended mode for providing high availability.
- The wireless mobility MAC is the MAC address used for mobility communication. In an SSO scenario, ensure that you explicitly configure the wireless mobility MAC address; otherwise, the mobility tunnel will go down after SSO.
- Device hardware and software must be identical for forming SSO pairs. Ex C9800-L-C cannot pair with C9800-L-F
- Configure using Redundant Management Interface (RMI) for dual active detection
- Set keep-alive retries to 5
- Set the higher priority (2) on the chassis that should be active.
- For RMI + RP (Redundancy Port) renumber the chassis prior to configuring to avoid Active-Active scenarios
- Link aggregation (LAG) mode is the preferred mode of operation as it provides redundancy and additional network bandwidth. If there are multiple physical links to the same uplink switch, then LAG should be used. Some important notes to remember:
    - All ports on the controller should have the same Layer 2 configuration matching the switch side.
    - For optimal load balancing among the physical ports in the port channel, use the src-dst-mixed-ip-port option. This must match on the controller as well as the neighbor switch.
    - The mode (Static mode On or (LACP) must be chosen on all interfaces partaking in the port channel group.

Figure 47 illustrates a recommended topology for high availability



In this design a single l2 port-channel on each box with dot1q enabled to carry multiple VLANs. The uplinks of the HA pair are spread across the VSS pair, and the Redundancy Port is connected back-to-back. The redundancy Management Interface is enabled for dual active detection. In addition to HA SSO, there are other methods of resiliency within the WLAN architecture such as rolling AP updates, Software Maintenance updates (SMU) , In service software Image upgrades and more. For more information, see the High Availability SSO Deployment Guide:
https://www.cisco.com/c/en/us/td/docs/wireless/controller/technotes/8-8/b_c9800_wireless_controller_ha_sso_dg.html

Key considerations when integrating Cisco Catalyst Center for Industrial Automation in Wireless

- Cisco Catalyst Center is recommended to be placed in the industrial zone with the following reasons:
  - Catalyst Center performs critical functions to maintain the operational status of the transit operation environment. Those critical functions include assurance and monitoring of the production network, guided remediation of identified problems and device replacement.
  - A separate instance for industrial environments helps ensure operational requirements are maintained. Industrial production environments have significantly higher and different operational requirements than an Enterprise system. A Catalyst Center instance that supports both Enterprise and Operation networks may lead to inadvertent changes or updates impacting the production system that could lead to downtime.
- Cisco Catalyst Center requires connectivity to all network devices it manages. That means that all devices that need to be discovered and monitored should have an IP address assigned that is routable and able to reach the Cisco Catalyst Center.
- Cisco Catalyst Center requires Internet connectivity to download software updates, licenses, device software, up-to-date map information and user feedback etc. It is recommended to use an IDMZ-based proxy service to provide this external communication. It is also recommended that you allow secure access via the proxy service only to URLs and fully qualified domain names required by Cisco Catalyst Center. For more details refer to Cisco Catalyst Center Security Best Practice Guide.

- If there is a firewall between Cisco Catalyst Center and managed devices, ensure that the required ports are allowed on the firewall. See “Required Network Ports” in Cisco DNA Center Second-Generation Appliance Installation Guide for a list of network ports that are required to be allowed by the firewall.
- Latency should be equal to or less than 100 milliseconds to achieve optimal performance for all solutions provided by Cisco Catalyst Center. The maximum supported latency is 200ms RTT. Latency between 100ms and 200ms is supported, although longer execution times could be experienced for certain functions including Inventory Collection and other processes that involve interactions with the managed devices.
- Cisco ISE must be deployed with a version compatible with Cisco Catalyst Center. Refer to the following link for compatibility information https://www.cisco.com/c/en/us/support/cloud-systems-management/dna-center/products-device-support-tables-list.html

Some known limitations of Cisco Catalyst Center include the following:

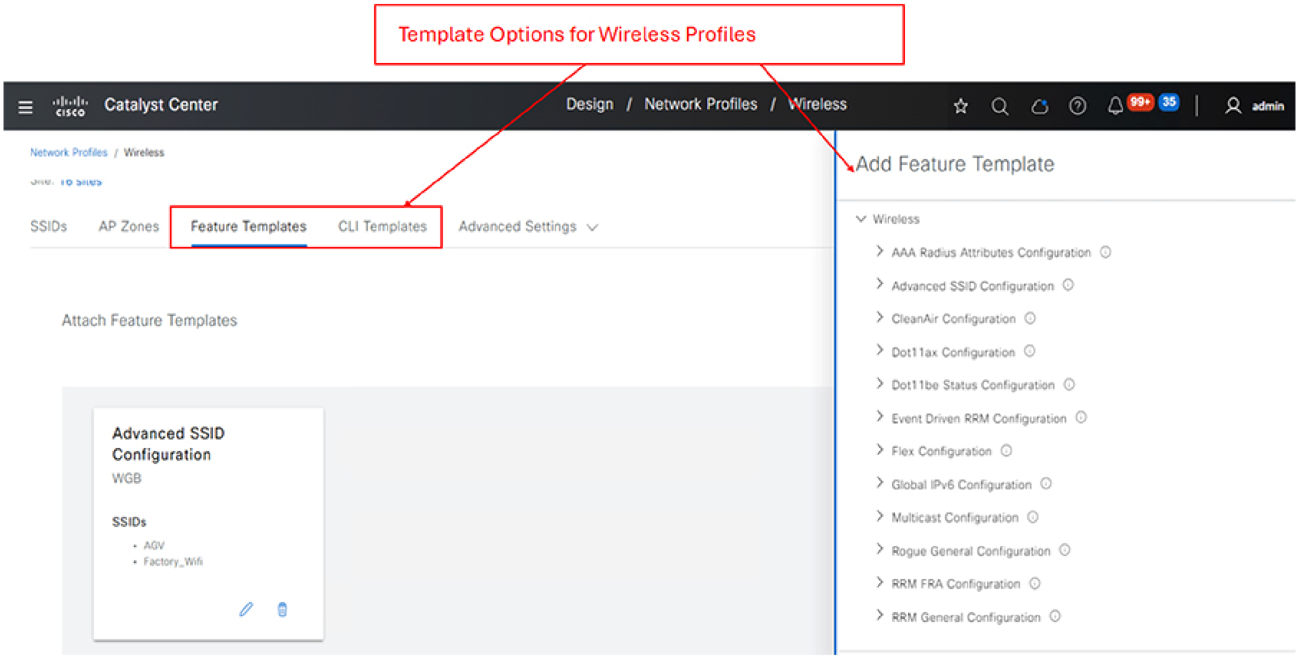
- Cisco Catalyst Center does not support managing network devices with management IP address behind a Network Address Translation (NAT) boundary.
- Firewalls running Firepower Threat Defense (FTD) software are not supported on Cisco Catalyst Center. However, devices that are connected behind an industrial firewall can be provisioned and managed by Cisco Catalyst Center.
- REP ring provisioning using REP automation workflow is supported by Cisco Catalyst Center. However, the ring of rings (also known as a subtended ring) REP provisioning is not automated in Cisco Catalyst Center workflows. We recommend using the Day-N template feature in Cisco Catalyst Center for day N configurations and Wireless LAN Controller provisioning.
- Cisco Catalyst Center cannot manage products from third-party vendors.

Day N Operations and Templates

Catalyst Center provides an interactive template hub to author CLI and feature templates. You can easily design templates with a predefined configuration by using parameterized elements or variables. Templates allow you to define a configuration of CLI commands that can be used to consistently configure multiple network devices, reducing deployment time. After creating a template, you can use the template and apply it to one or more network profiles. Network profiles allow you to configure settings and apply them to a specific site or group of sites.

The following diagram displays the wireless feature templates that can be deployed within Catalyst Center.

Figure 48 Wireless Feature templates



Additionally, network devices lifecycle management, which includes running configuration compliance checks, guided troubleshooting, software image management (SWIM), and network assurance features are included in day N operations and management.

See following guides for more information about Cisco Catalyst Center in Industrial IoT applications:

- Renewable Energy – Offshore Wind Farm Design Guide
- Renewable Energy – Offshore Wind Farm Implementation Guide
- Industrial Automation Network Design Guide

Summary

Industrial wireless networking can provide lower-cost and faster deployment options, can better monitor assets and operations, makes possible autonomous and remote-controlled robots and vehicles, and improves collaboration between mobile workers. Organizations considering using wireless in their operations are advised to evaluate available technologies and pick the one that best fits their particular use case. Apart from the technical criteria, such as bandwidth and latency, they should consider availability, cost of ownership, and the projected evolution in making their decision, while understanding that they will likely need more than one technology for their operations. The Cisco multiaccess networking portfolio provides the full breadth of wireless equipment for its foundation.

Acronyms – Initialisms

Term	Definition
AAA	Authentication, Authorization, and Accounting
ACL	Access Control List
AVC	Application Visibility and Control
CCTV	Closed Circuit Television
CTS	Cisco TrustSec
CVD	Cisco Validated Design
DC	Data Center
DHCP	Dynamic Host Configuration Protocol
DMZ	De-militarized Zone
DNS	Domain Name System
FEC	Forwarding Equivalence Class
FTD	Firepower Threat Defense
FW	Firewall
GE	Gigabit Ethernet
HA	High Availability
IA	Industry Automation
IACS	Industrial Automation and Control Systems
IDS	Intrusion Detection System
IE	Industrial Ethernet
IoT OD	IOT Operations Dashboard
IPS	Intrusion Prevention System
ISE	Identity Services Engine
LER	Label Edge Router
L2TP	Layer 2 Tunneling Protocol

ICS	Industrial Control System
LSP	Label Switched Path
LSR	Label Switched Router
IT	Information Technology
MAB	MAC Authentication Bypass
MAC	Media Access Control
ME	Mesh End
MP	Mesh Point
MPLS	Multi-protocol Label Switching
MTTD	Mean Time To Detect
MTTR	Mean Time To Repair
NIST	National Institute of Standards and Technology
NAT	Network Address Translation
NTP	Network Time Protocol
OCC	Operational Control Center
OT	Operational Technology
PnP	Plug and Play
pxGrid	Platform eXchange Grid
RADIUS	Remote Authentication Dial-In User Service
REP	Resilient Ethernet Protocol
SD-Access	Software-defined Access
SDN	Software Defined Networking
SEA	Secure Equipment Access
SGACL	Security Group-based Access Control List
SGTs	Scalable Group Tags

SLA	Service Level Agreement
SSL	Secure Sockets Layer
SSID	Service Set Identifier
STCN	Segment Topology Change Notification
STP	Spanning Tree Protocol
SVI	Switch Virtual Interface
SVL	StackWise Virtual Link
SWIM	Software Image Management
SXP	SGT eXchange Protocol
TLS	Transport Layer Security
UCS	Cisco Unified Computing System
UDP	User Datagram Protocol
URWB	Ultra-Reliable Wireless Backhaul
VIP	Virtual IP Address
VLAN	Virtual Local Area Network
WLC	Wireless LAN Controller
WLAN	Wireless Local Area Network
ZTNA	Zero Trust Network Access